

ImPRESS — Biomarker Model Fit Assessment

ImPRESS Statistical Team

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1 Introduction

This document provides a pre-unblinding assessment of model fit for the **33 ADLB analytes** from Bio-Plex 19-plex and 14-plex serum panels in **54 subjects** (cohort 2).

The primary analysis (`make_rdlb.R`) fits two competing LMMs per analyte:

Model	Response	Interpretation
Linear	<code>chg = aval - base</code>	Mean change from Screening baseline (pg/mL)
Log-ratio	<code>log(aval/base)</code> <code>log_chg</code>	Mean log-fold change from Screening baseline

For both models the fixed effects are: baseline value, stepped-wedge dose (`trtcd`), and visit day (`avisitn`), with a random intercept per subject. AIC selects the preferred model per analyte.

This report assesses:

1. Data completeness and out-of-range proportions
2. Concentration distributions (raw and log scale)
3. Change-score distributions
4. Residual diagnostics (both models) per analyte
5. AIC comparison summary and recommendations

2 Data overview

Table 2: Table 1: Data completeness by analyte (analysis cohort)

PARAMCD	Parameter	Panel	N obs	N subjects	% missing AVAL	N >ULOQ	N <LLOQ
FRACKTN	Fractalkine	14- PLEX	191	54	0.0	0	0
GZMB	Granzyme B	14- PLEX	191	54	0.0	0	0
IFNG	IFN-g	14- PLEX	191	54	0.0	0	0
IL1B	IL-1b	14- PLEX	191	54	0.0	0	0
IL6	IL-6	14- PLEX	191	54	0.0	0	0
IL8	IL-8	14- PLEX	191	54	0.0	0	0
MCP1	MCP-1	14- PLEX	191	54	0.0	0	0
MMP10	MMP-10	14- PLEX	191	54	0.0	0	0
PDGFDD	PDGF-DD	14- PLEX	191	54	0.0	0	0
PECAM1	PECAM-1	14- PLEX	191	54	0.5	0	0
TGFA	TGF-alpha	14- PLEX	191	54	0.0	0	0
THBS2	Thrombospondin-2	14- PLEX	191	54	0.0	0	0
VEGFA	VEGF-A	14- PLEX	191	54	0.0	0	0
VEGFC	VEGF-C	14- PLEX	191	54	0.0	0	0
ANGPT1	Angiopoietin-1	19- PLEX	191	54	0.0	0	0
ANGPT2	Angiopoietin-2	19- PLEX	191	54	0.0	0	0
ANGPTL4	ANGPTL4	19- PLEX	191	54	0.0	0	0
CA9	CA9	19- PLEX	191	54	0.0	0	0
CD163	CD163	19- PLEX	191	54	0.0	0	0
COLIVA1	Collagen IV alpha 1	19- PLEX	191	54	0.5	0	0
EGF	EGF	19- PLEX	191	54	0.0	0	0
ERBB2	ErbB2	19- PLEX	191	54	0.0	0	0
FGFB	FGF Basic	19- PLEX	191	54	0.0	0	0
IL12P70	IL-12 p70	19- PLEX	191	54	0.0	0	0
IL2	IL-2	19- PLEX	191	54	0.0	0	0

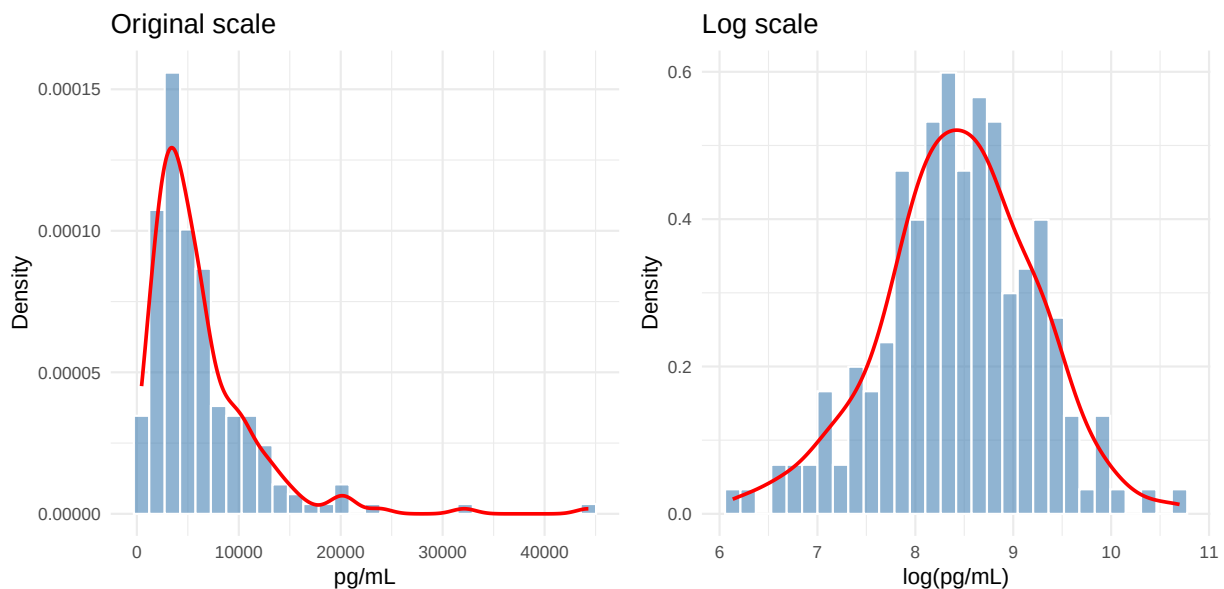
PARAMCD	Parameter	Panel	N obs	N subjects	% missing AVAL	N >ULOQ	N <LLOQ
IP10	IP-10 (CXCL10)	19- PLEX	191	54	0.5	0	0
ITAC	I-TAC (CXCL11)	19- PLEX	191	54	0.0	0	0
MIG	MIG (CXCL9)	19- PLEX	191	54	0.0	0	0
PIGF	PIGF	19- PLEX	191	54	0.0	0	0
TENASC	Tenascin C	19- PLEX	191	54	0.0	0	0
TNFRSF18	TNFRSF18 (GITR)	19- PLEX	191	54	0.0	0	0
TNFSF6	TNFSF6 (FasL)	19- PLEX	191	54	0.0	0	0
VEGFR2	VEGFR2	19- PLEX	191	54	0.0	0	0

Note: >ULOQ values are set to NA in **aval**. If >ULOQ exceeds ~20% of observations for an analyte, consider a Tobit (censored) mixed model for that analyte.

3 Concentration distributions

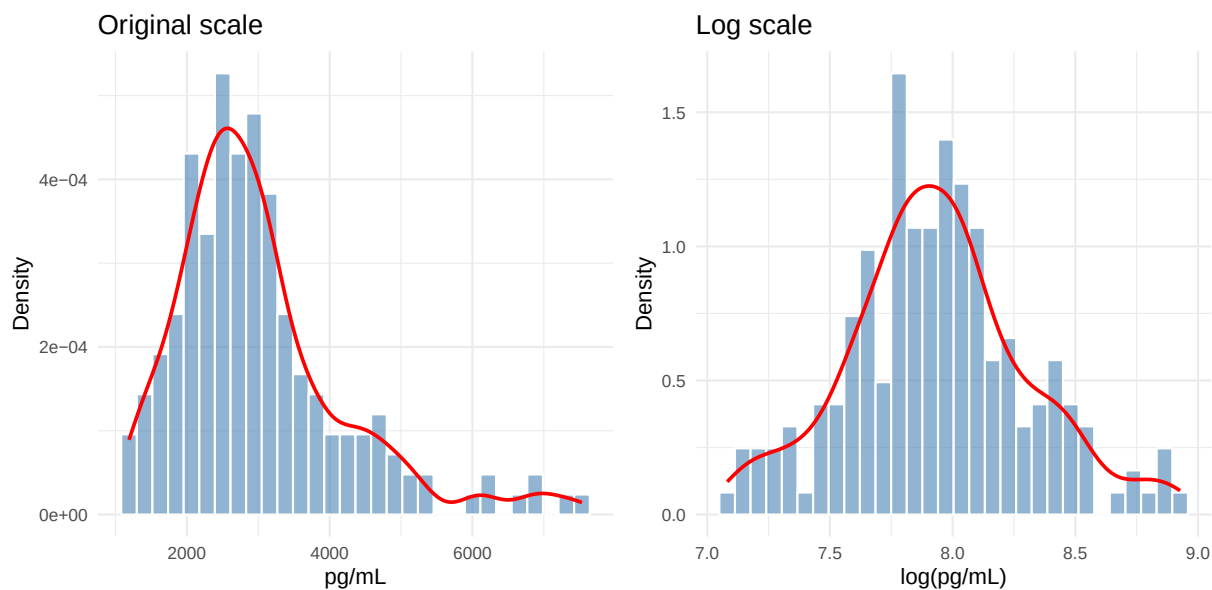
3.1 Angiopoietin-1 (ANGPT1)

Skewness: original scale = 3.02 | log scale = -0.19



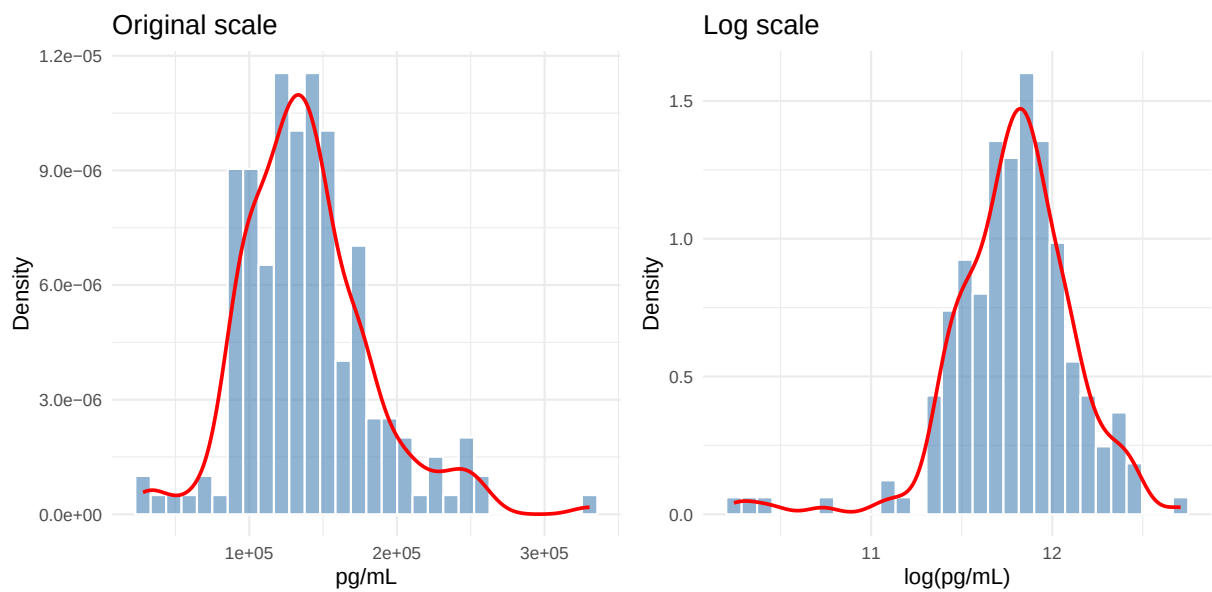
3.2 Angiopoietin-2 (ANGPT2)

Skewness: original scale = 1.39 | log scale = 0.21



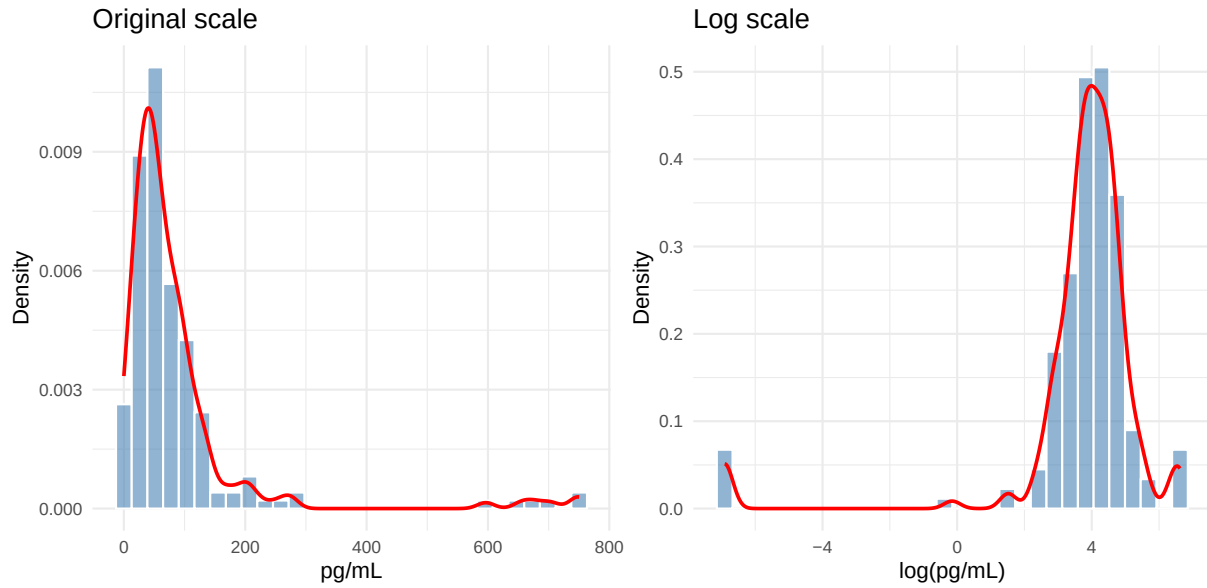
3.3 ANGPTL4 (ANGPTL4)

Skewness: original scale = 0.81 | log scale = -1.1



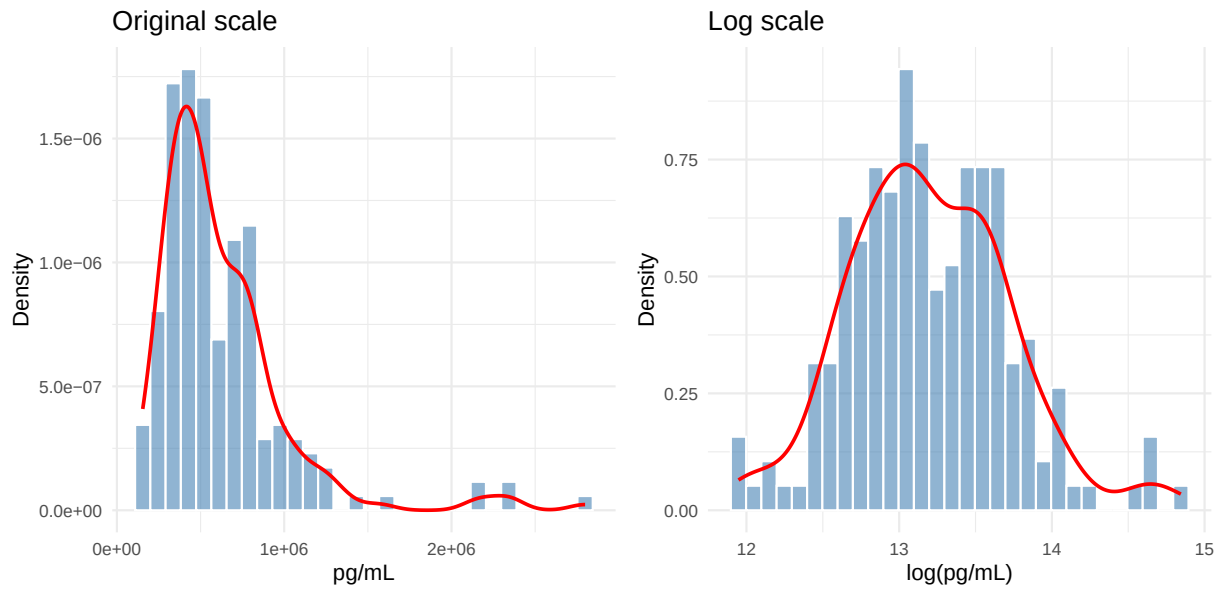
3.4 CA9 (CA9)

Skewness: original scale = 4.08 | log scale = -3.89



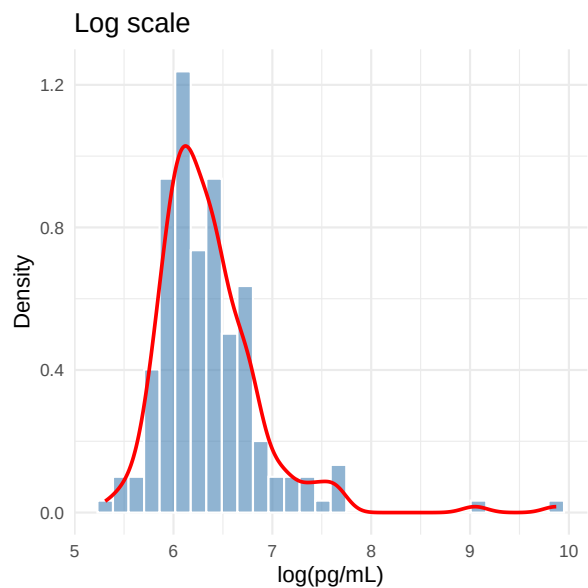
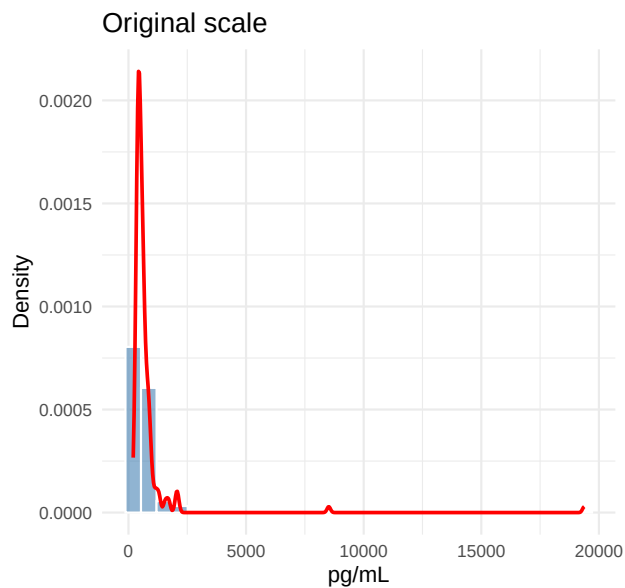
3.5 CD163 (CD163)

Skewness: original scale = 2.55 | log scale = 0.32



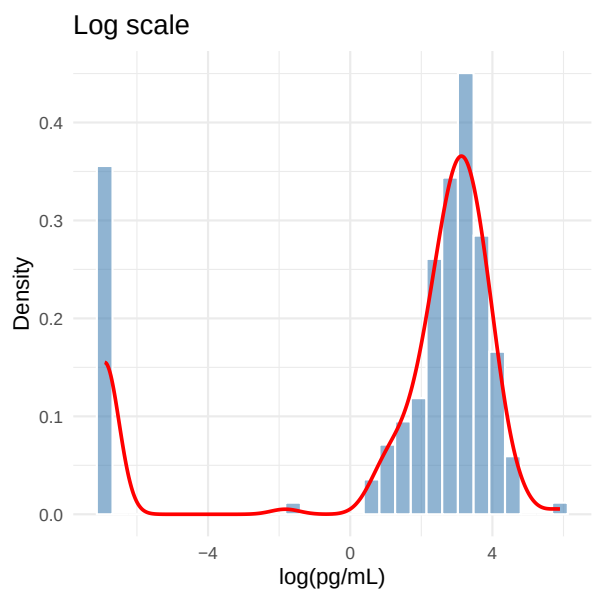
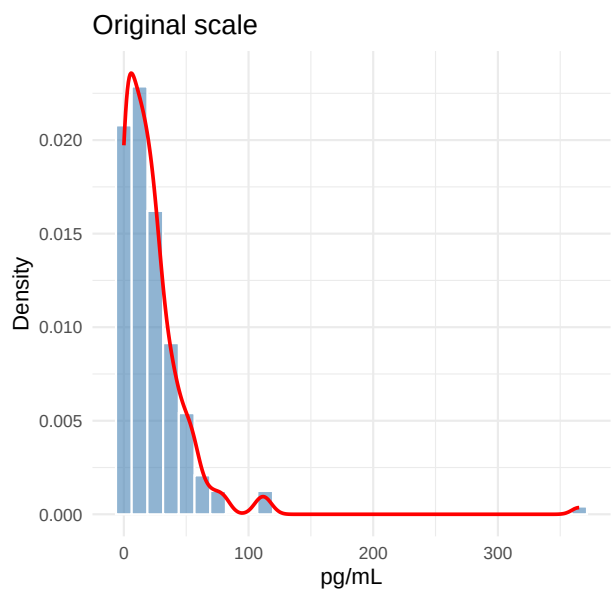
3.6 Collagen IV alpha 1 (COL1A1)

Skewness: original scale = 10.56 | log scale = 2.44



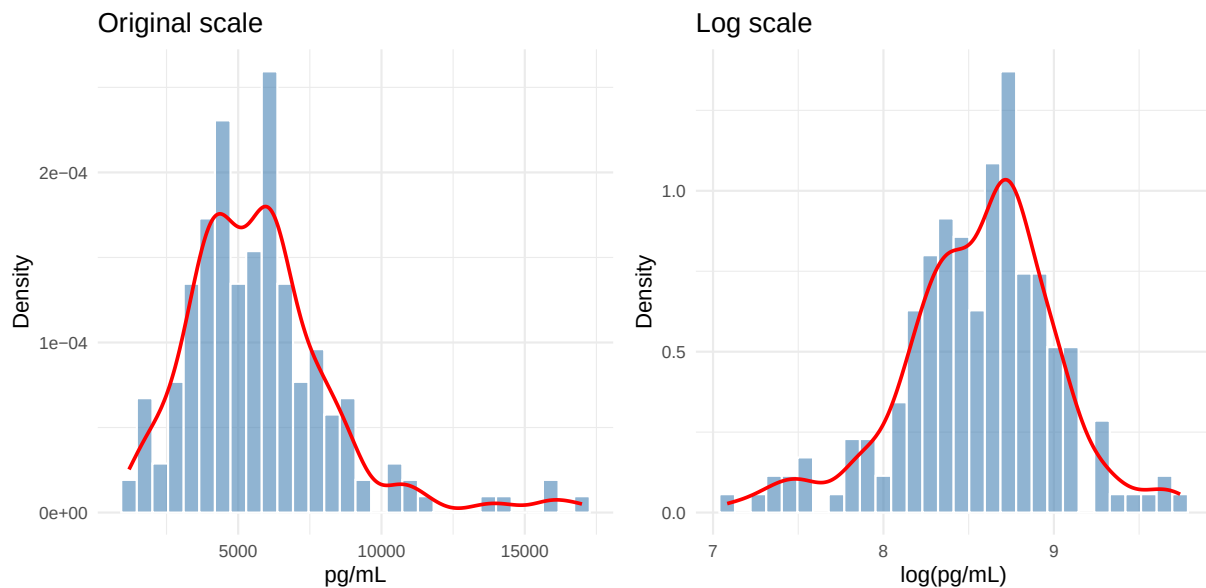
3.7 EGF (EGF)

Skewness: original scale = 6.35 | log scale = -1.65



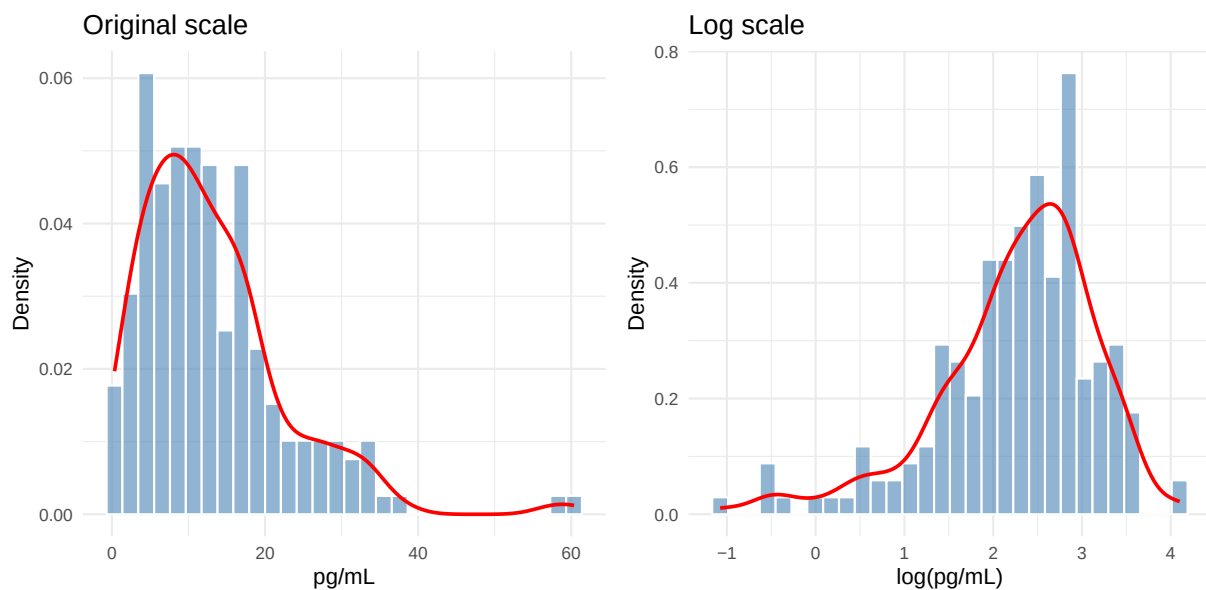
3.8 ErbB2 (ERBB2)

Skewness: original scale = 1.43 | log scale = -0.44



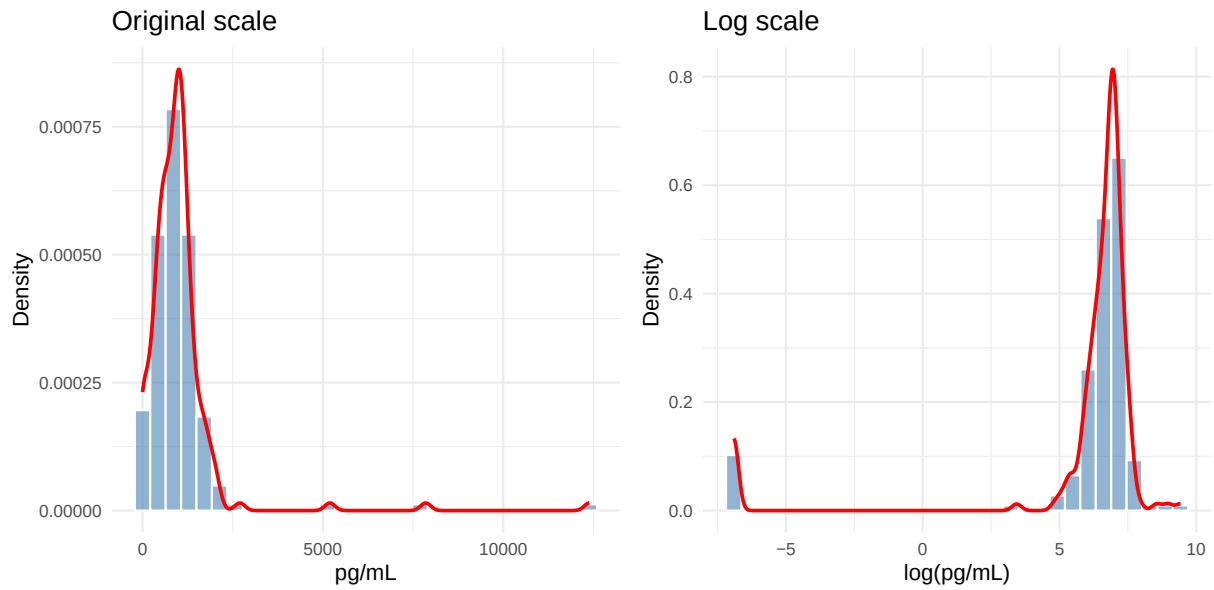
3.9 FGF Basic (FGFB)

Skewness: original scale = 1.59 | log scale = -1.04



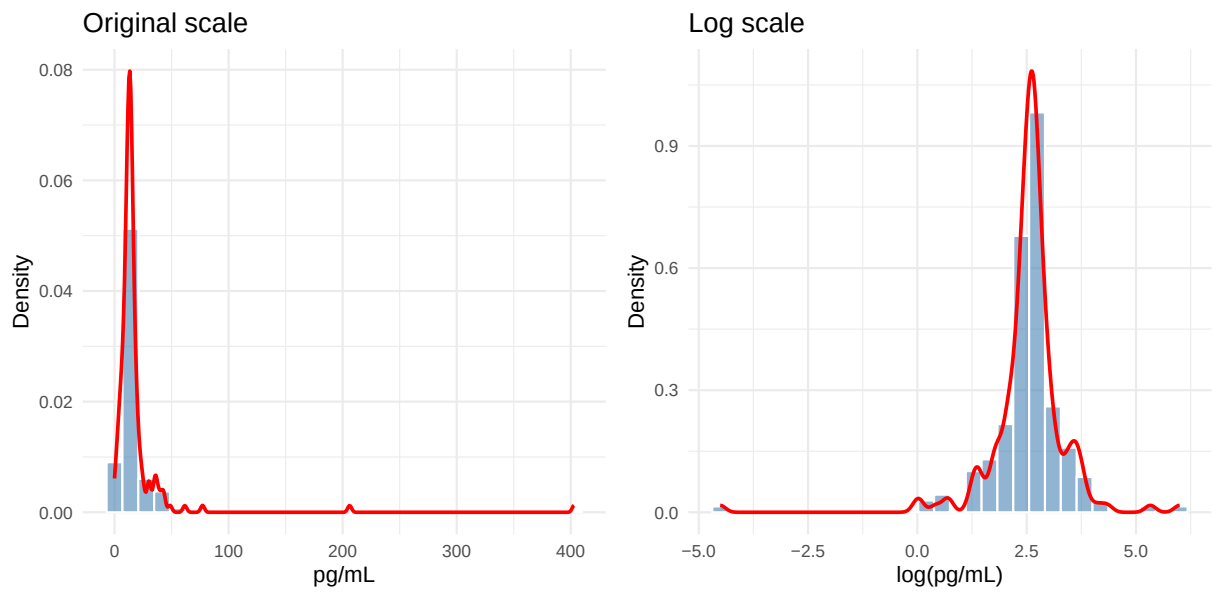
3.10 Fractalkine (FRACTKN)

Skewness: original scale = 6.82 | log scale = -3.53



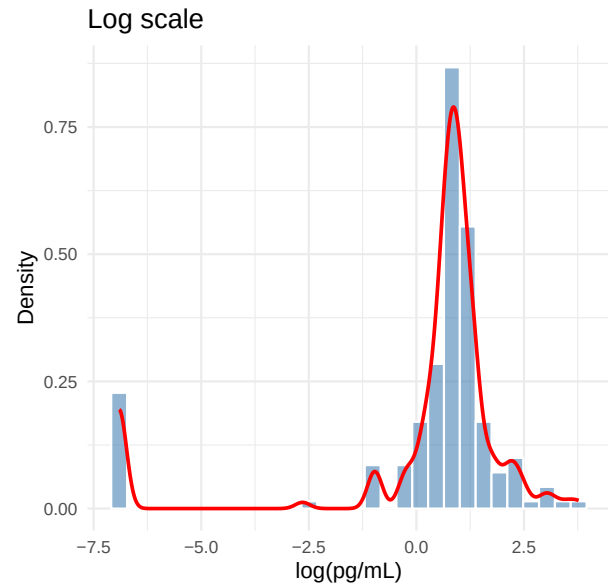
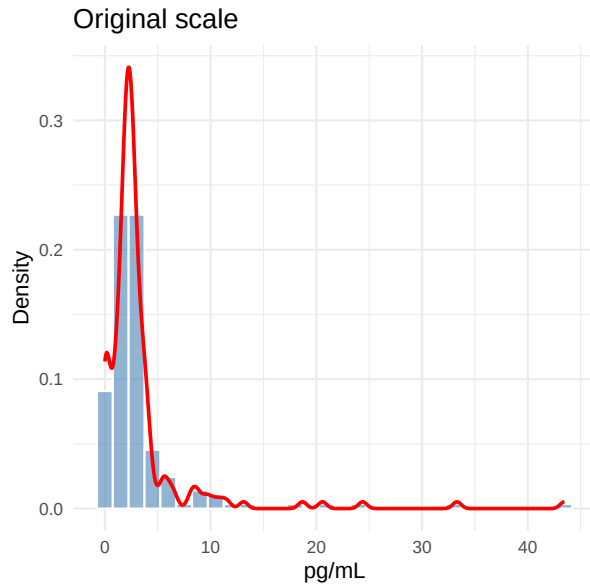
3.11 Granzyme B (GZMB)

Skewness: original scale = 9.45 | log scale = -2.56



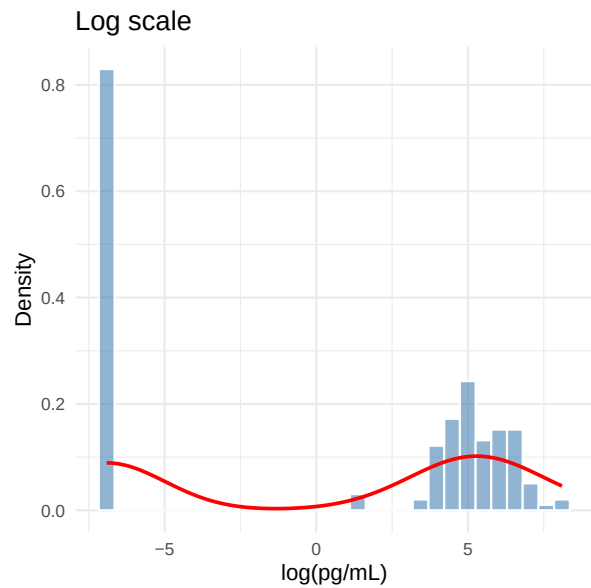
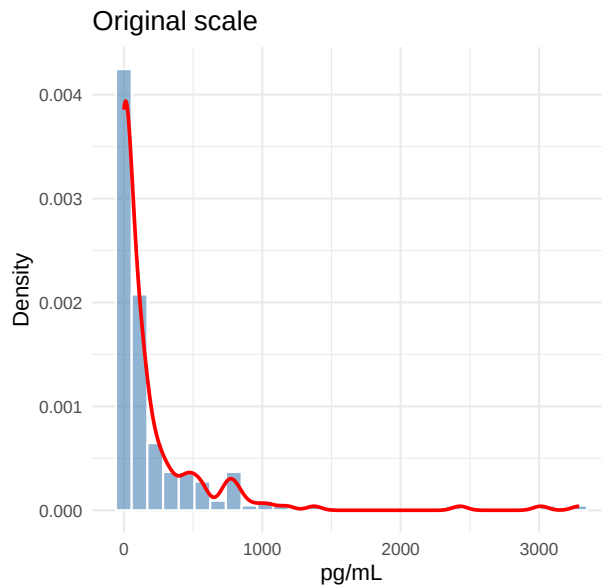
3.12 IFN-g (IFNG)

Skewness: original scale = 5.16 | log scale = -2.41



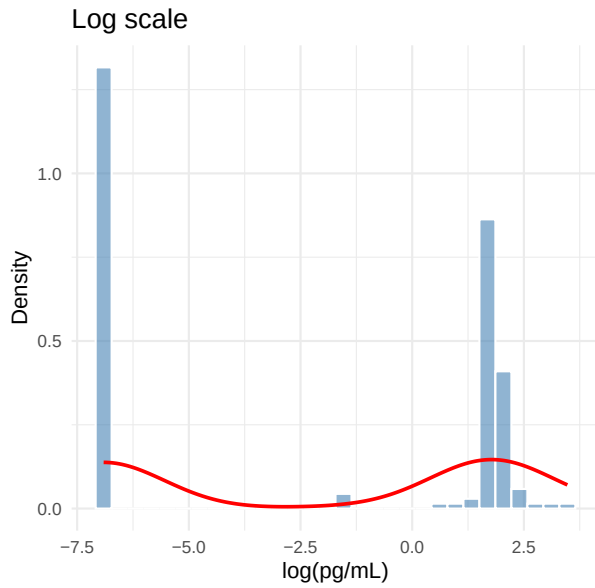
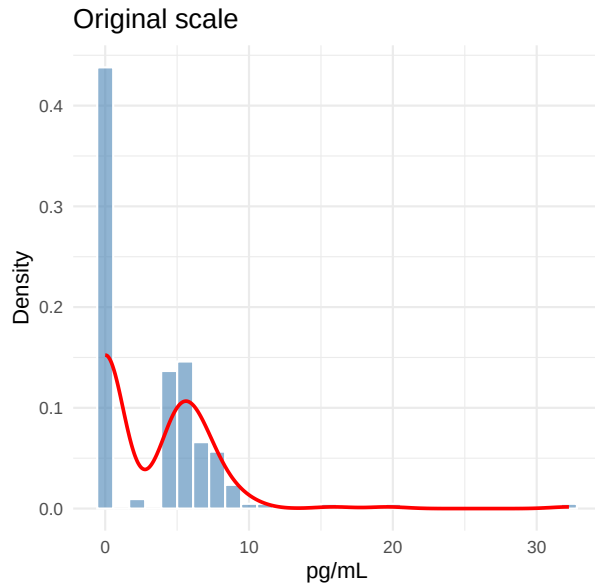
3.13 IL-12 p70 (IL12P70)

Skewness: original scale = 4.4 | log scale = -0.22



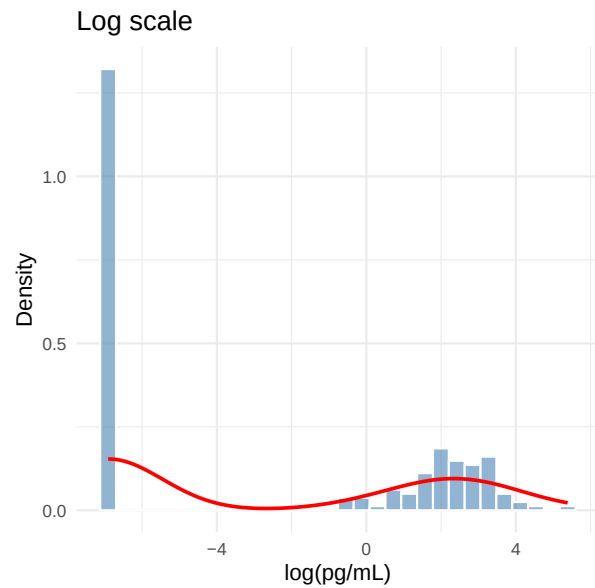
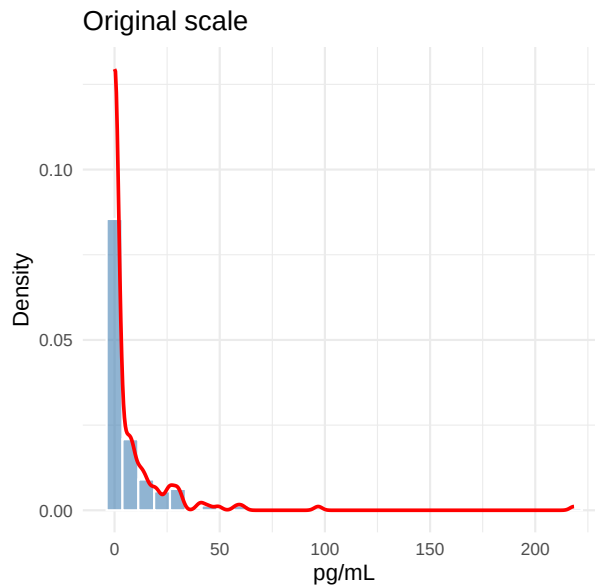
3.14 IL-1b (IL1B)

Skewness: original scale = 2.38 | log scale = -0.08



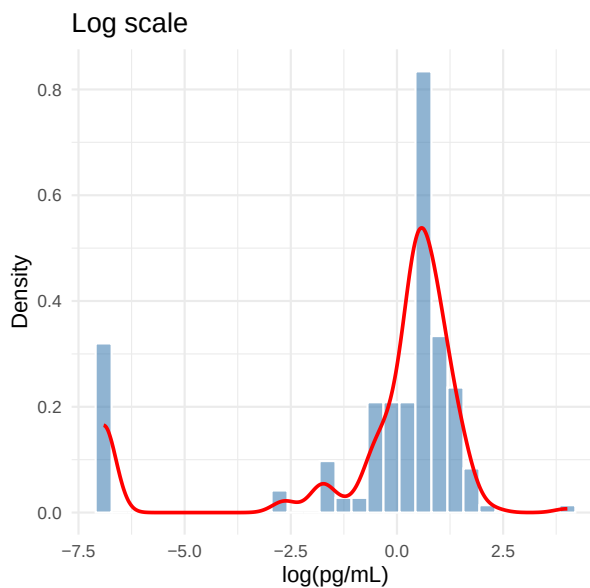
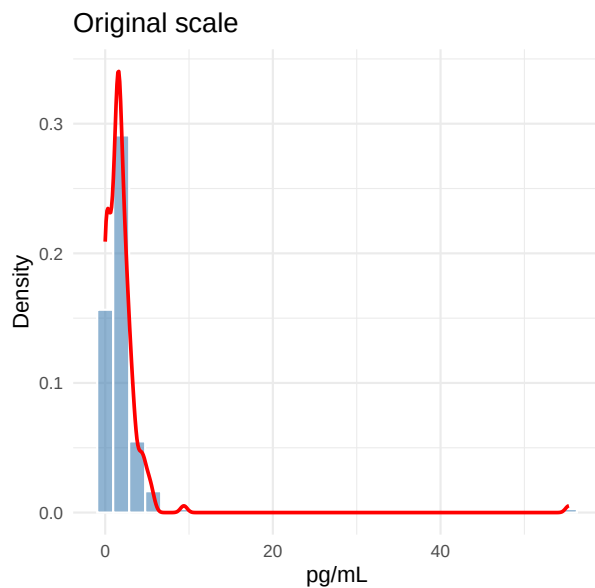
3.15 IL-2 (IL2)

Skewness: original scale = 6.86 | log scale = 0.32



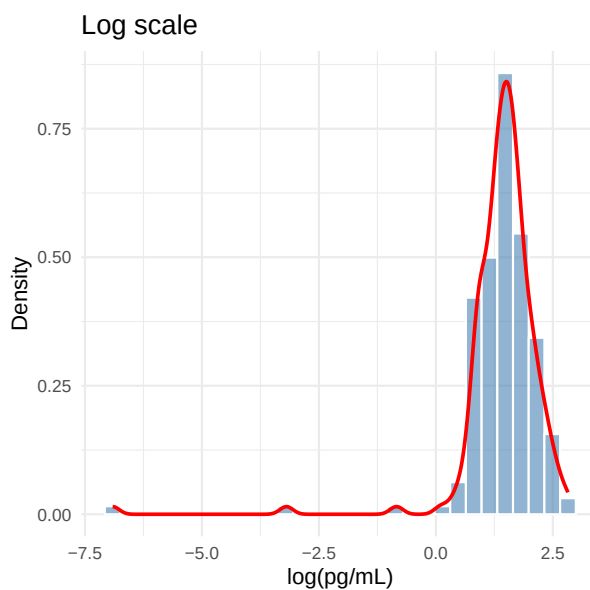
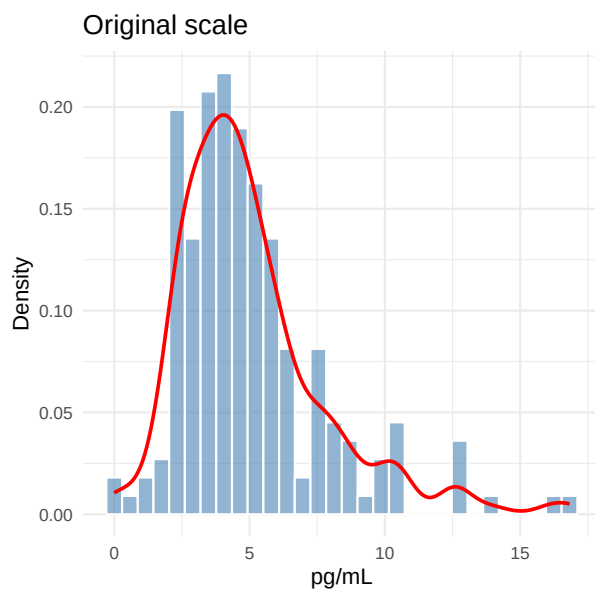
3.16 IL-6 (IL6)

Skewness: original scale = 11.36 | log scale = -1.86



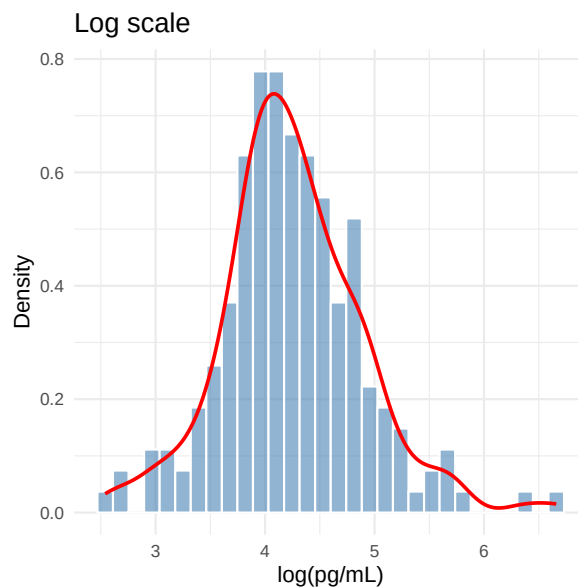
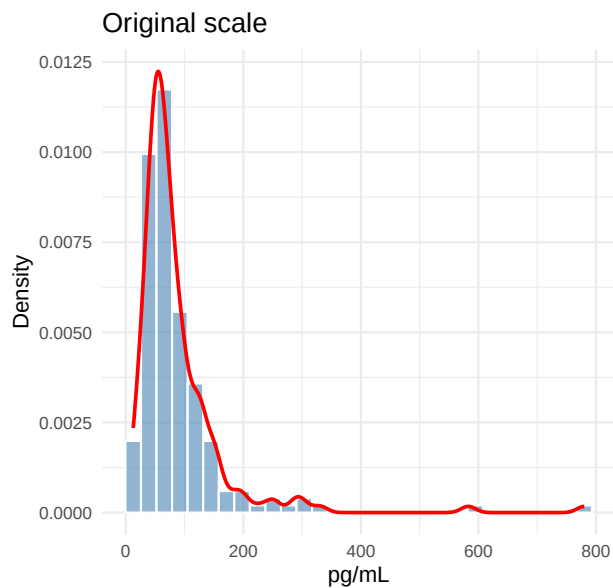
3.17 IL-8 (IL8)

Skewness: original scale = 1.47 | log scale = -5.5



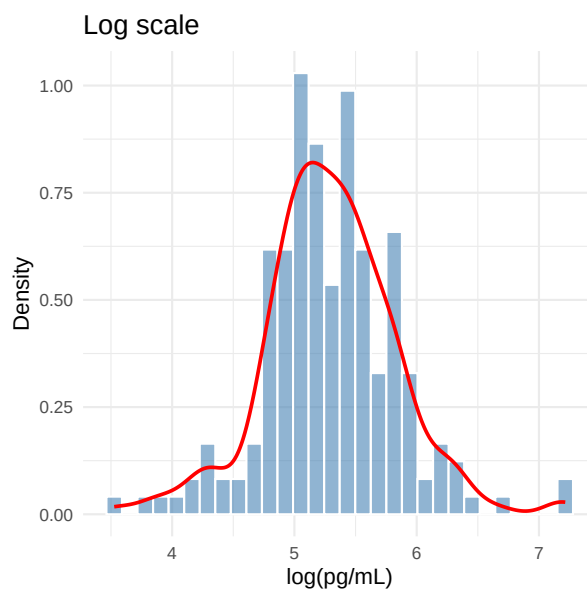
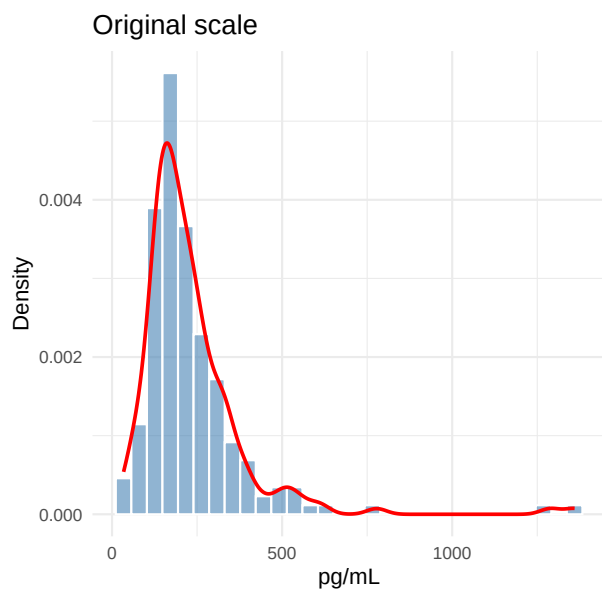
3.18 IP-10 (CXCL10) (IP10)

Skewness: original scale = 4.71 | log scale = 0.4



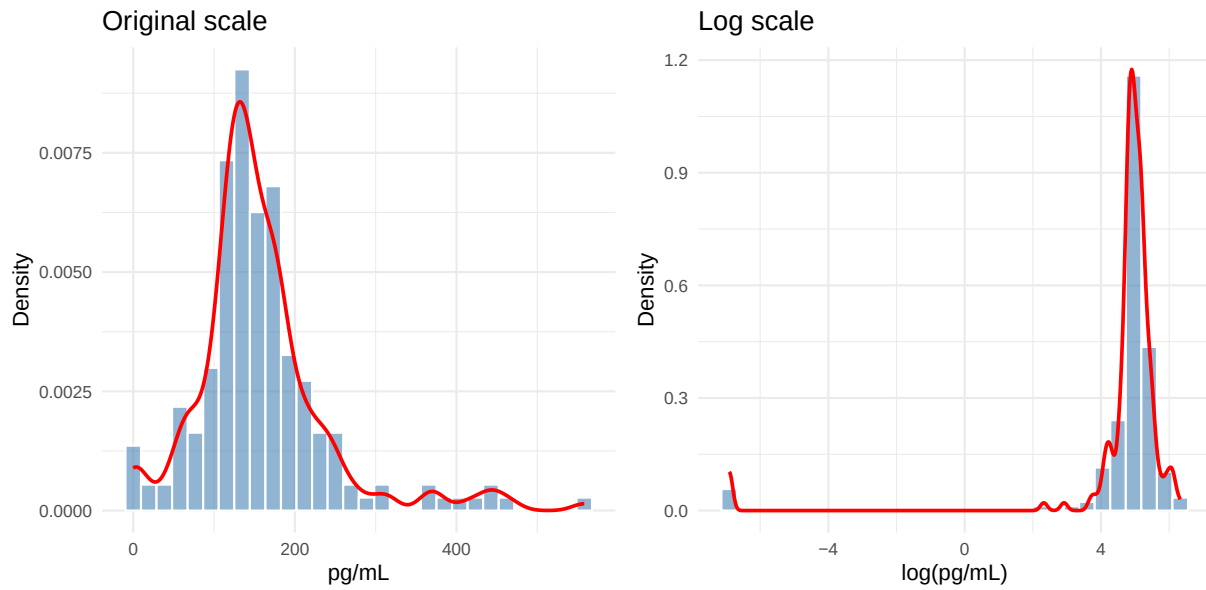
3.19 I-TAC (CXCL11) (ITAC)

Skewness: original scale = 3.74 | log scale = 0.11



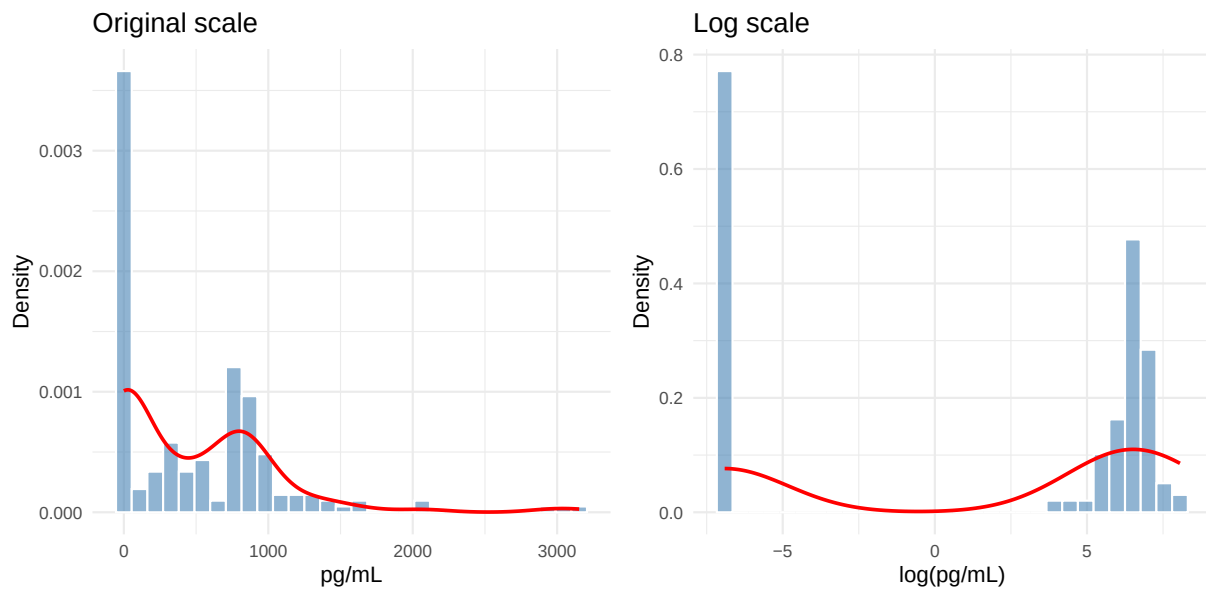
3.20 MCP-1 (MCP1)

Skewness: original scale = 1.65 | log scale = -5.34



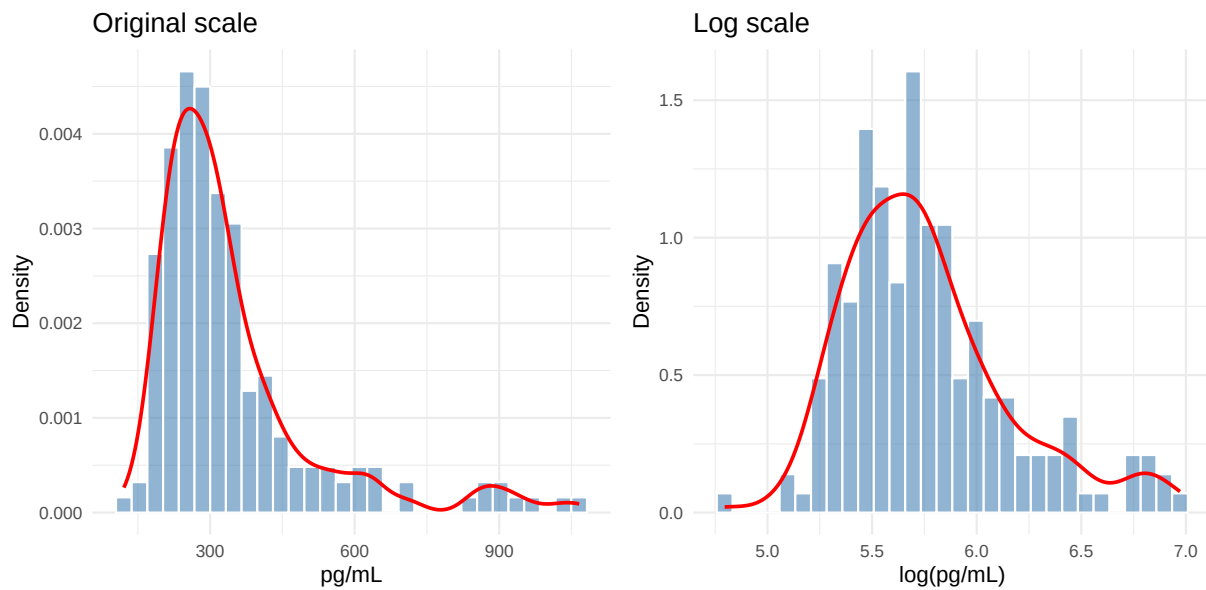
3.21 MIG (CXCL9) (MIG)

Skewness: original scale = 1.74 | log scale = -0.39



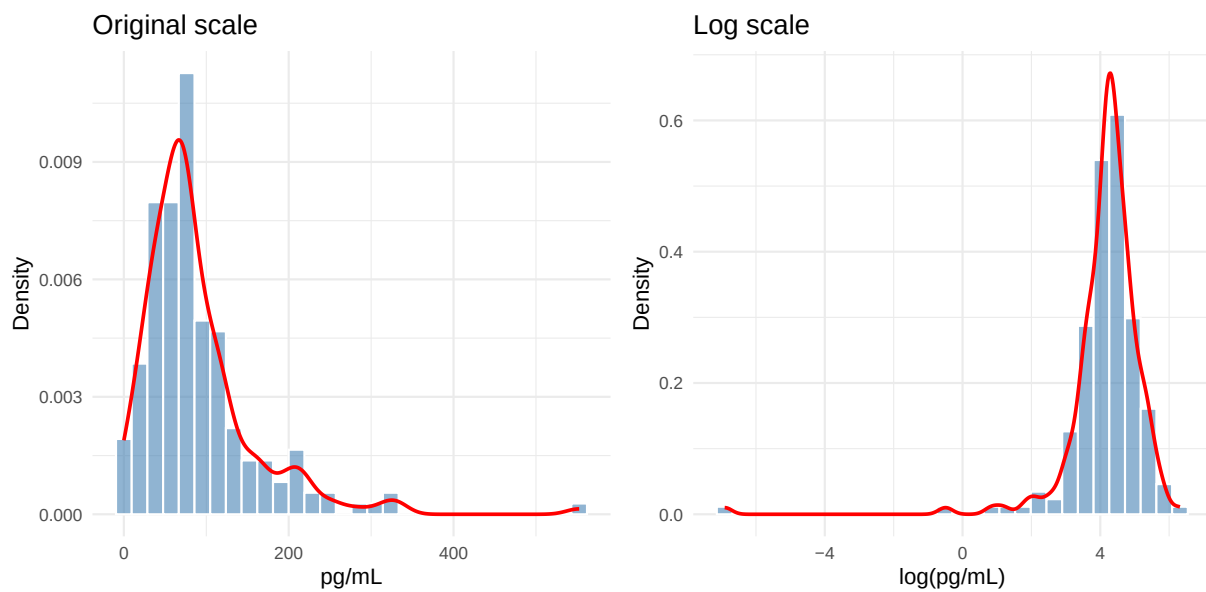
3.22 MMP-10 (MMP10)

Skewness: original scale = 2.08 | log scale = 0.93



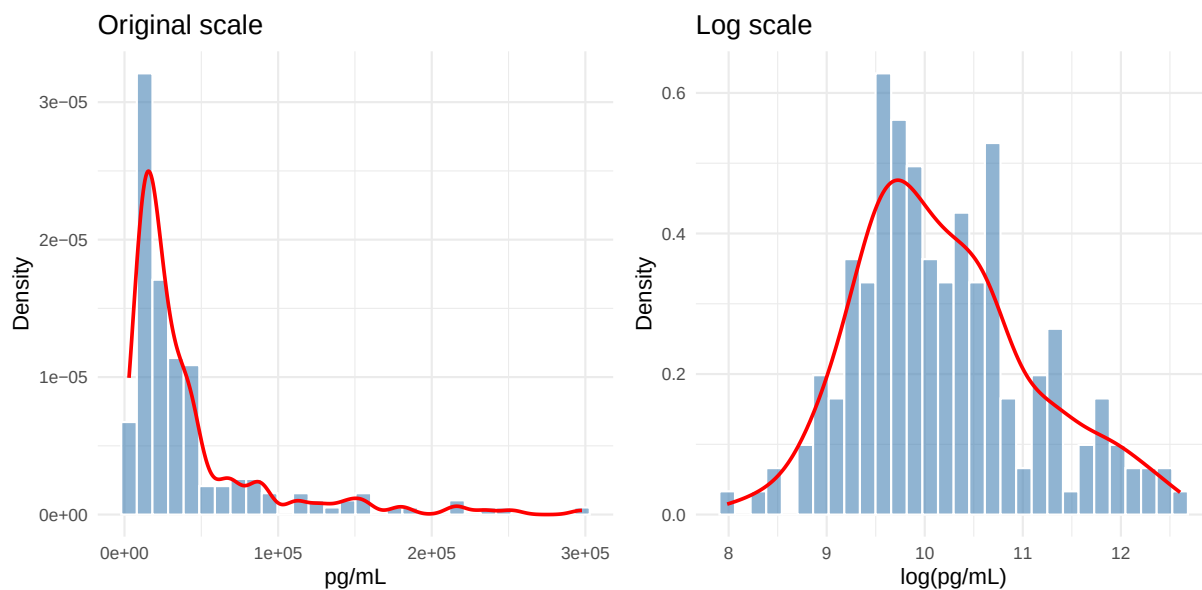
3.23 PDGF-DD (PDGFDD)

Skewness: original scale = 2.48 | log scale = -4.88



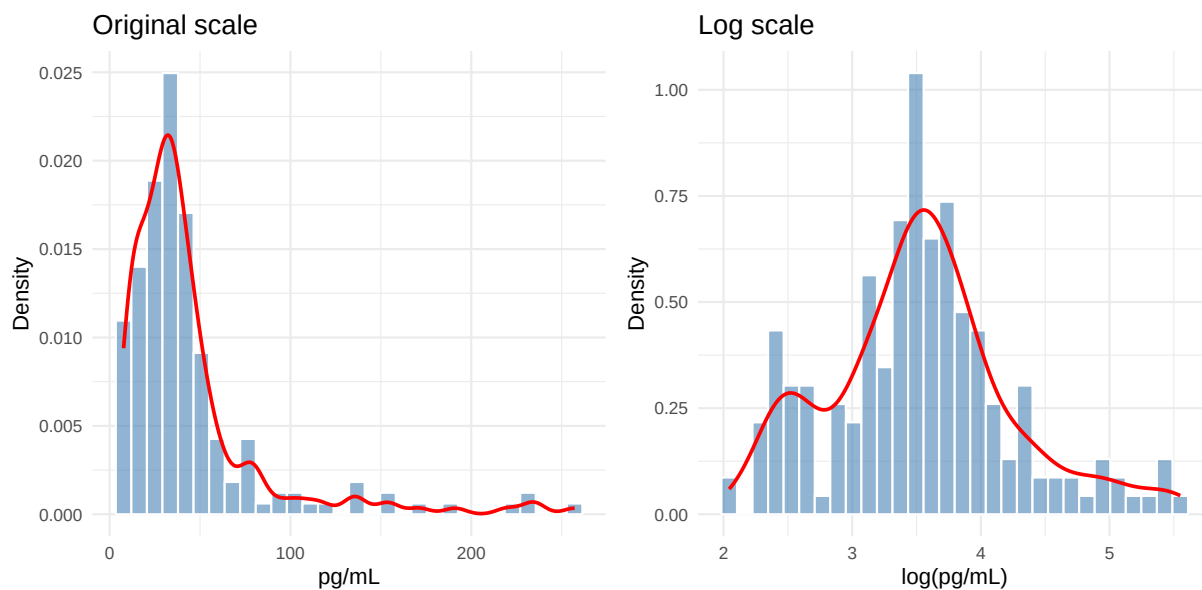
3.24 PECAM-1 (PECAM1)

Skewness: original scale = 2.64 | log scale = 0.48



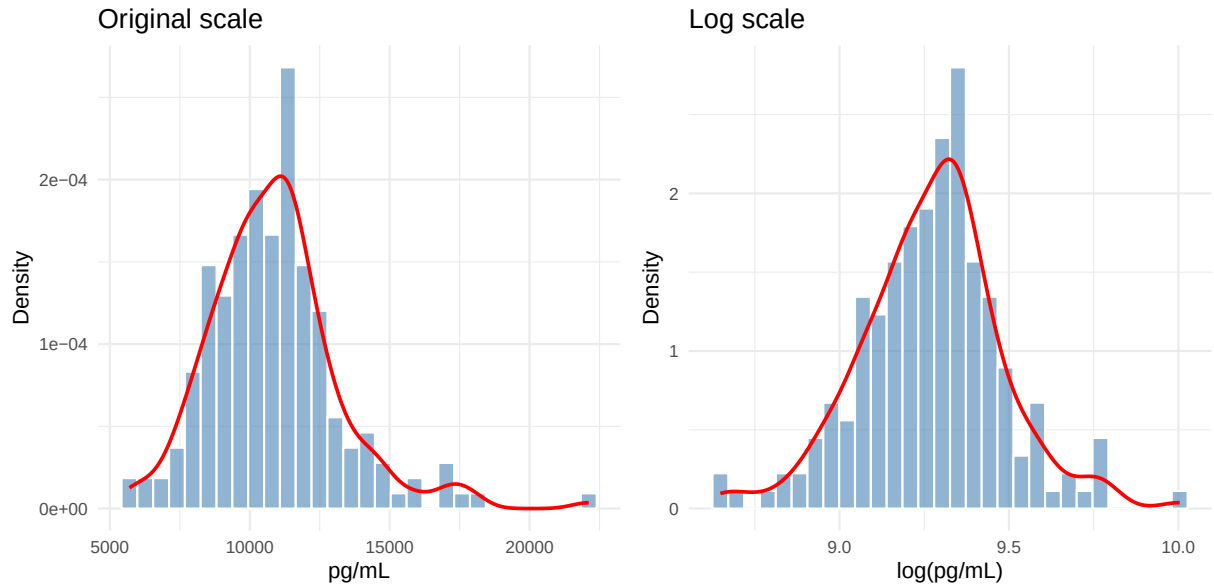
3.25 PIGF (PIGF)

Skewness: original scale = 2.83 | log scale = 0.4



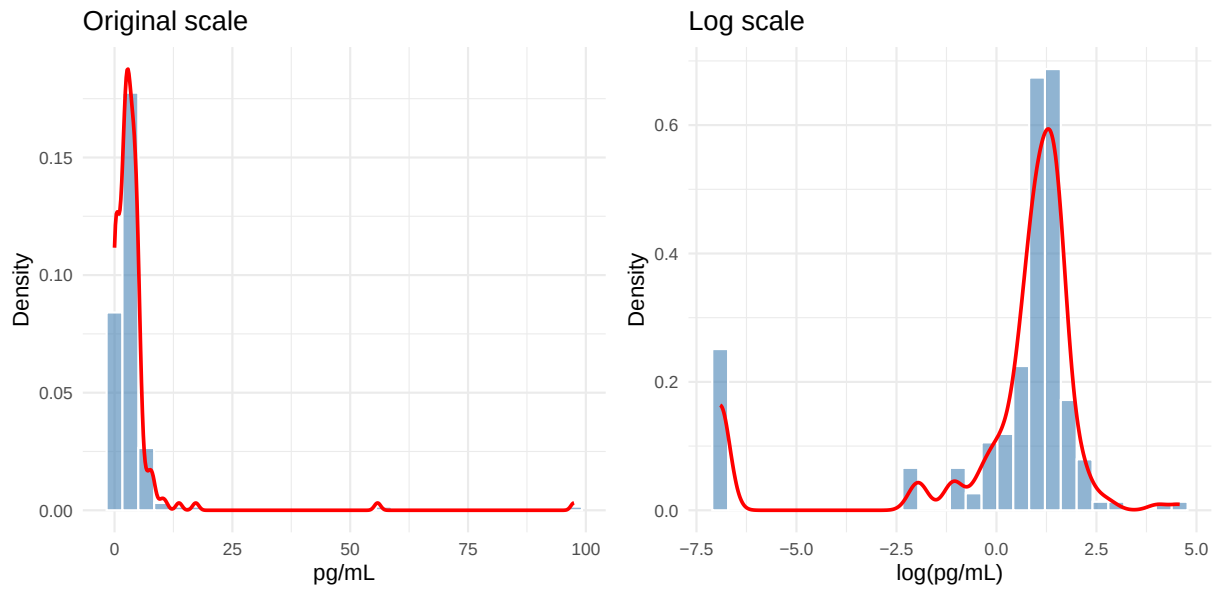
3.26 Tenascin C (TENASC)

Skewness: original scale = 1.01 | log scale = 0.01



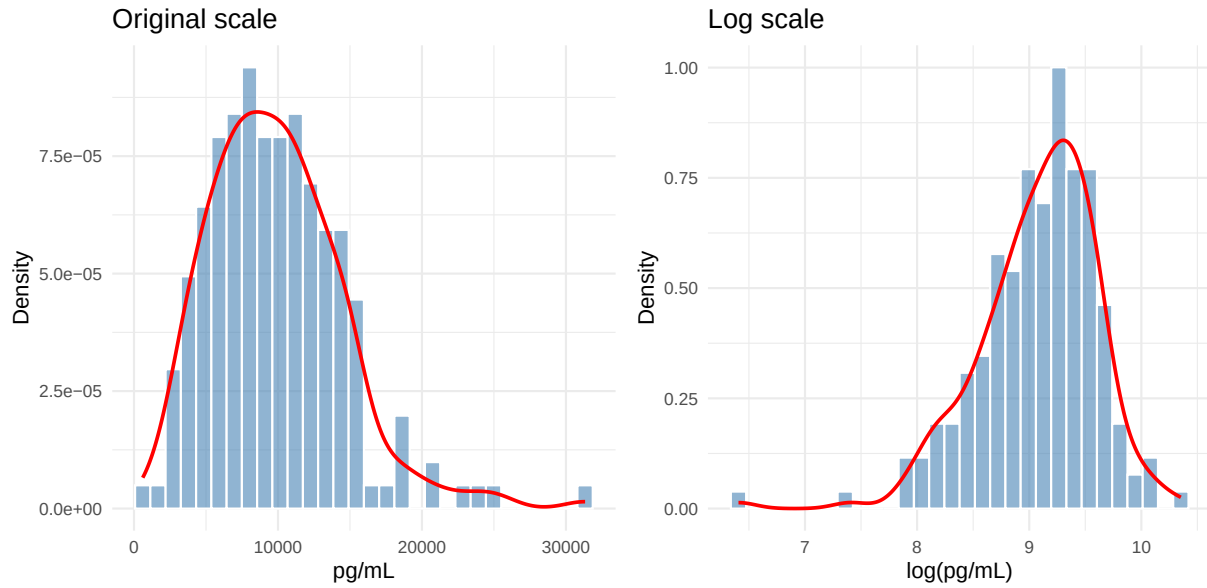
3.27 TGF-alpha (TGFA)

Skewness: original scale = 9.27 | log scale = -2.11



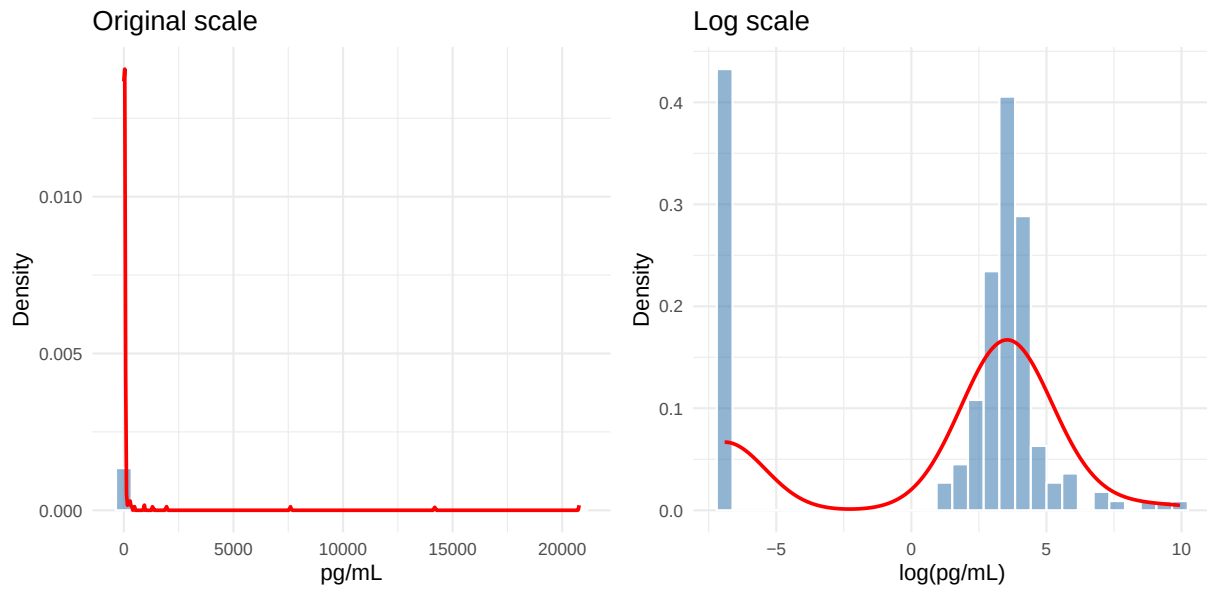
3.28 Thrombospondin-2 (THBS2)

Skewness: original scale = 1.02 | log scale = -0.99



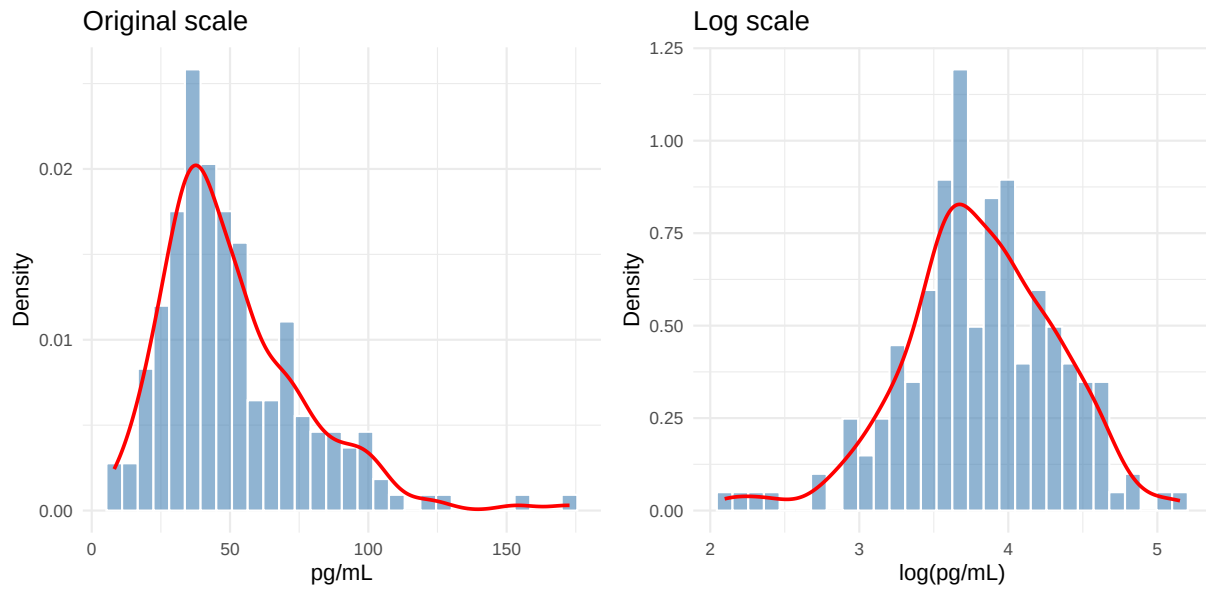
3.29 TNFRSF18 (GITR) (TNFRSF18)

Skewness: original scale = 8.99 | log scale = -0.92



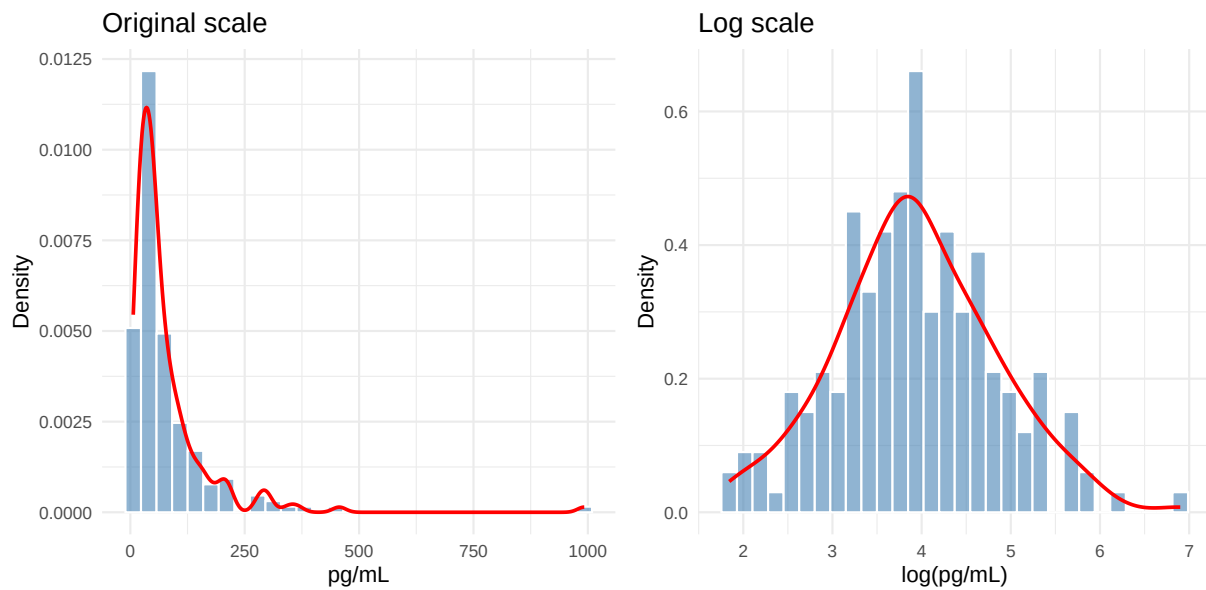
3.30 TNFSF6 (FasL) (TNFSF6)

Skewness: original scale = 1.35 | log scale = -0.36



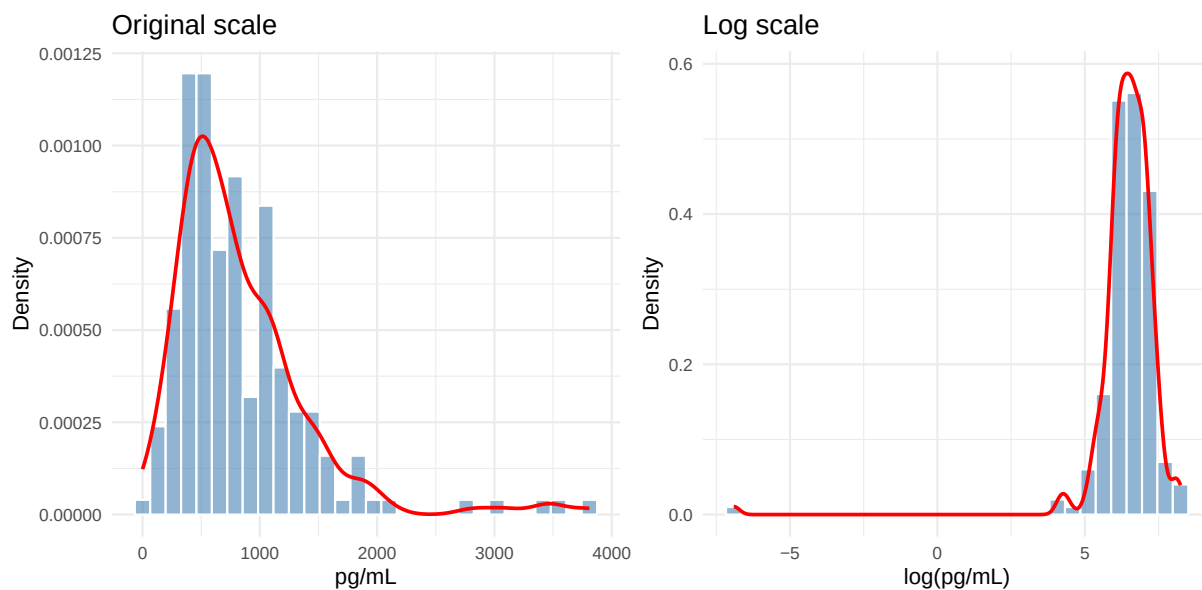
3.31 VEGF-A (VEGFA)

Skewness: original scale = 5.07 | log scale = 0.16



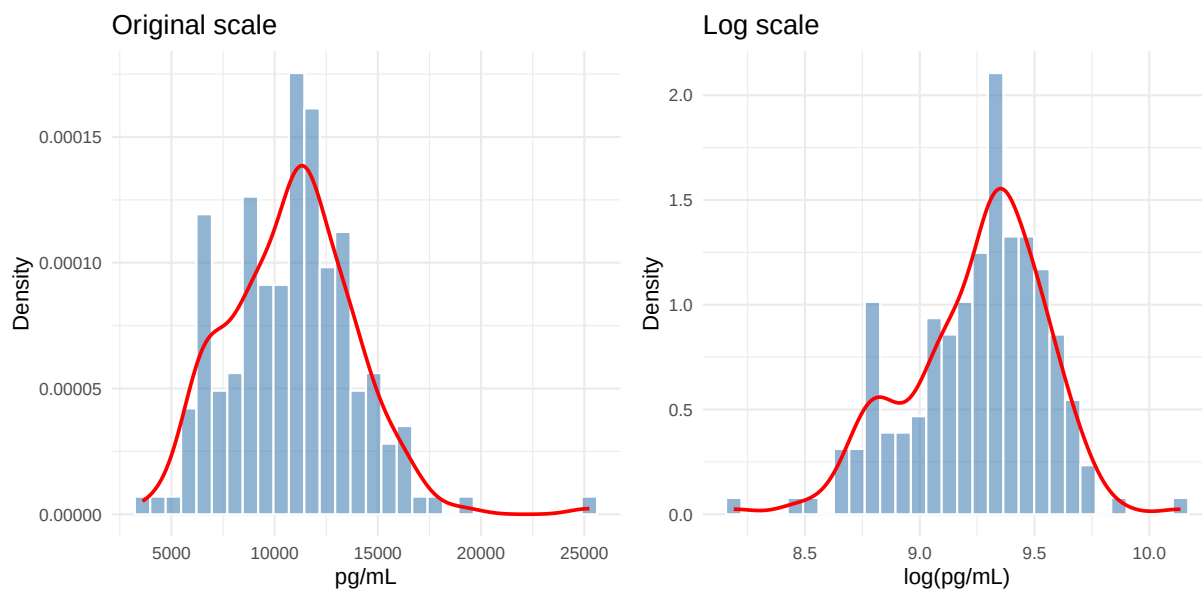
3.32 VEGF-C (VEGFC)

Skewness: original scale = 2.23 | log scale = -7.72



3.33 VEGFR2 (VEGFR2)

Skewness: original scale = 0.52 | log scale = -0.52

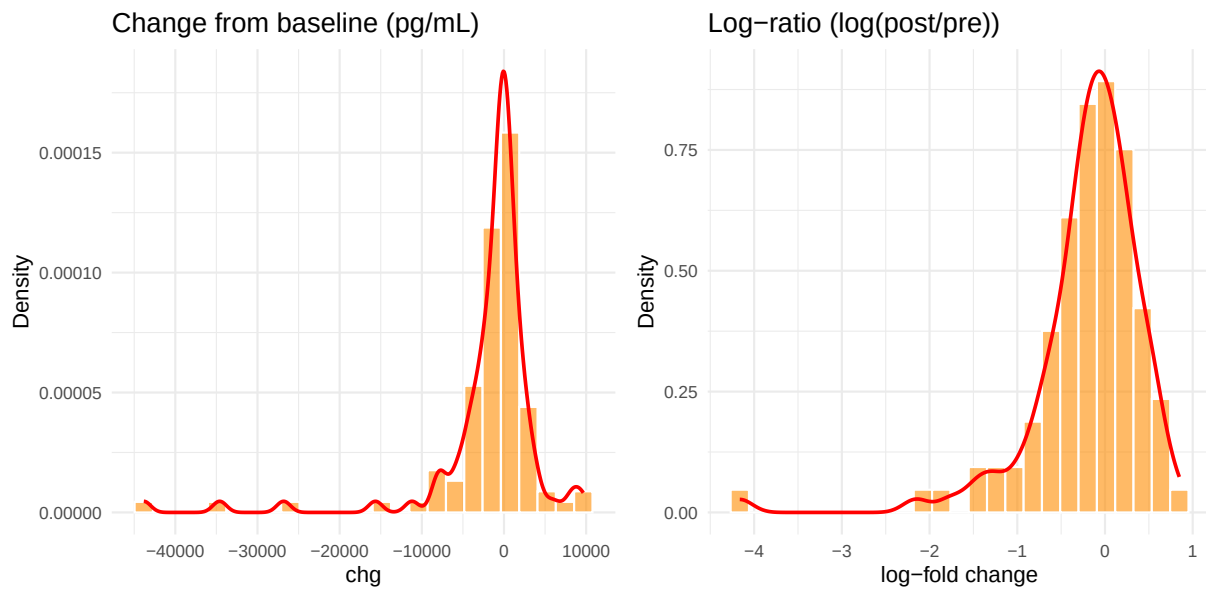


4 Change-score distributions

Change scores ($\text{chg} = \text{aval} - \text{base}$) are the response variable in the linear model. The log-ratio model uses $\log(\text{aval}/\text{base}) \sim \log(\text{aval} + \text{offset}) - \log(\text{base} + \text{offset})$.

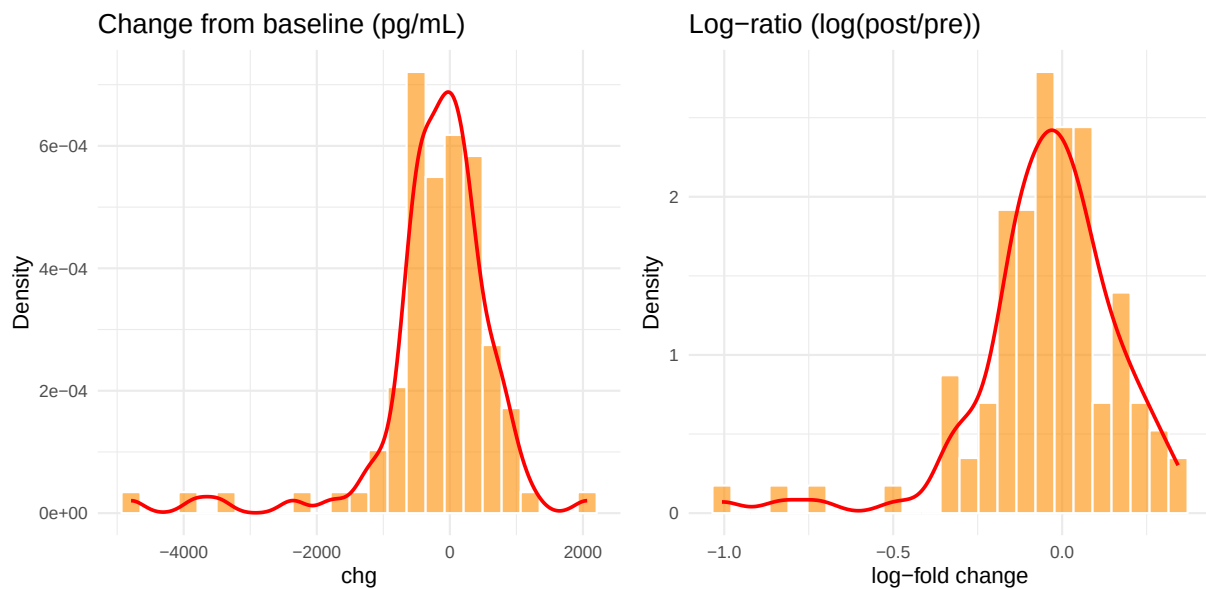
4.1 Angiopoietin-1

Linear chg: skewness = -3.69, excess kurtosis = 17.48 | Log-ratio skewness = -2.62



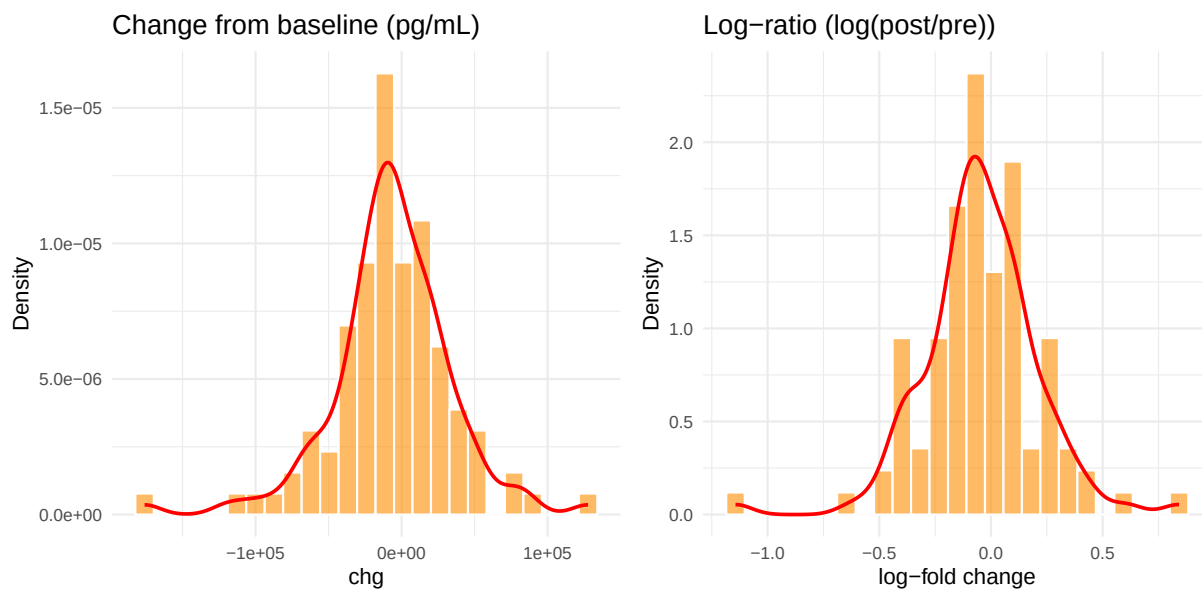
4.2 Angiopoietin-2

Linear chg: skewness = -2.24, excess kurtosis = 8.6 | Log-ratio skewness = -1.53



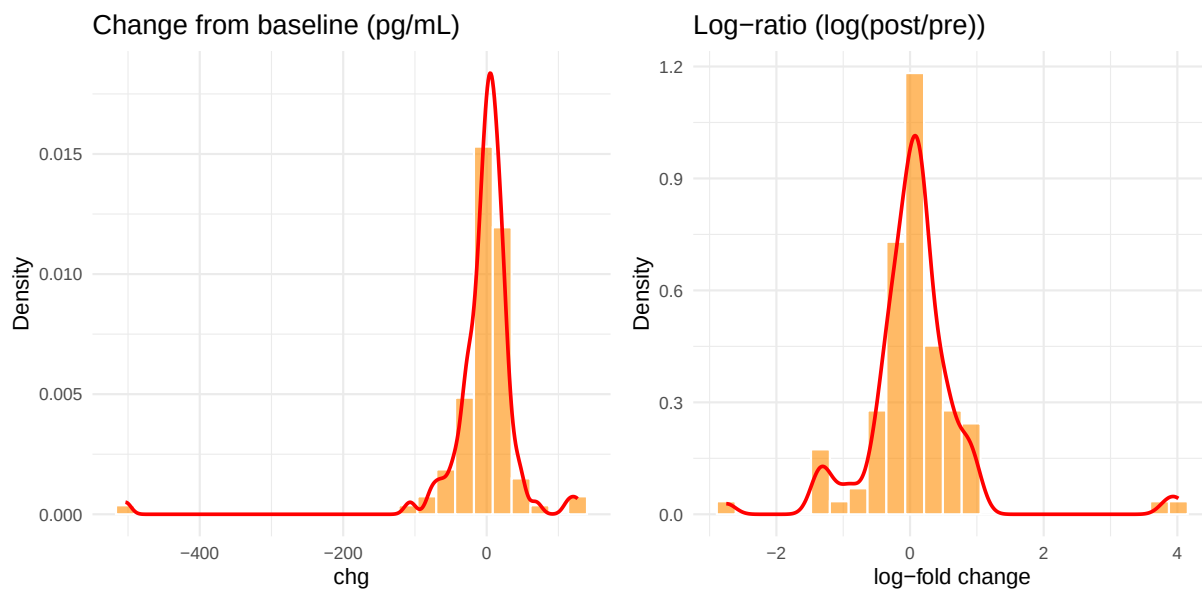
4.3 ANGPTL4

Linear chg: skewness = -0.46, excess kurtosis = 2.89 | Log-ratio skewness = -0.27



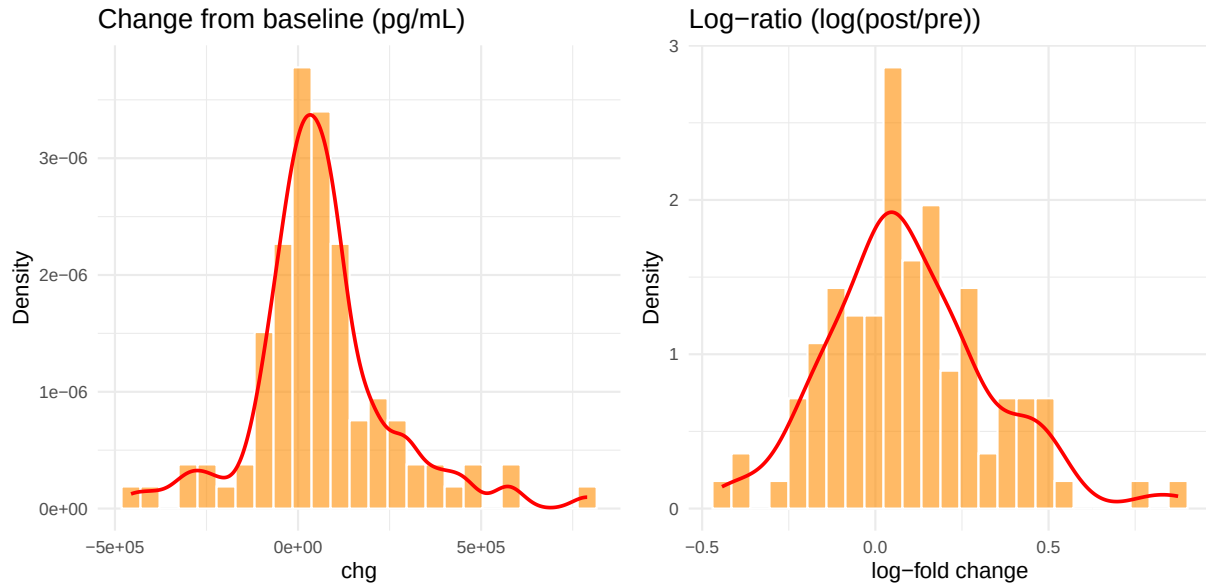
4.4 CA9

Linear chg: skewness = -5.72, excess kurtosis = 46.71 | Log-ratio skewness = 1.65



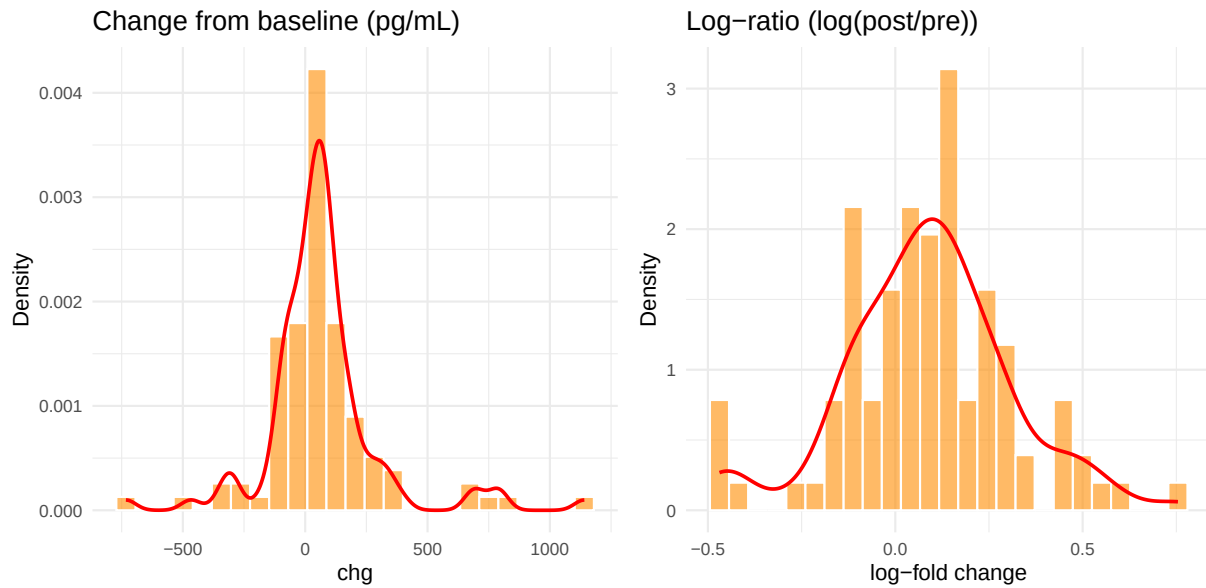
4.5 CD163

Linear chg: skewness = 0.67, excess kurtosis = 2.4 | Log-ratio skewness = 0.52



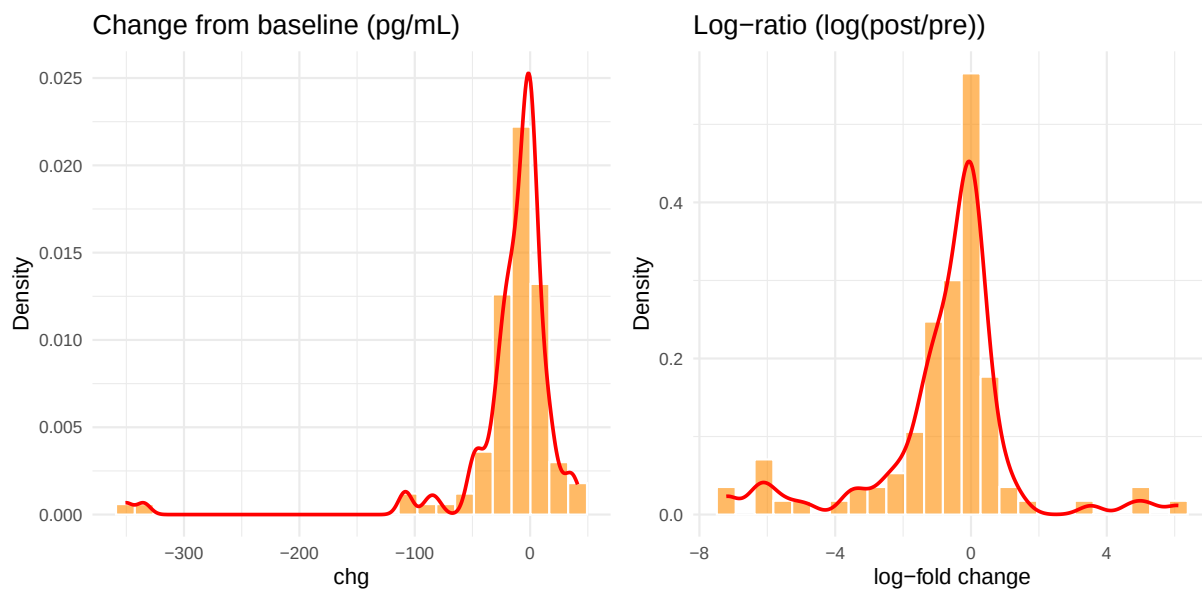
4.6 Collagen IV alpha 1

Linear chg: skewness = 1.19, excess kurtosis = 5.7 | Log-ratio skewness = -0.03



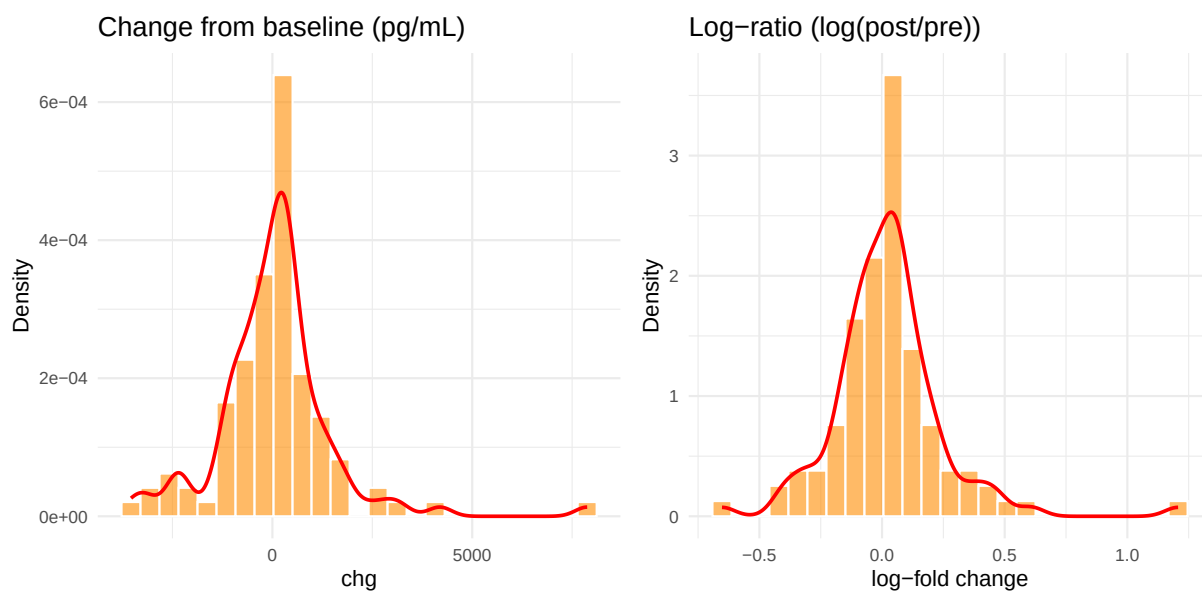
4.7 EGF

Linear chg: skewness = -4.72, excess kurtosis = 26.08 | Log-ratio skewness = -0.58



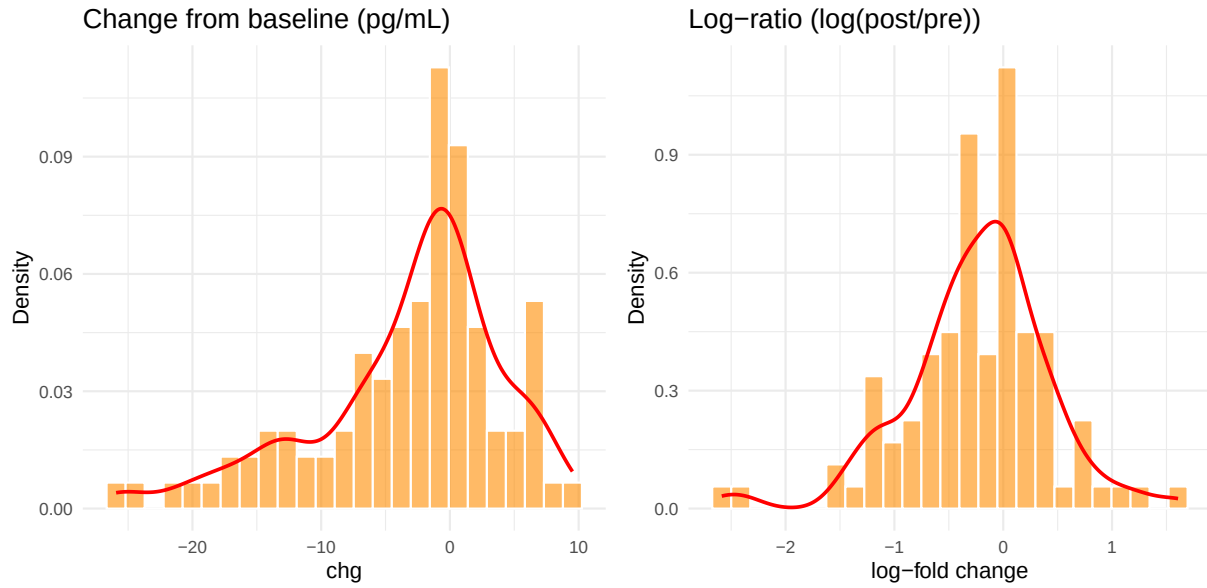
4.8 ErbB2

Linear chg: skewness = 1.37, excess kurtosis = 7.37 | Log-ratio skewness = 1.27



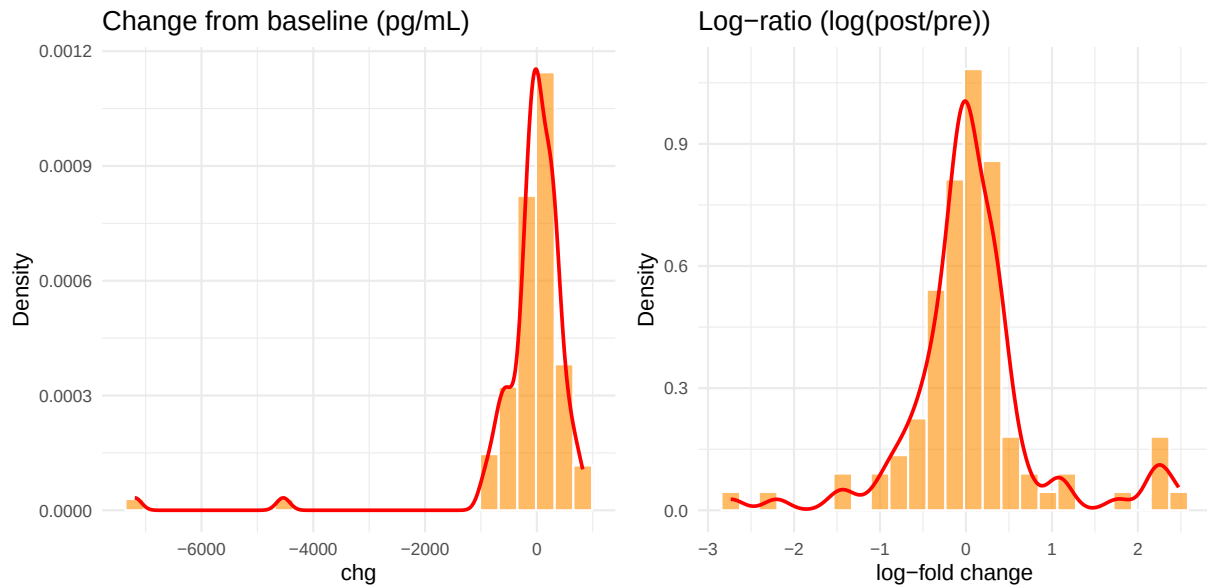
4.9 FGF Basic

Linear chg: skewness = -0.94, excess kurtosis = 0.63 | Log-ratio skewness = -0.52



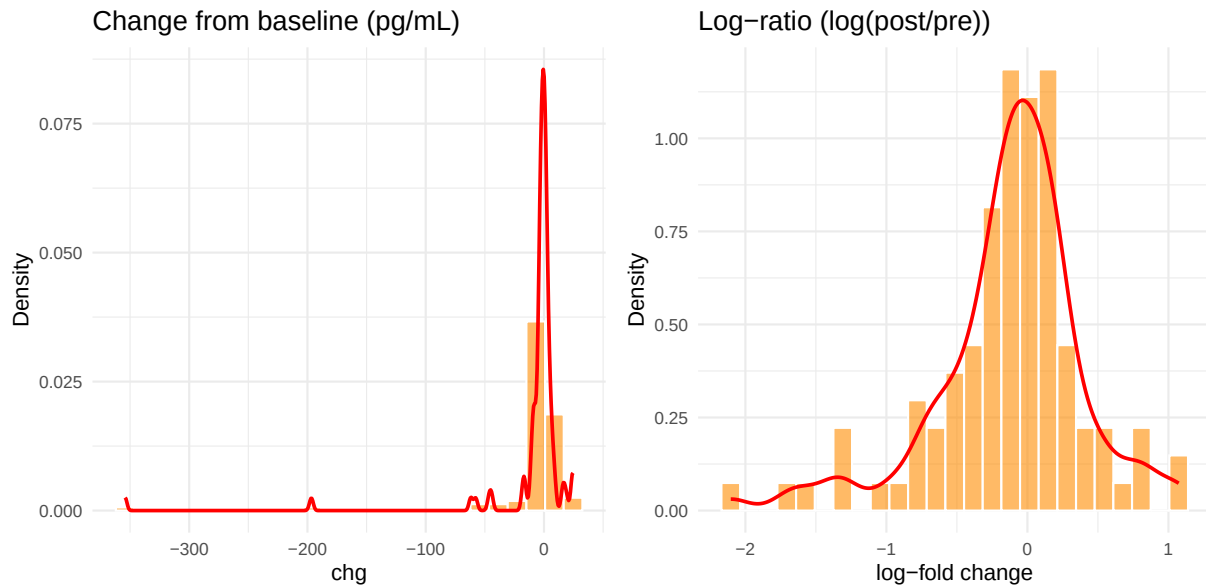
4.10 Fractalkine

Linear chg: skewness = -5.6, excess kurtosis = 37.51 | Log-ratio skewness = 0.39



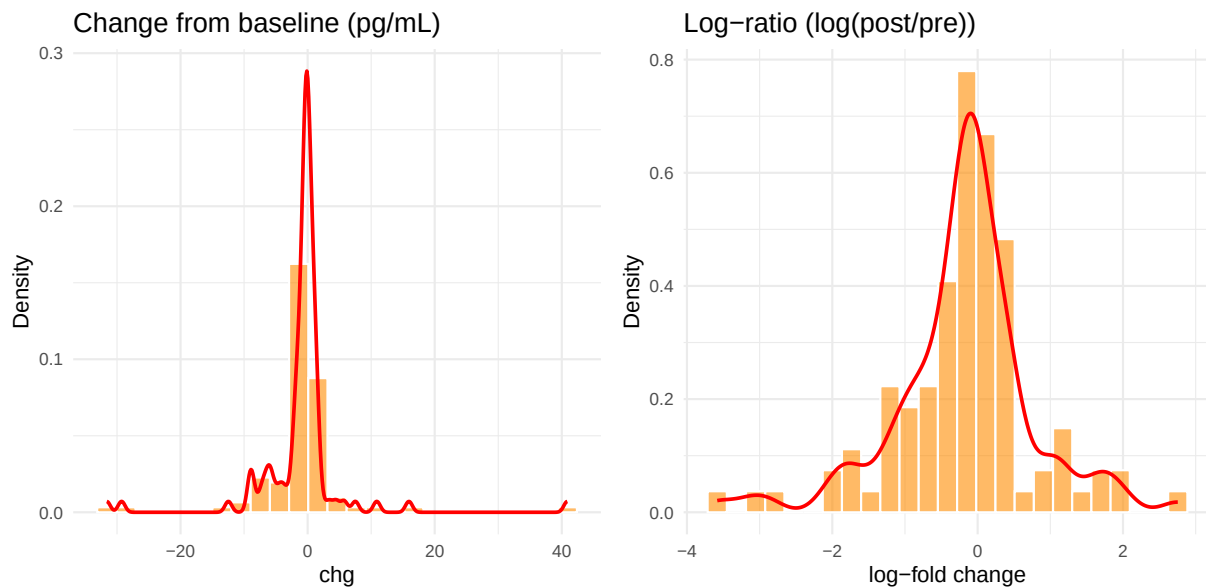
4.11 Granzyme B

Linear chg: skewness = -6.62, excess kurtosis = 48.53 | Log-ratio skewness = -0.92



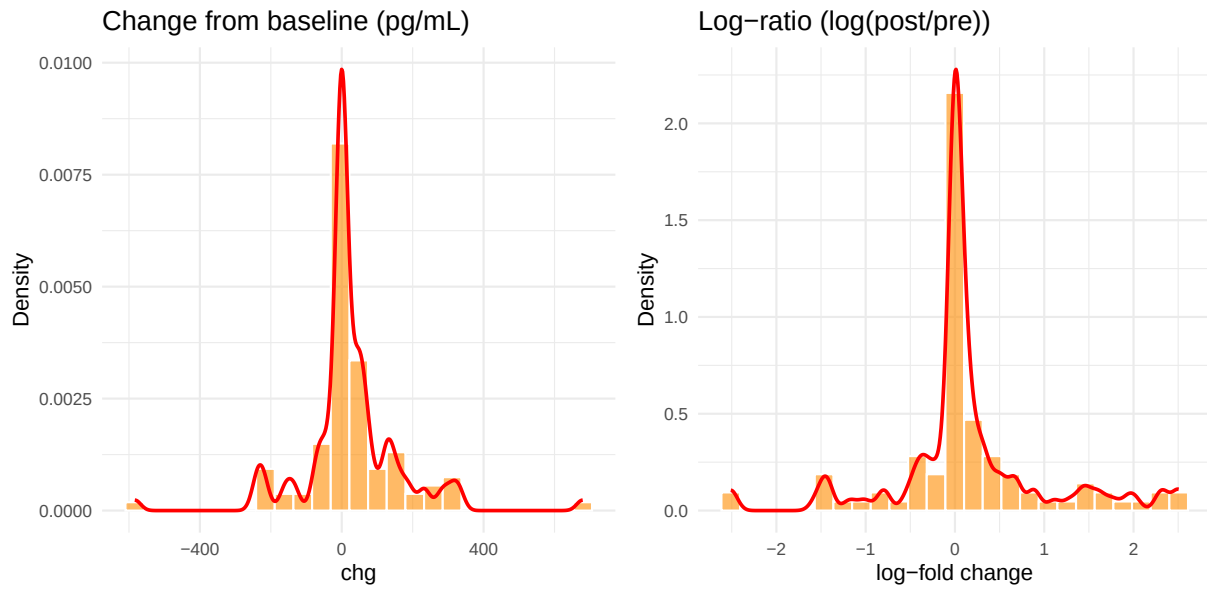
4.12 IFN-g

Linear chg: skewness = 0.72, excess kurtosis = 17.28 | Log-ratio skewness = -0.43



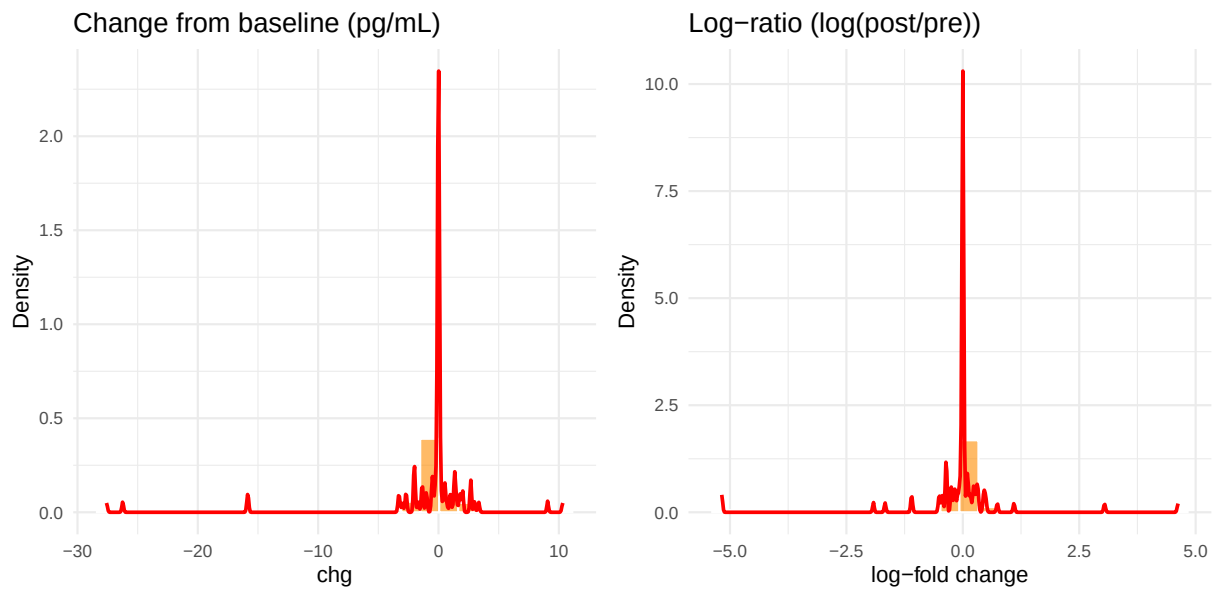
4.13 IL-12 p70

Linear chg: skewness = 0.34, excess kurtosis = 6.49 | Log-ratio skewness = 0.26



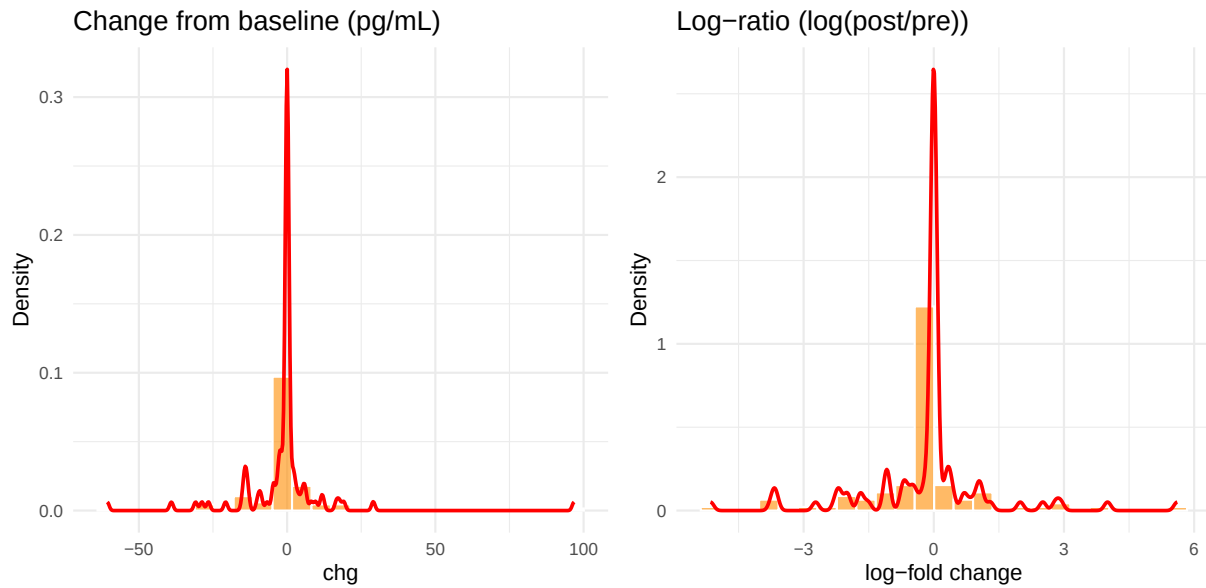
4.14 IL-1b

Linear chg: skewness = -3.82, excess kurtosis = 18.59 | Log-ratio skewness = -1.46



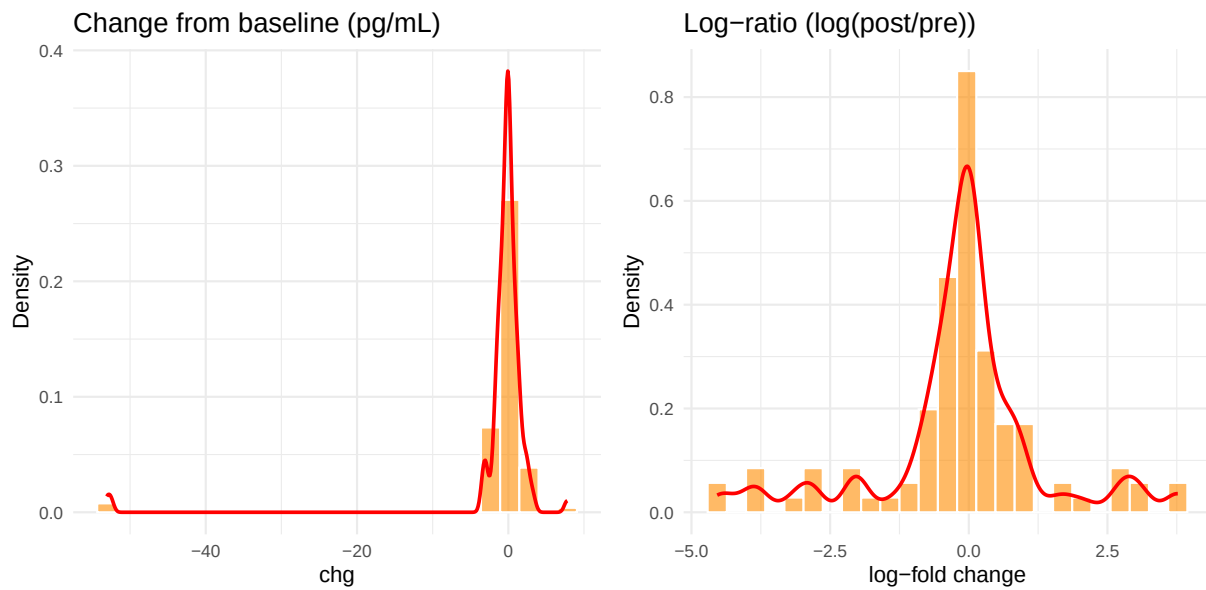
4.15 IL-2

Linear chg: skewness = 2.01, excess kurtosis = 20.61 | Log-ratio skewness = 0.11



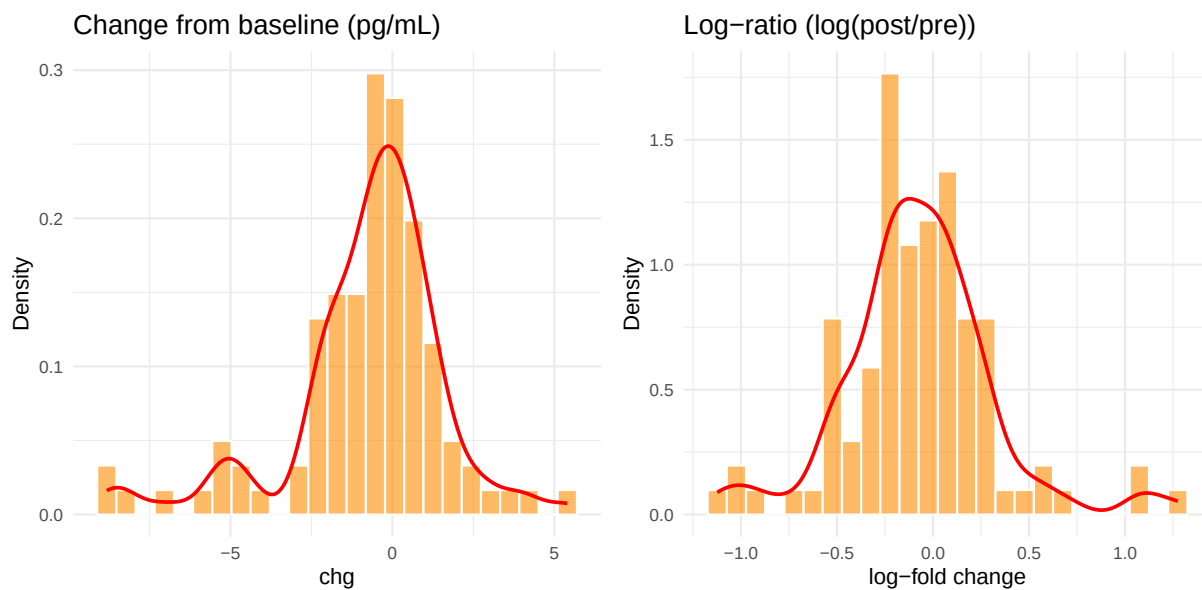
4.16 IL-6

Linear chg: skewness = -6.41, excess kurtosis = 41.45 | Log-ratio skewness = -0.36



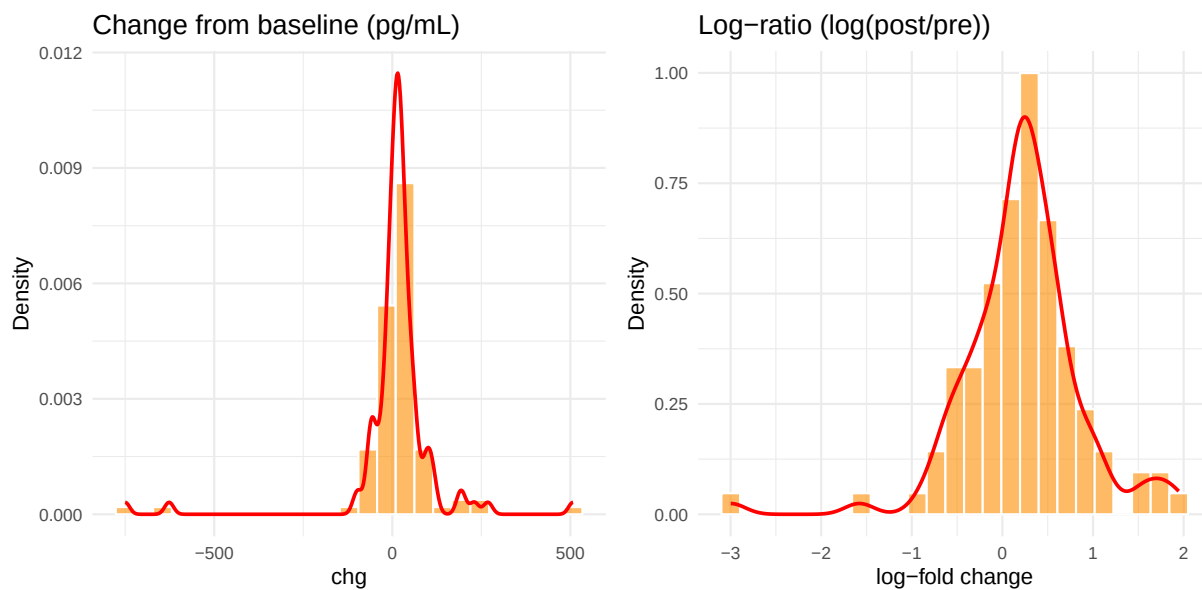
4.17 IL-8

Linear chg: skewness = -1.04, excess kurtosis = 2.2 | Log-ratio skewness = 0.42



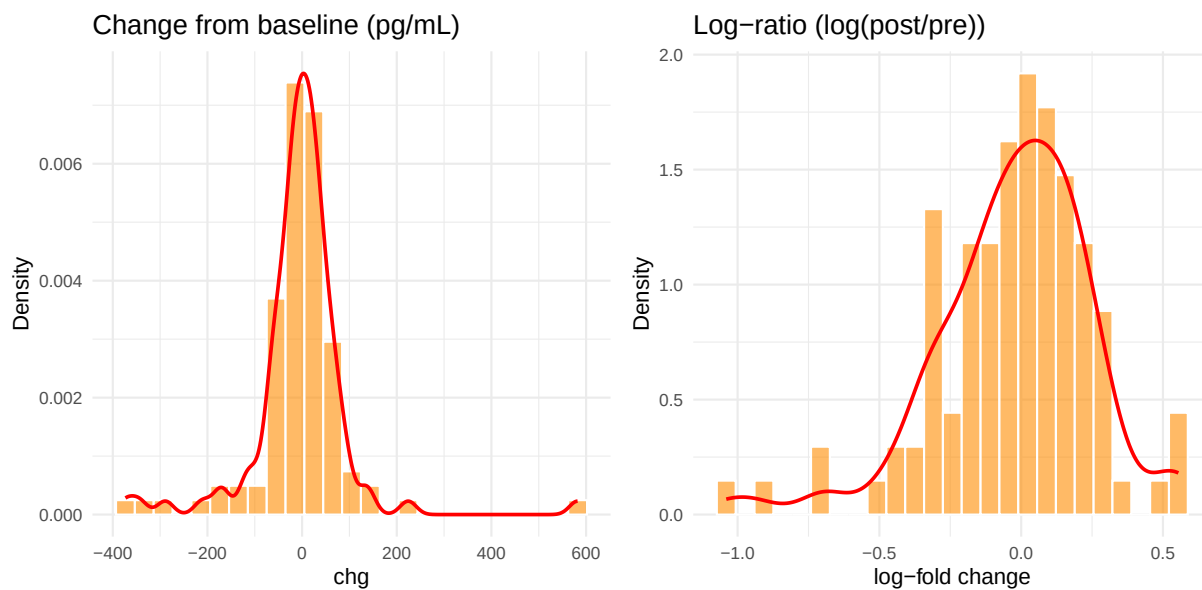
4.18 IP-10 (CXCL10)

Linear chg: skewness = -2.74, excess kurtosis = 20.04 | Log-ratio skewness = -0.85



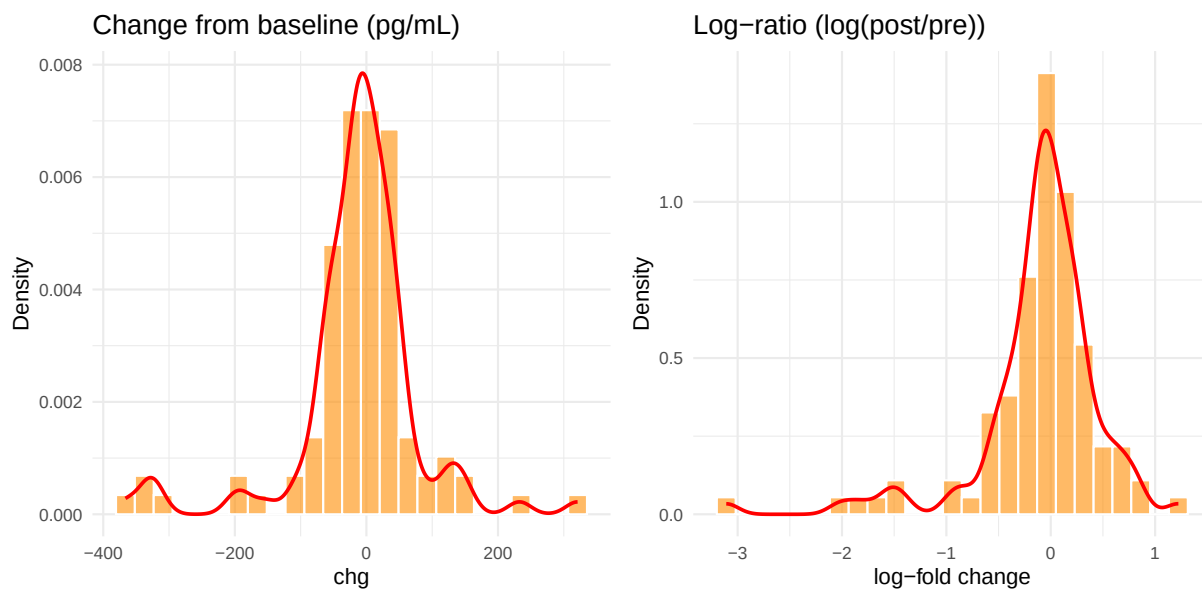
4.19 I-TAC (CXCL11)

Linear chg: skewness = 0.81, excess kurtosis = 11.55 | Log-ratio skewness = -0.84



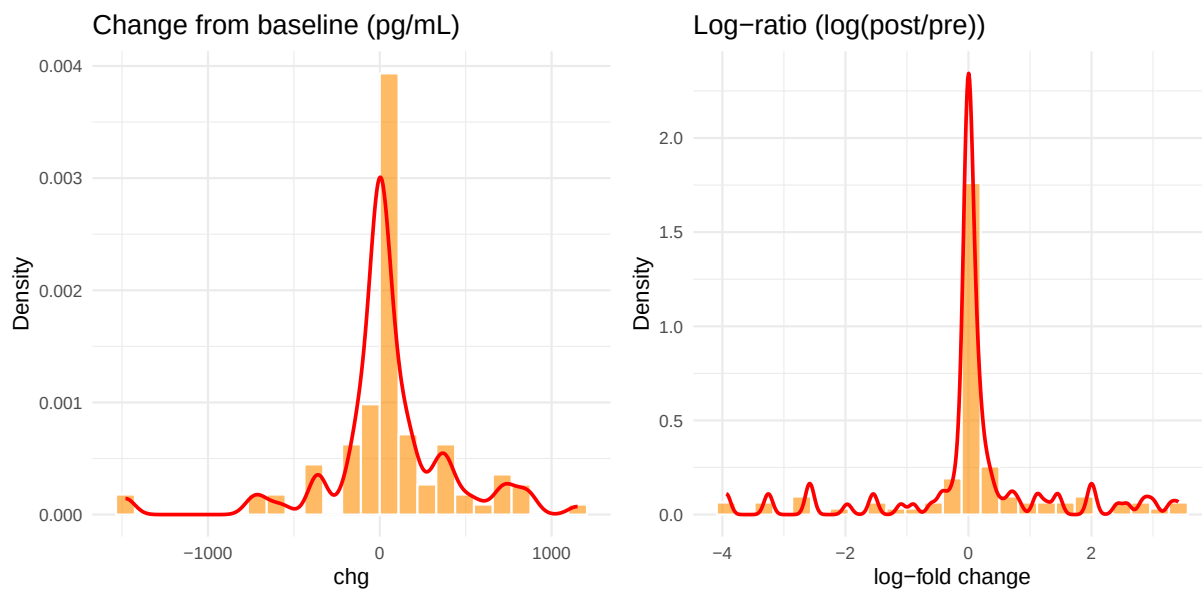
4.20 MCP-1

Linear chg: skewness = -0.91, excess kurtosis = 4.65 | Log-ratio skewness = -1.94



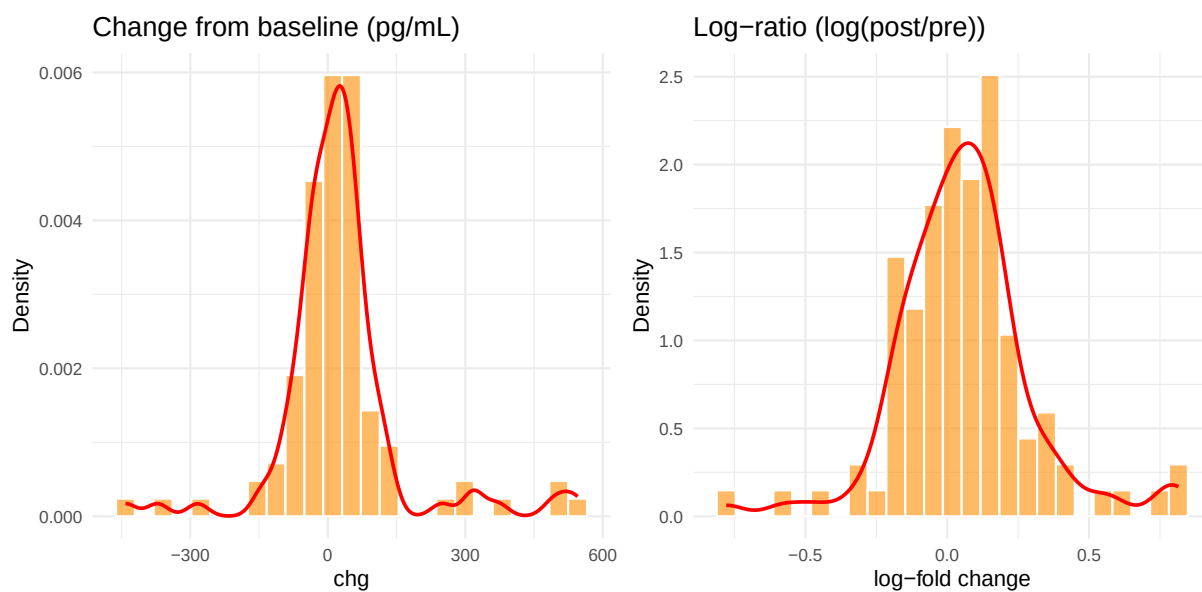
4.21 MIG (CXCL9)

Linear chg: skewness = -0.73, excess kurtosis = 4.29 | Log-ratio skewness = -0.36



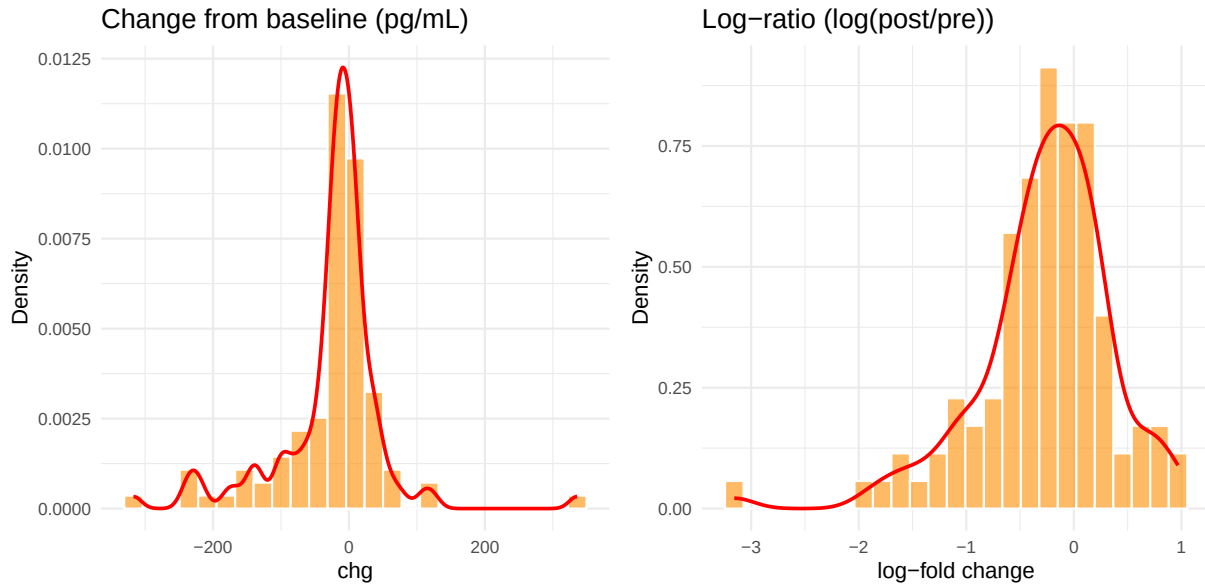
4.22 MMP-10

Linear chg: skewness = 0.95, excess kurtosis = 5.38 | Log-ratio skewness = 0.29



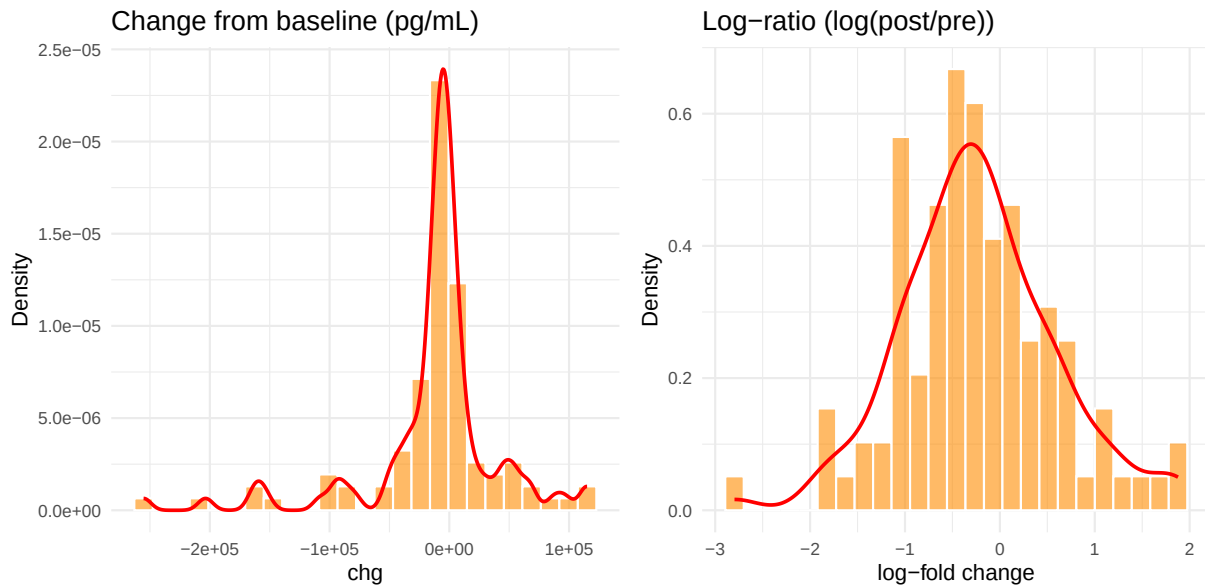
4.23 PDGF-DD

Linear chg: skewness = -0.3, excess kurtosis = 5.36 | Log-ratio skewness = -1.2



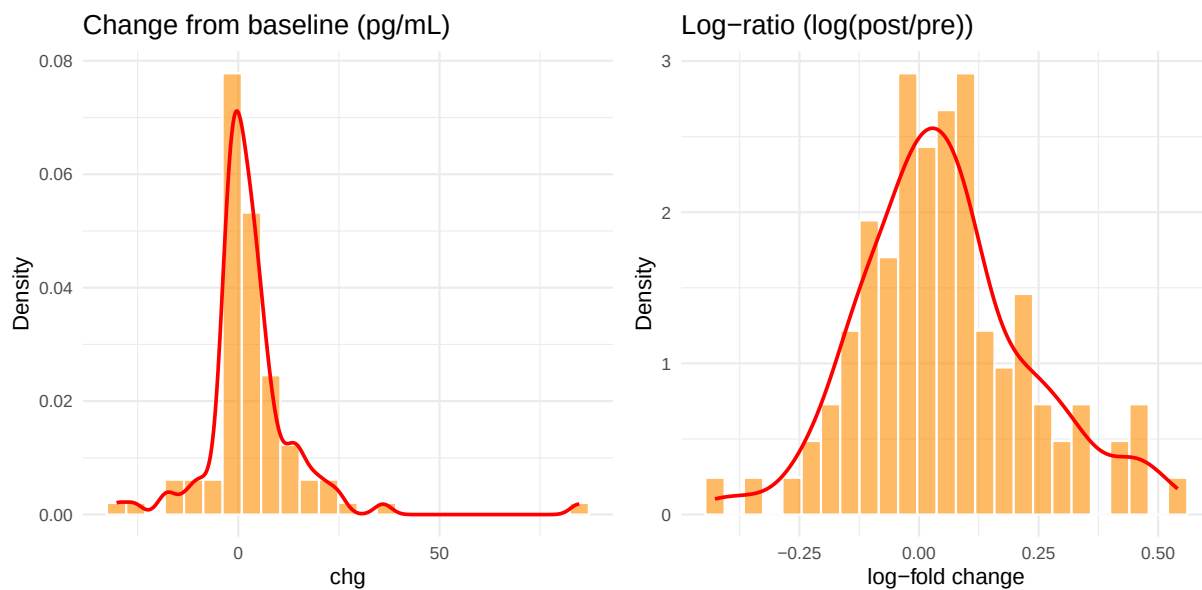
4.24 PECAM-1

Linear chg: skewness = -1.49, excess kurtosis = 4.93 | Log-ratio skewness = 0.11



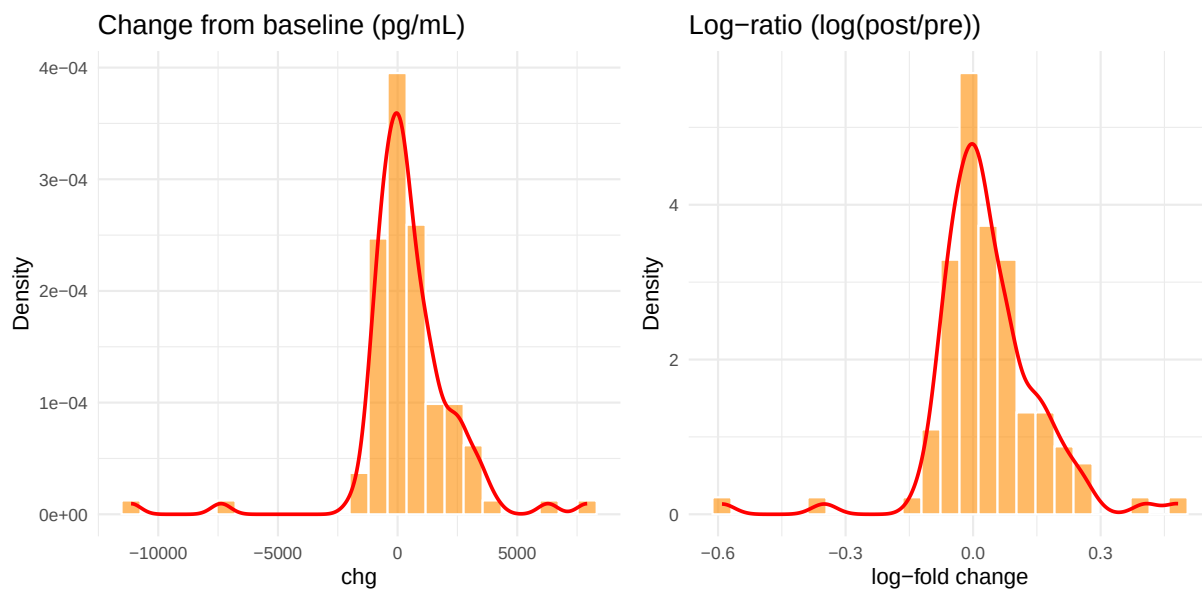
4.25 PIGF

Linear chg: skewness = 2.71, excess kurtosis = 17.16 | Log-ratio skewness = 0.36



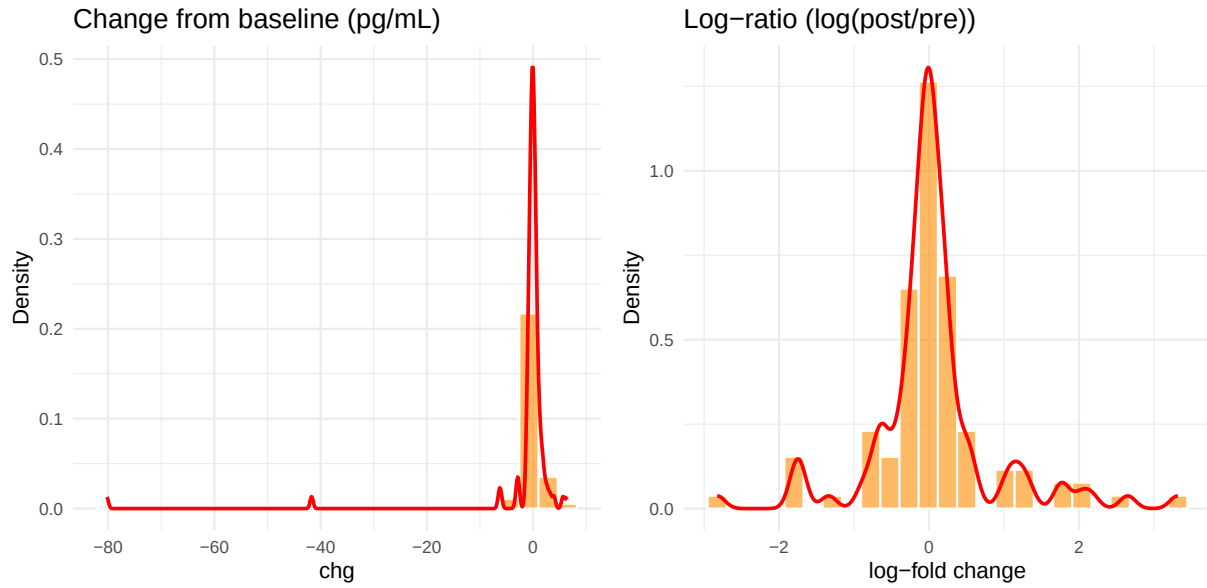
4.26 Tenascin C

Linear chg: skewness = -1.36, excess kurtosis = 11.25 | Log-ratio skewness = -0.5



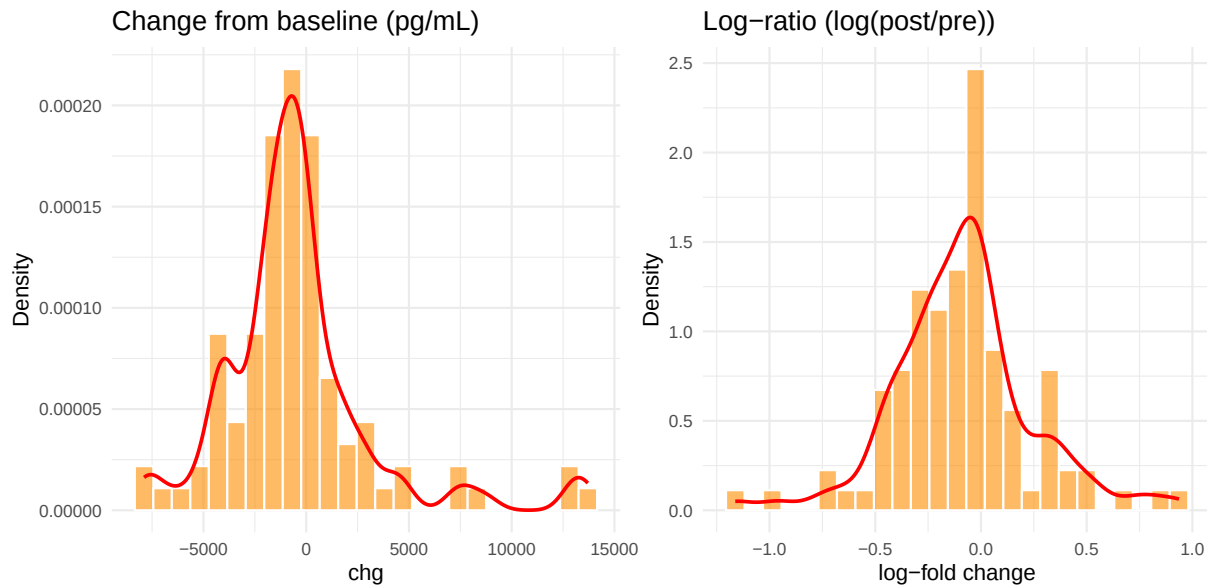
4.27 TGF-alpha

Linear chg: skewness = -7.37, excess kurtosis = 57.72 | Log-ratio skewness = 0.58



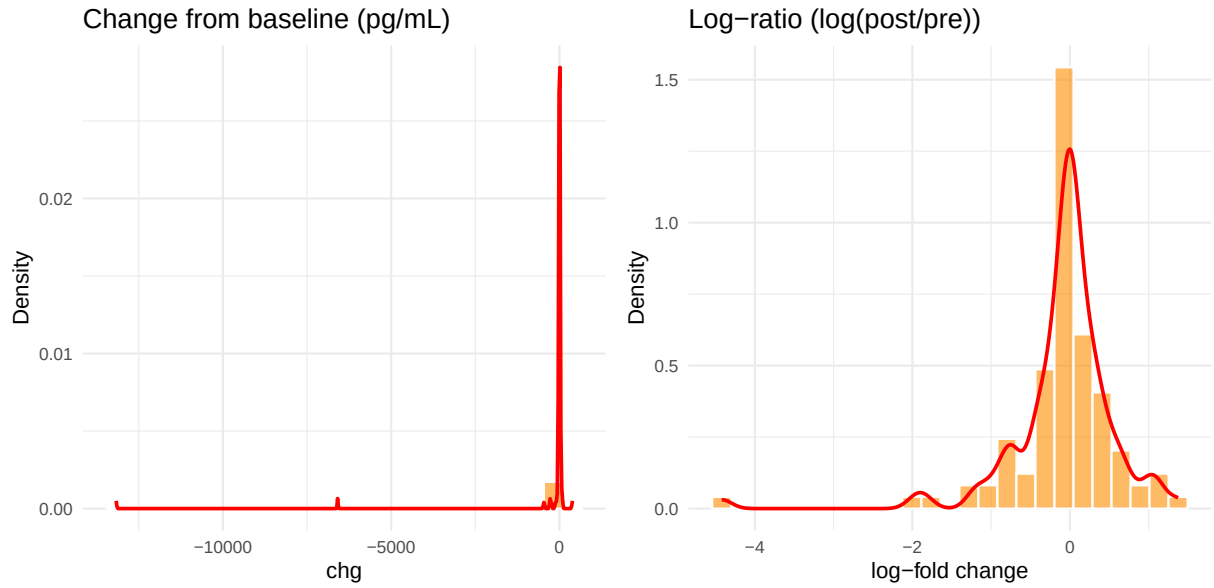
4.28 Thrombospondin-2

Linear chg: skewness = 1.53, excess kurtosis = 4.13 | Log-ratio skewness = 0.15



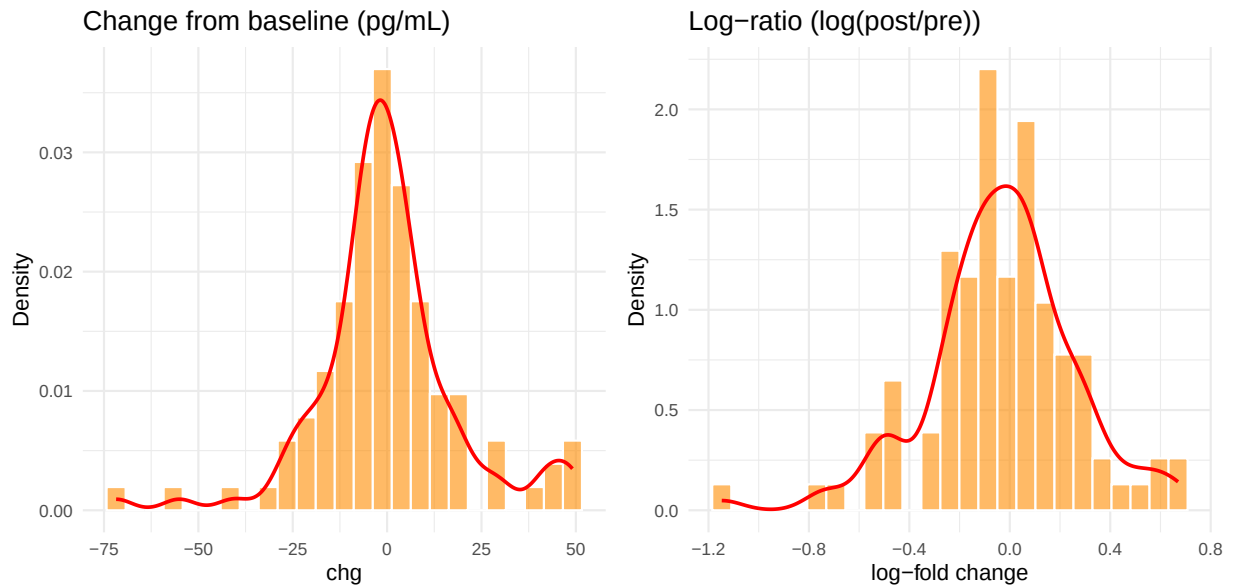
4.29 TNFRSF18 (GITR)

Linear chg: skewness = -7.8, excess kurtosis = 62.85 | Log-ratio skewness = -2.65



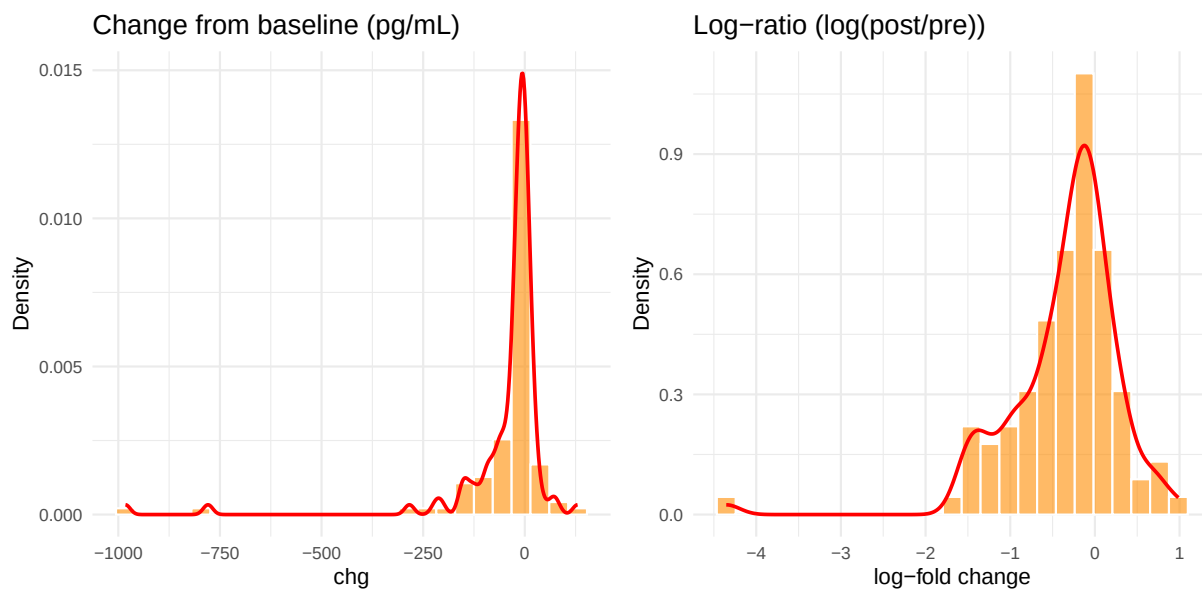
4.30 TNFSF6 (FasL)

Linear chg: skewness = -0.08, excess kurtosis = 2.51 | Log-ratio skewness = -0.45



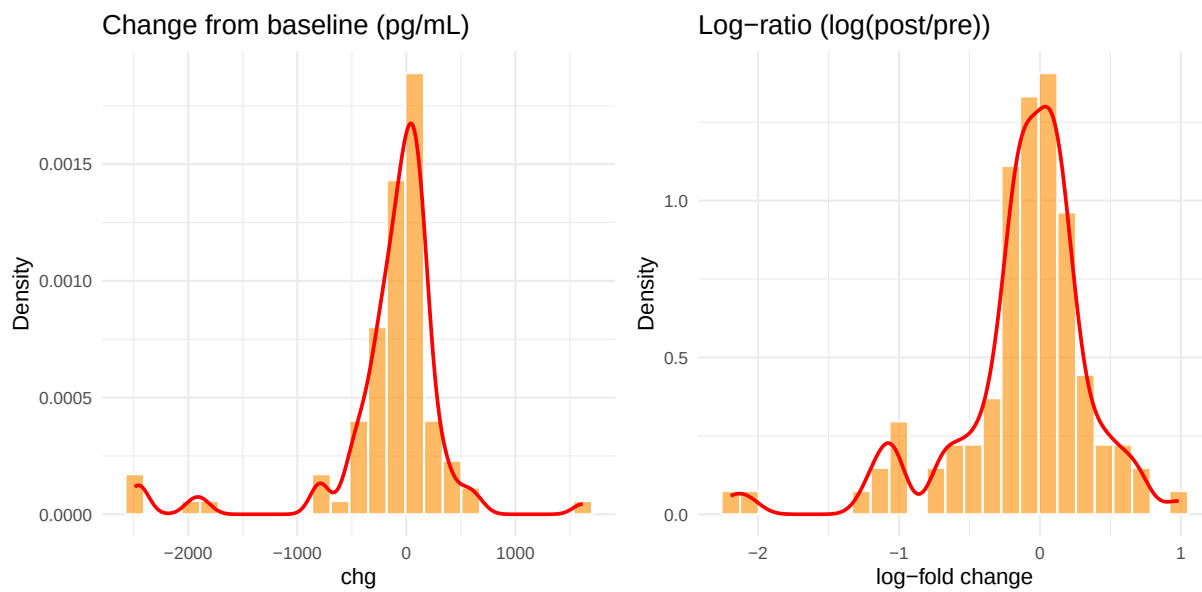
4.31 VEGF-A

Linear chg: skewness = -5.14, excess kurtosis = 30.45 | Log-ratio skewness = -2.17



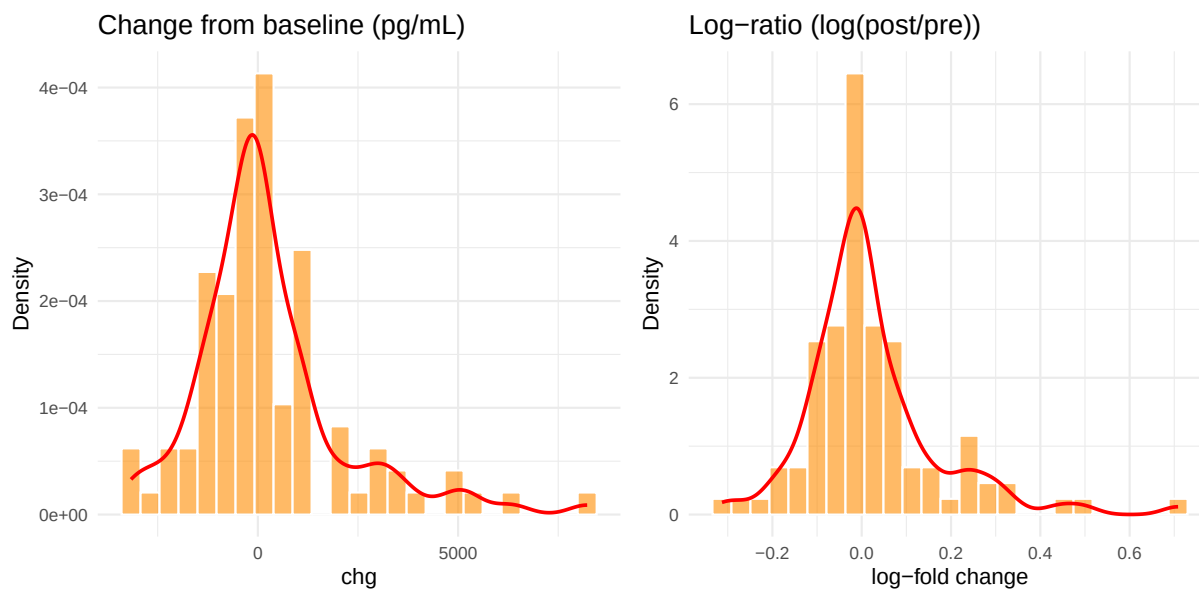
4.32 VEGF-C

Linear chg: skewness = -2.21, excess kurtosis = 7.88 | Log-ratio skewness = -1.49



4.33 VEGFR2

Linear chg: skewness = 1.44, excess kurtosis = 3.2 | Log-ratio skewness = 1.46

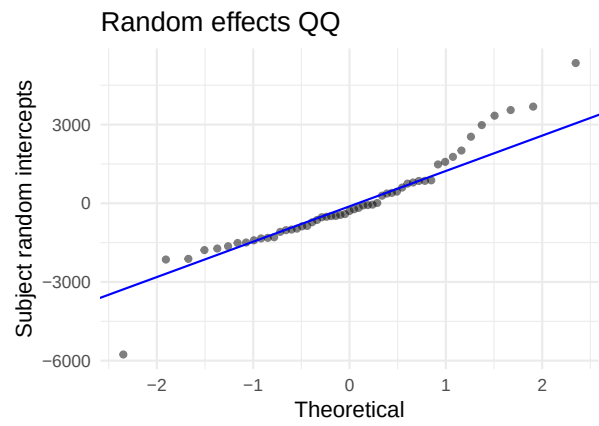
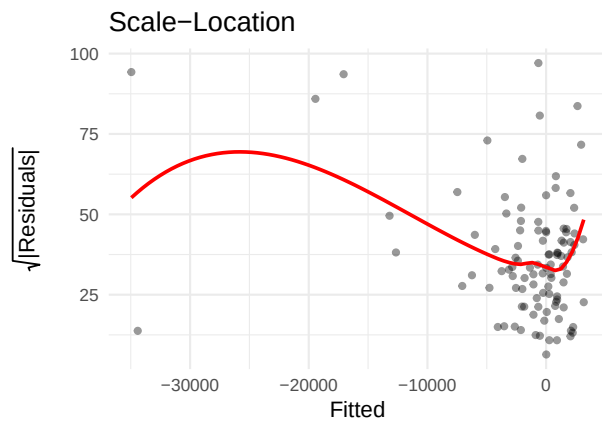
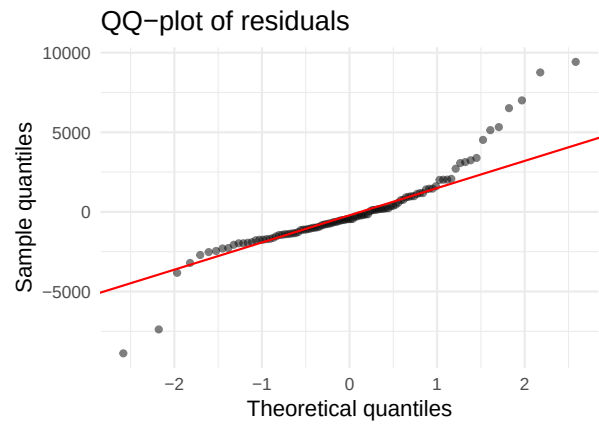
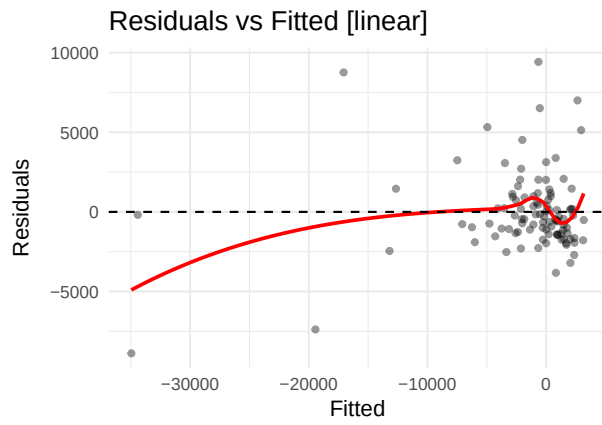


5 Residual diagnostics per analyte

For each analyte both LMMs are fitted on early visits (Days 1–43). Four diagnostic panels are shown per model: **Residuals vs Fitted** (homoscedasticity), **QQ-plot** (normality), **Scale-Location** (trend in residual variance), and **Random effects QQ** (normality of subject-level intercepts). A Shapiro-Wilk test is reported when the residual sample size is between 5 and 5000.

5.1 Angiopoietin-1 (ANGPT1)

AIC linear: 1917.7 | **AIC log-ratio:** 197.6 | **Preferred:** log



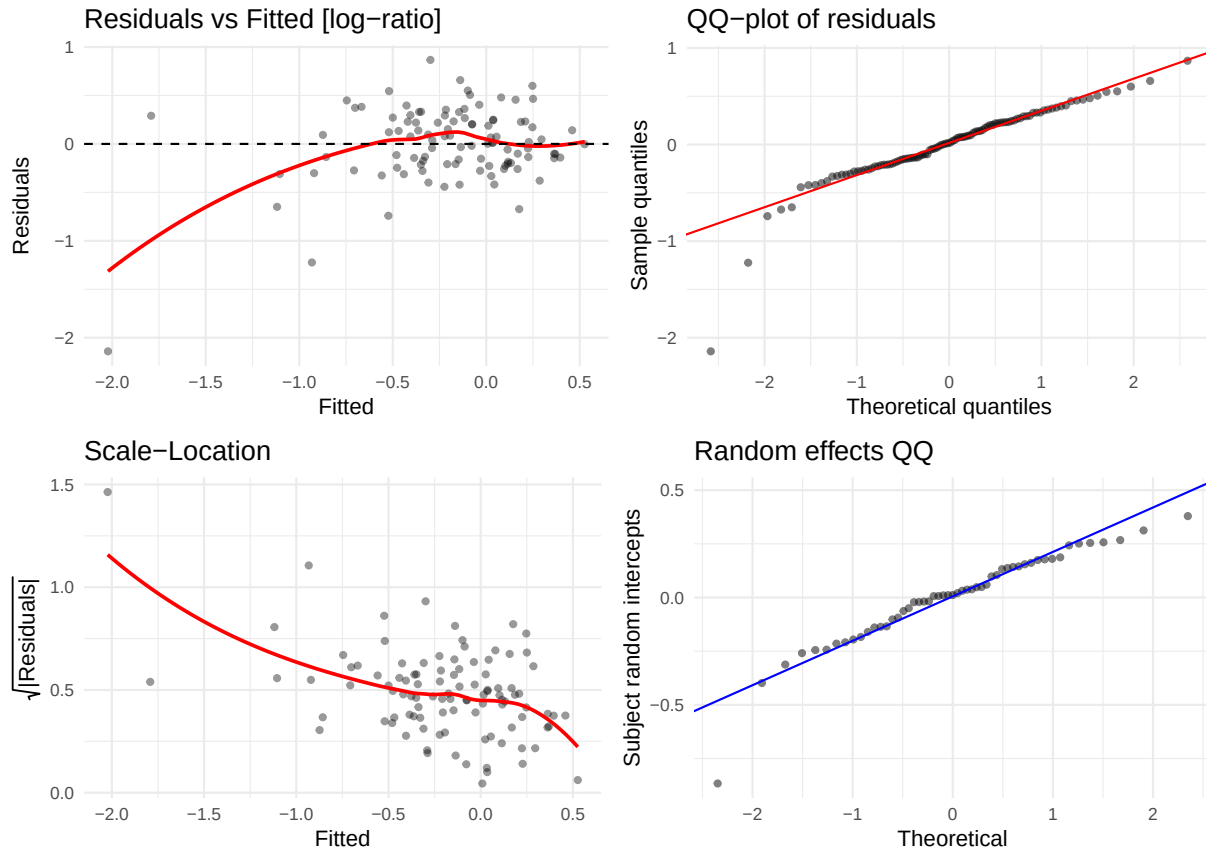
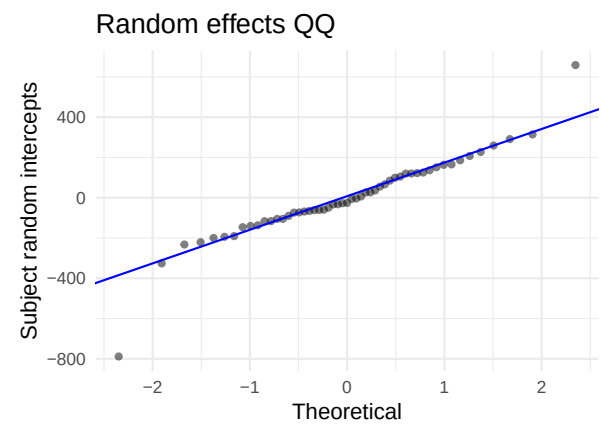
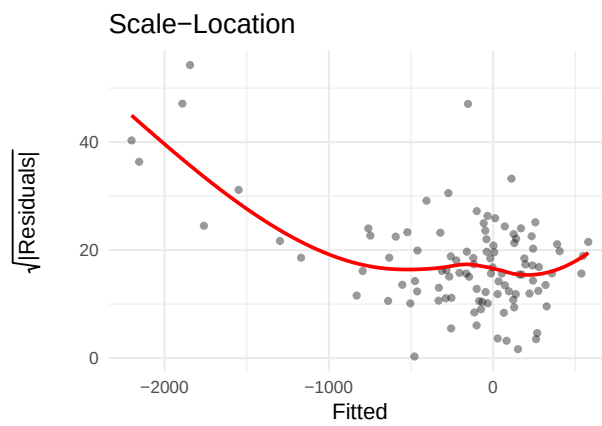
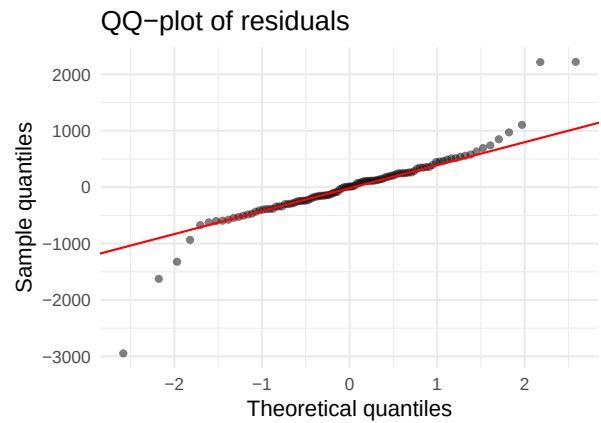
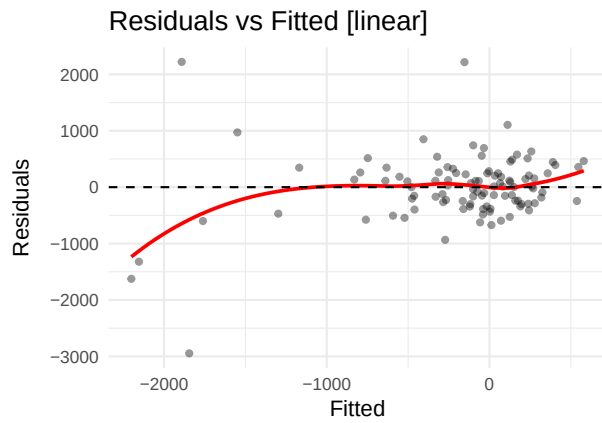


Table 3: Shapiro-Wilk — Angiopoietin-1

Scale	W	p-value	Interpretation
linear	0.881	0	Strong departure from normality
log-ratio	0.884	0	Strong departure from normality

5.2 Angiopoietin-2 (ANGPT2)

AIC linear: 1604.4 | AIC log-ratio: -7 | Preferred: log



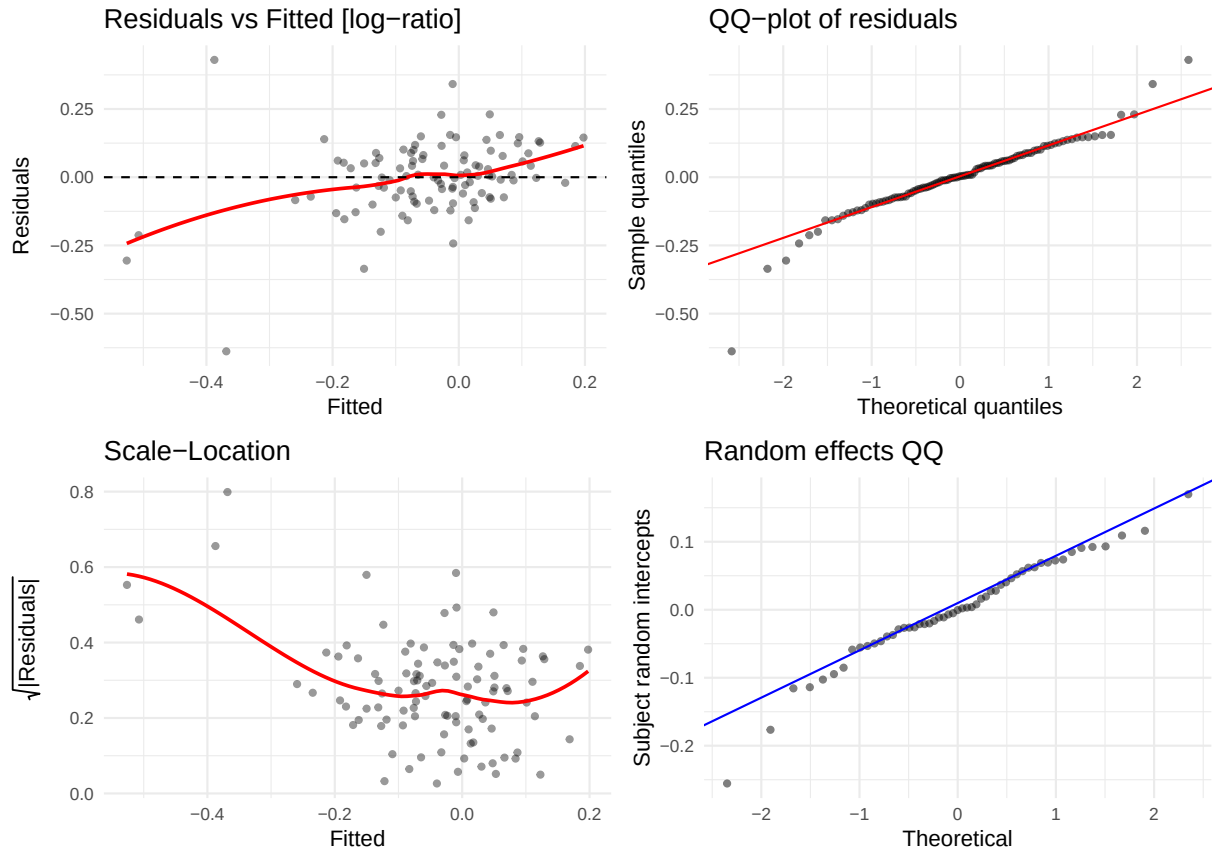
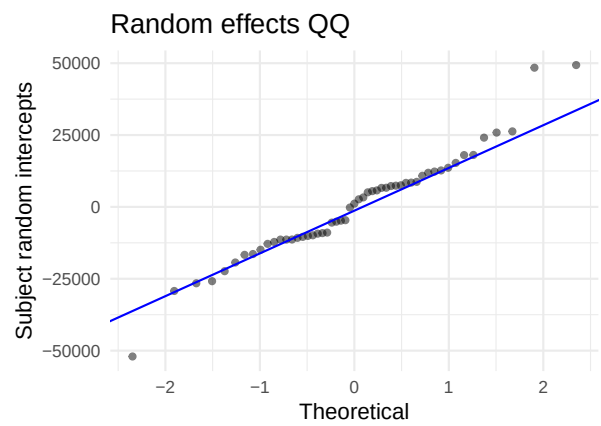
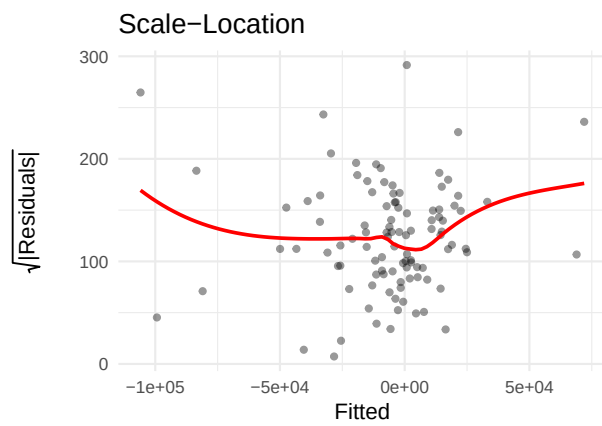
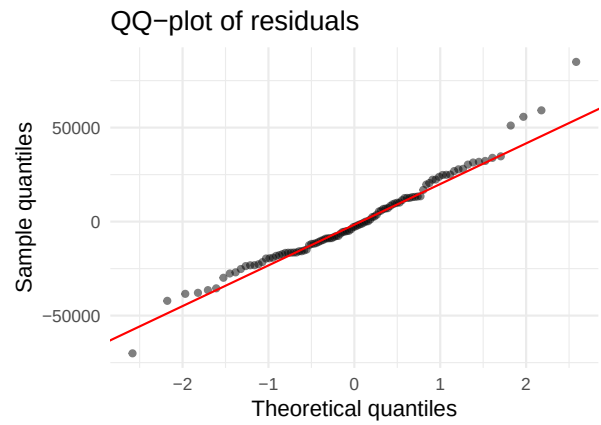
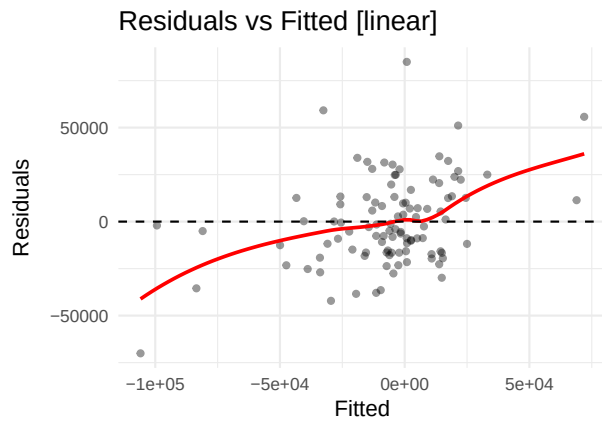


Table 4: Shapiro-Wilk — Angiopoietin-2

Scale	W	p-value	Interpretation
linear	0.8698	0	Strong departure from normality
log-ratio	0.9285	0	Strong departure from normality

5.3 ANGPTL4 (ANGPTL4)

AIC linear: 2349.9 | AIC log-ratio: 31.9 | Preferred: log



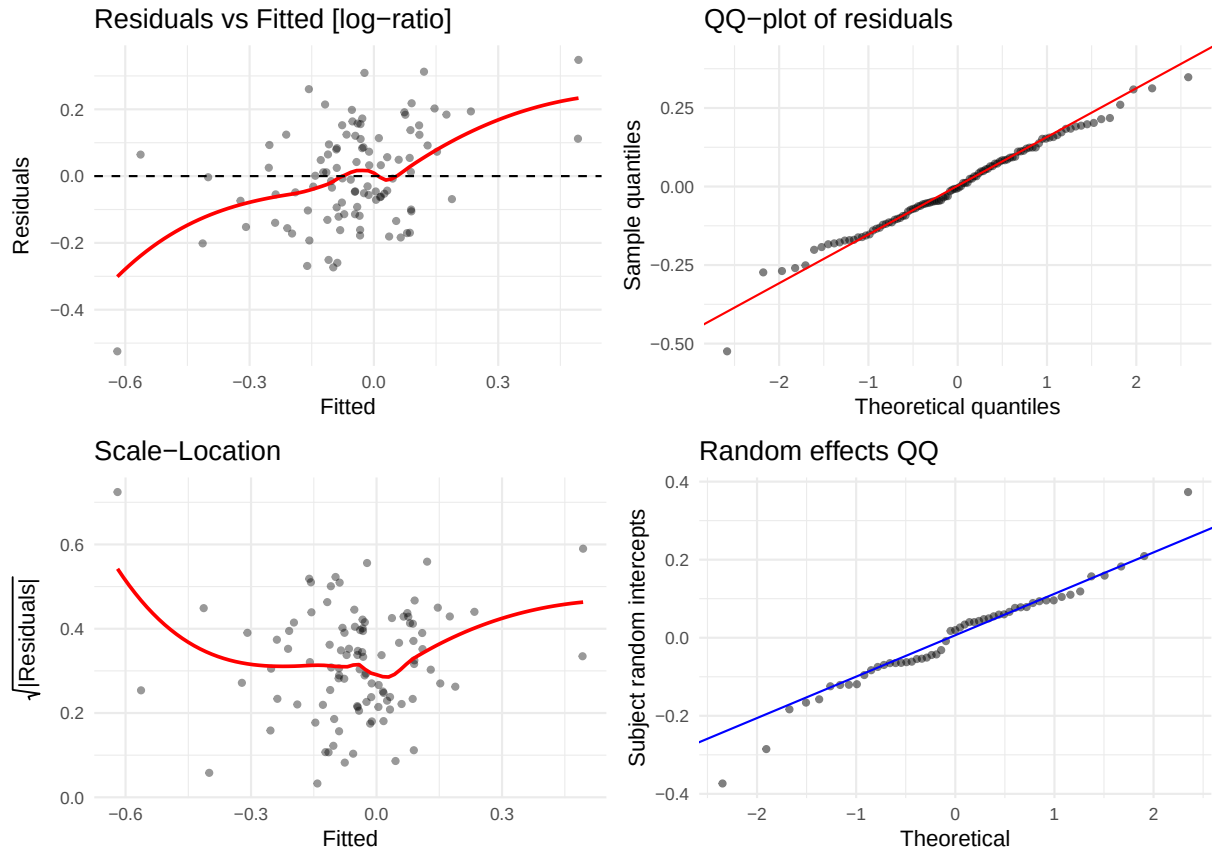
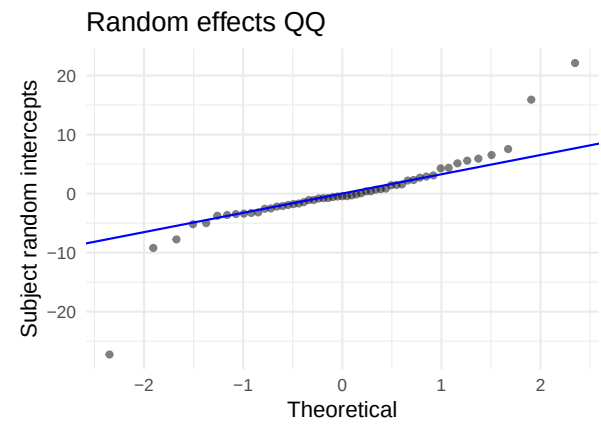
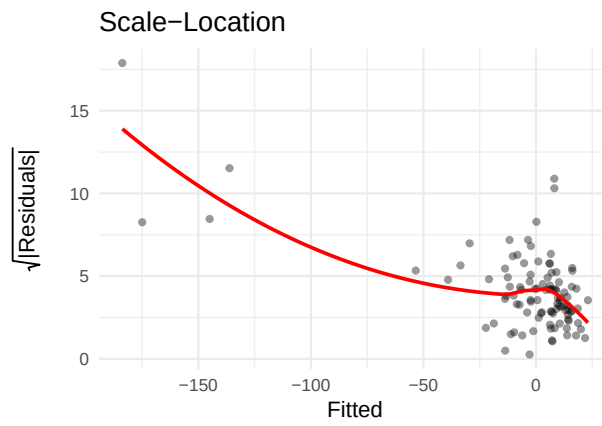
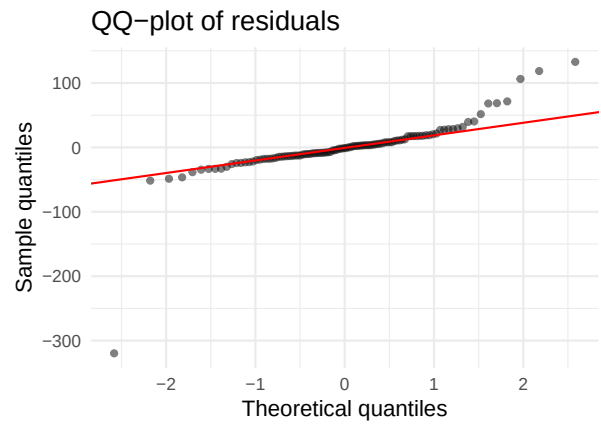
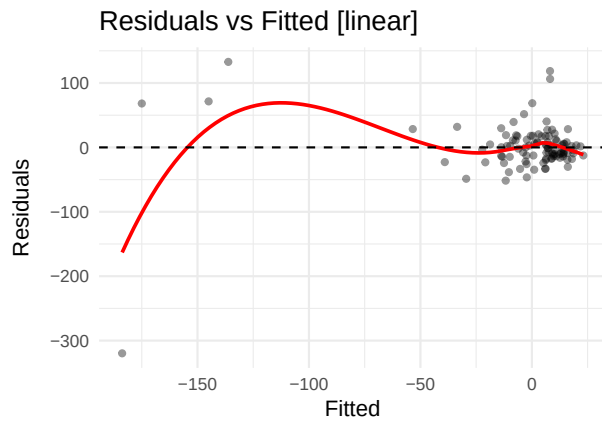


Table 5: Shapiro-Wilk — ANGPTL4

Scale	W	p-value	Interpretation
linear	0.9723	0.0303	Some departure from normality
log-ratio	0.9862	0.3691	No evidence of non-normality

5.4 CA9 (CA9)

AIC linear: 1075.6 | **AIC log-ratio:** 231.5 | **Preferred:** log



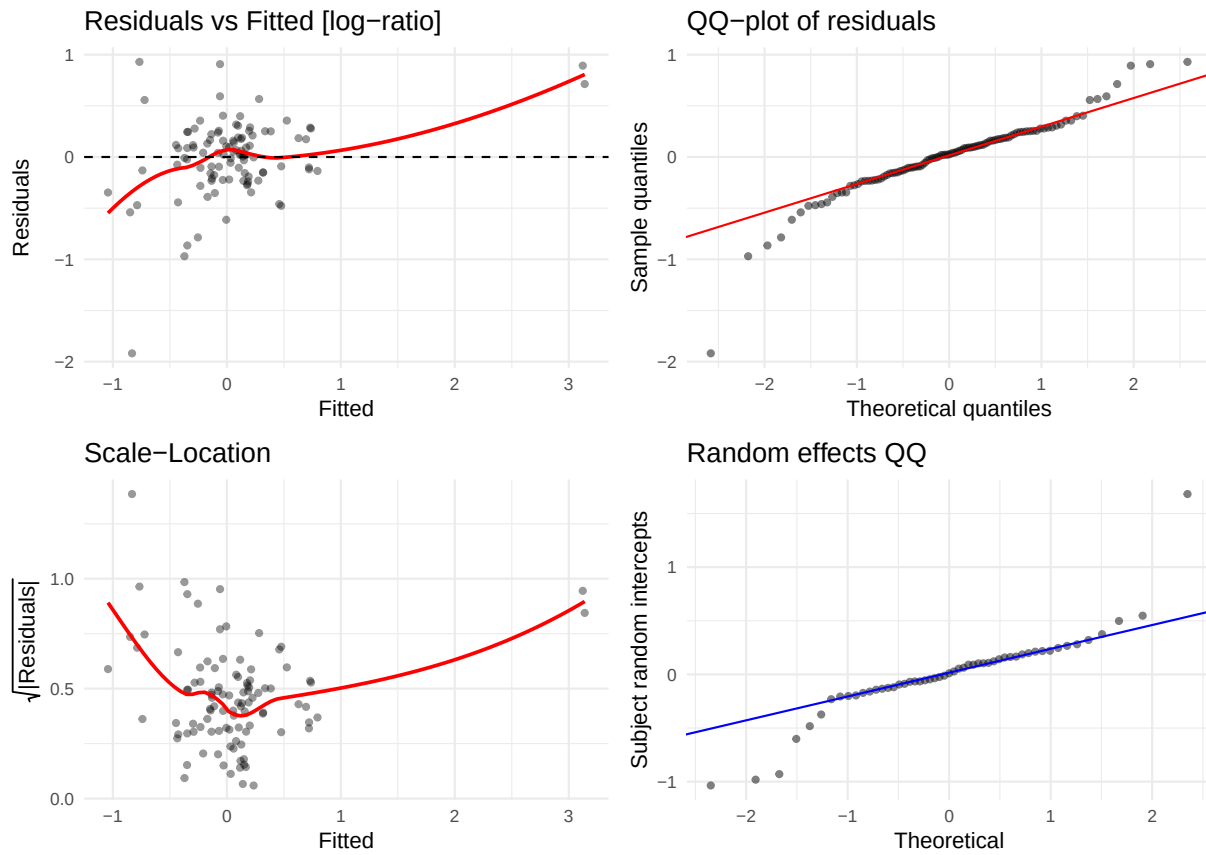
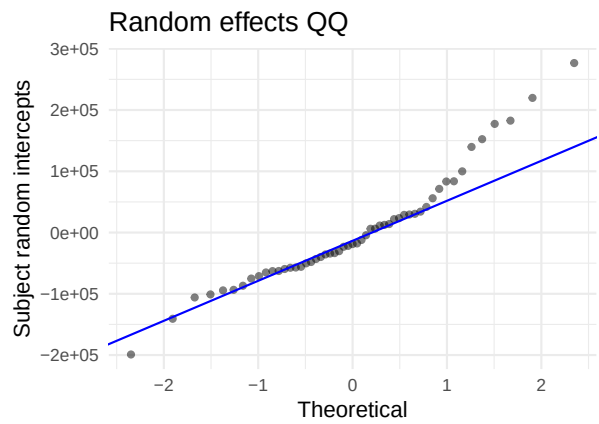
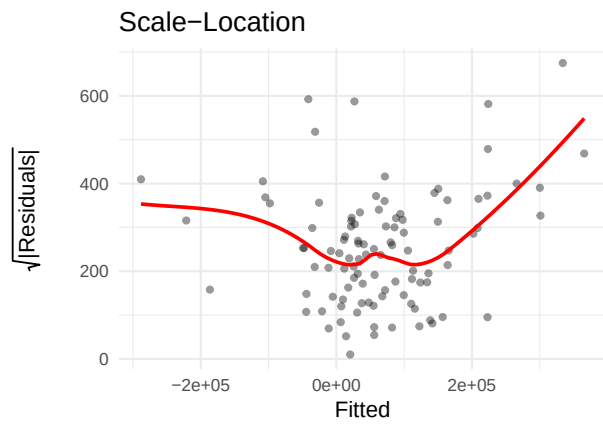
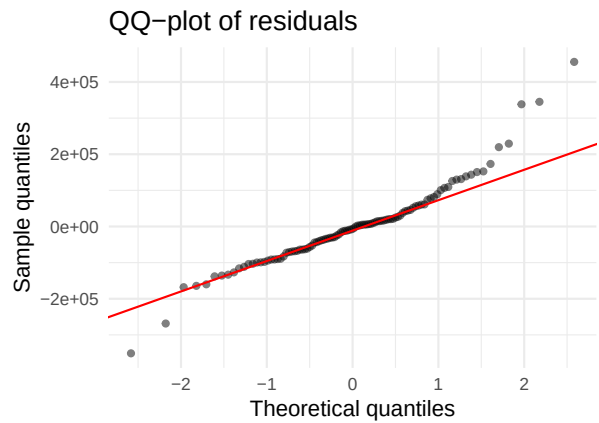
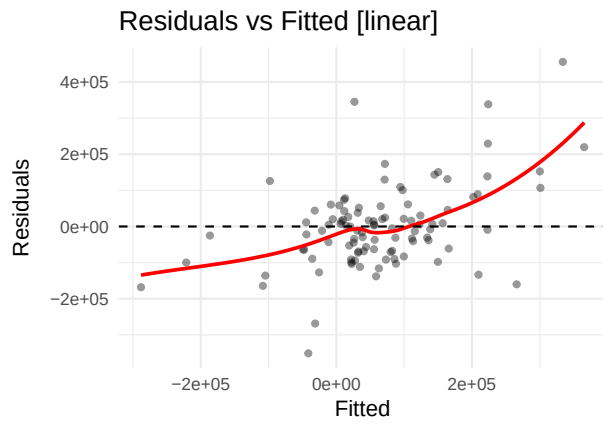


Table 6: Shapiro-Wilk — CA9

Scale	W	p-value	Interpretation
linear	0.6674	0	Strong departure from normality
log-ratio	0.9118	0	Strong departure from normality

5.5 CD163 (CD163)

AIC linear: 2660.7 | AIC log-ratio: 22.6 | Preferred: log



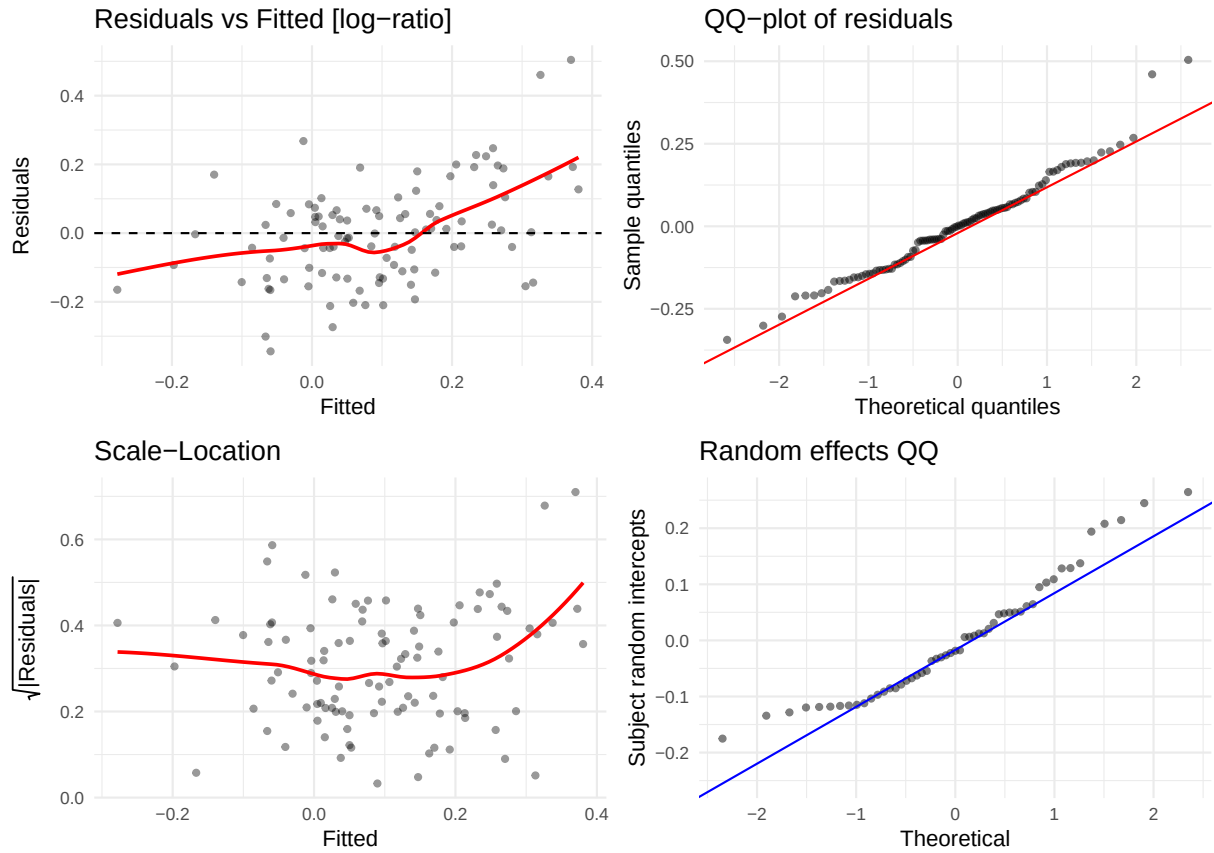
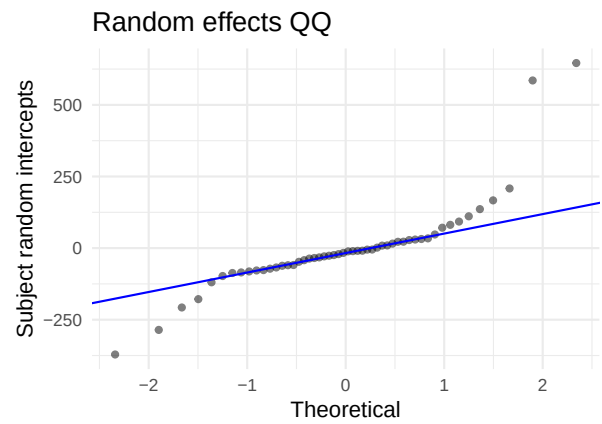
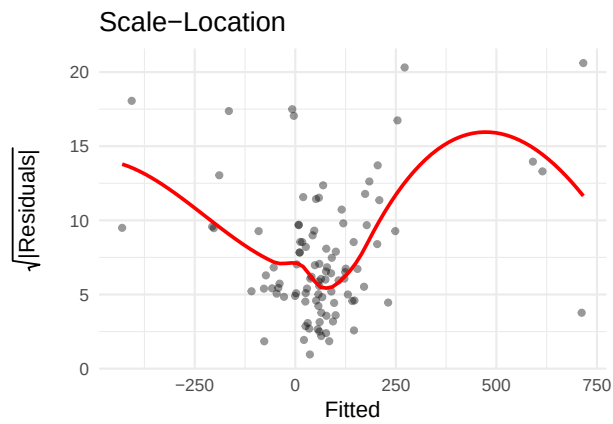
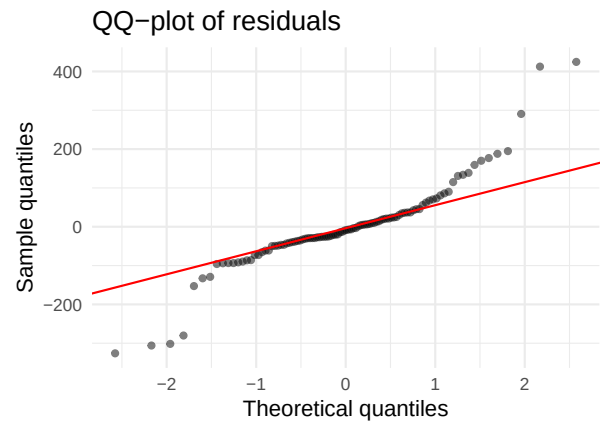
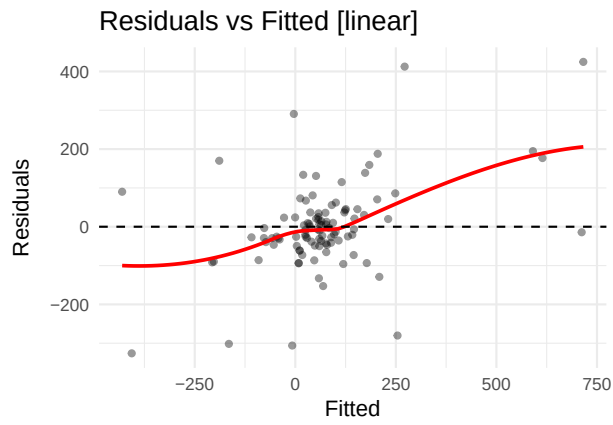


Table 7: Shapiro-Wilk — CD163

Scale	W	p-value	Interpretation
linear	0.9347	0.0001	Strong departure from normality
log-ratio	0.9738	0.0402	Some departure from normality

5.6 Collagen IV alpha 1 (COL1A1)

AIC linear: 1328.6 | AIC log-ratio: 11.3 | Preferred: log



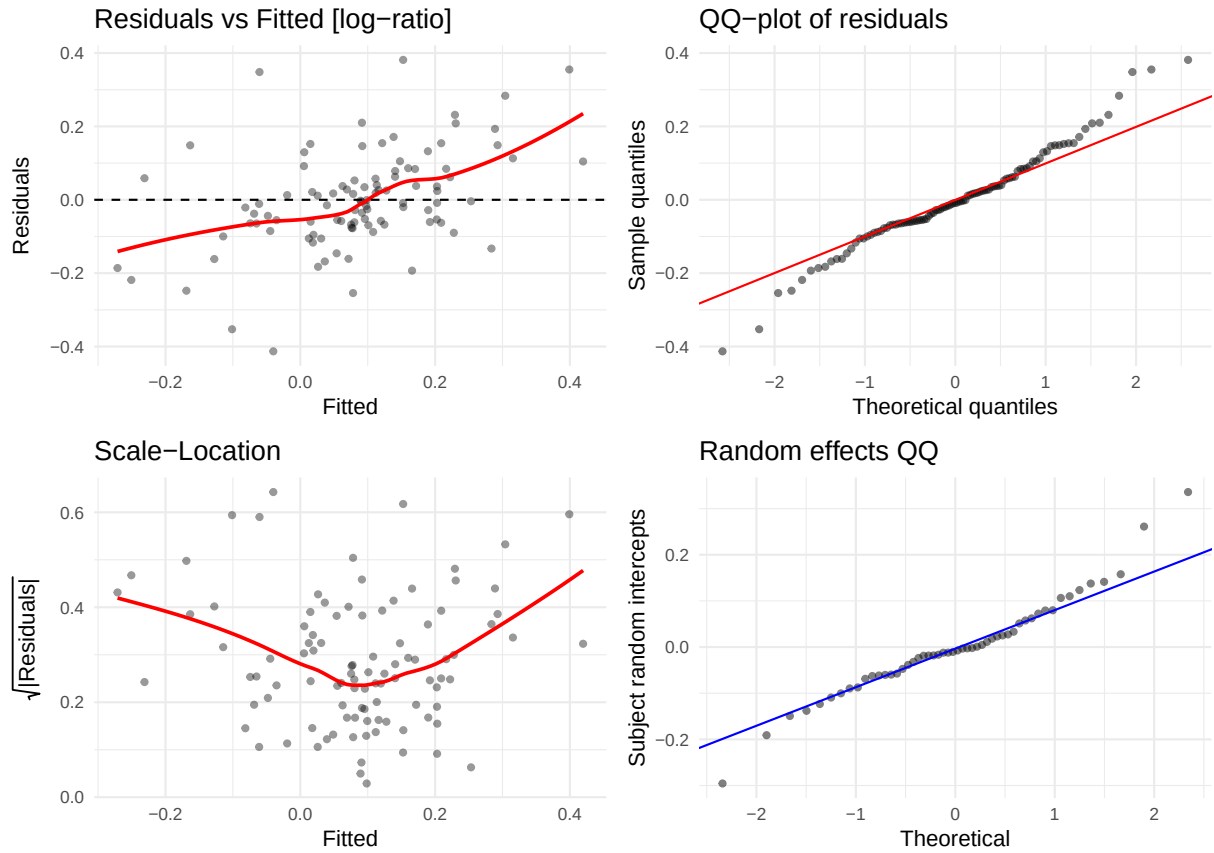
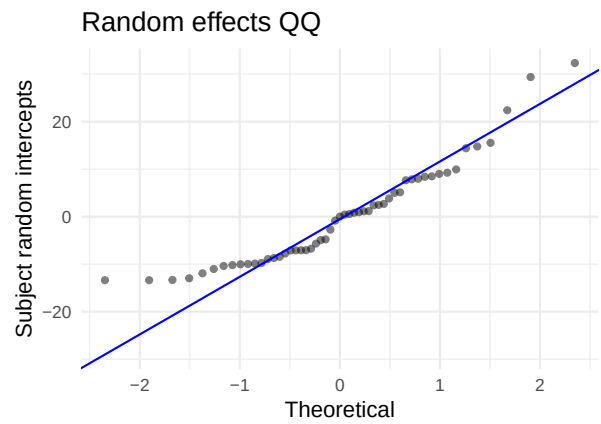
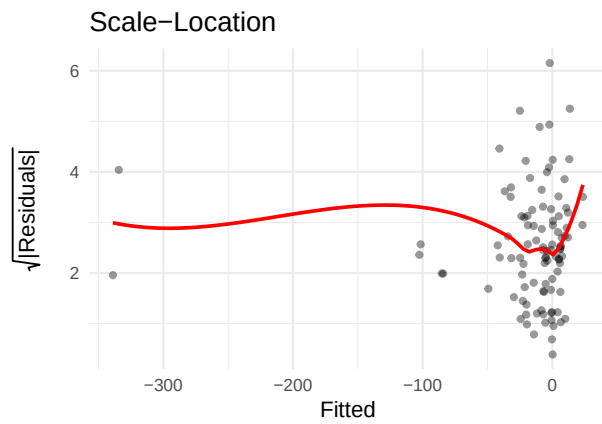
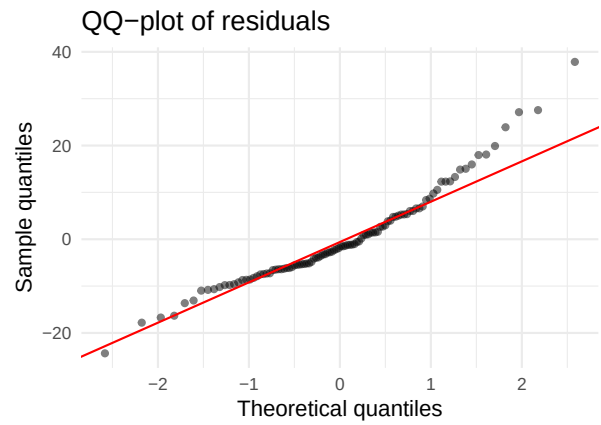
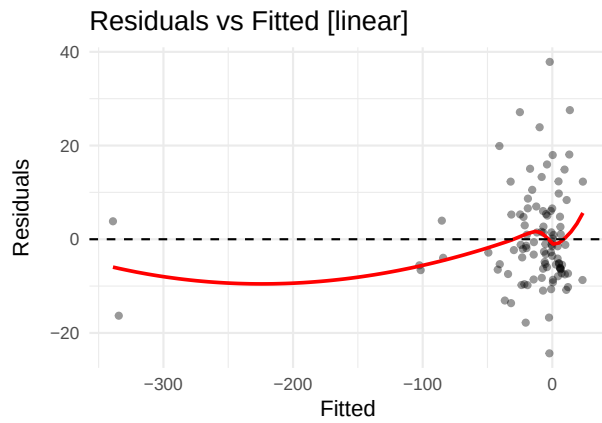


Table 8: Shapiro-Wilk — Collagen IV alpha 1

Scale	W	p-value	Interpretation
linear	0.8860	0.0000	Strong departure from normality
log-ratio	0.9752	0.0563	Borderline

5.7 EGF (EGF)

AIC linear: 869.9 | AIC log-ratio: 430.9 | Preferred: log



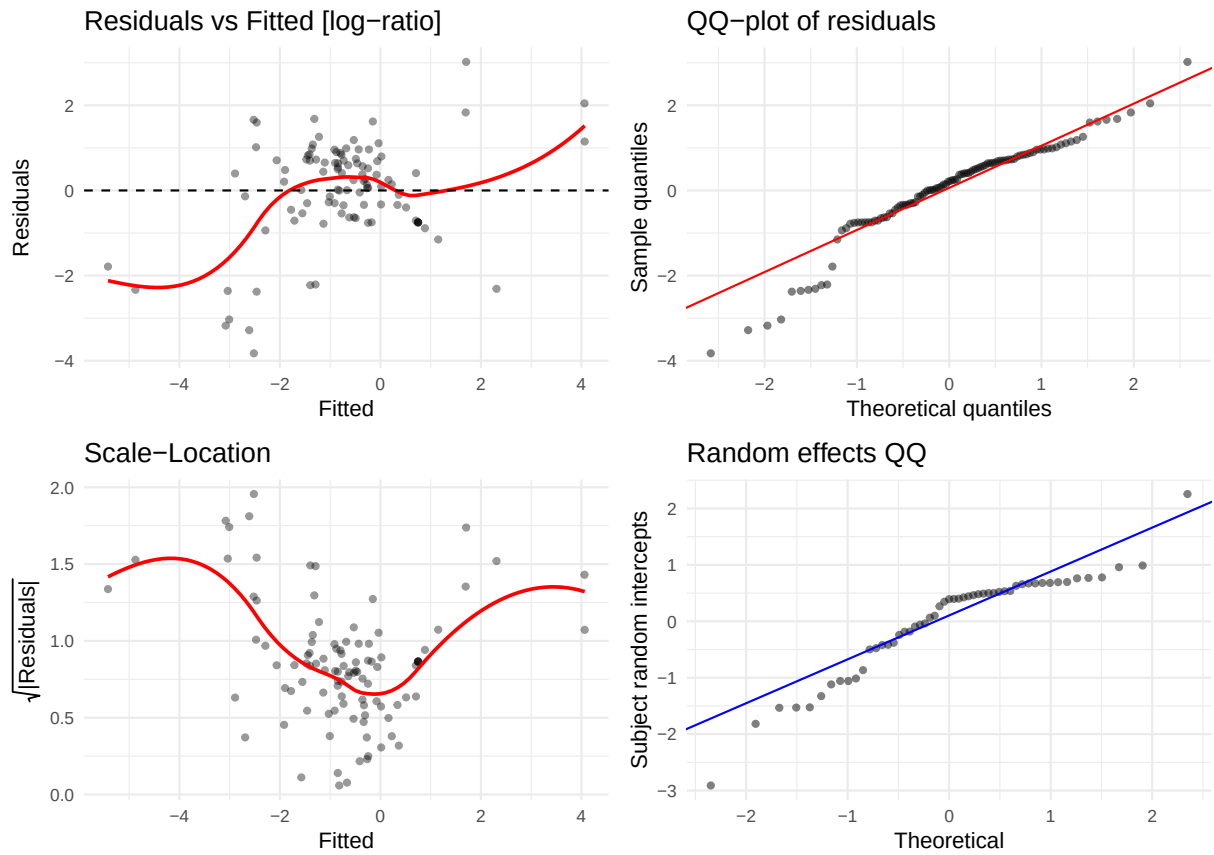
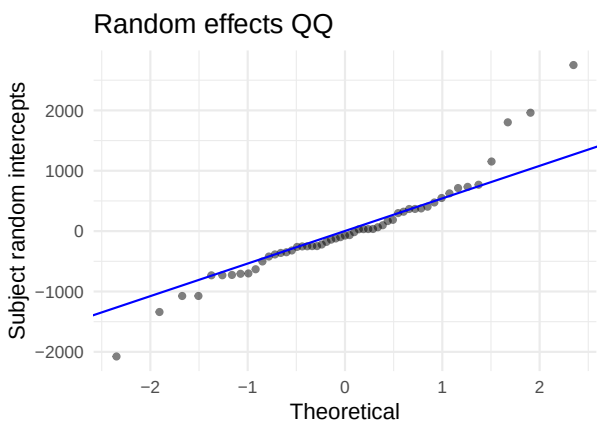
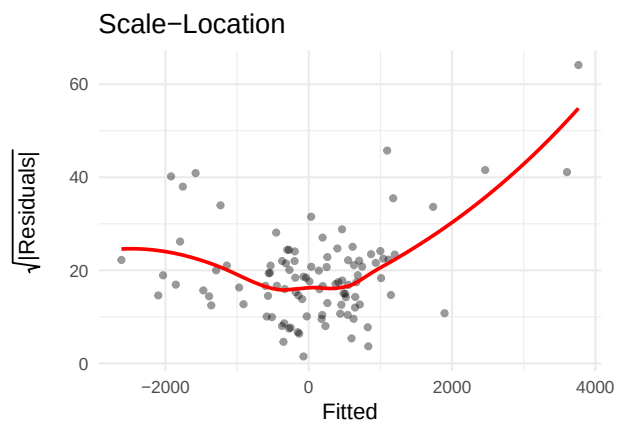
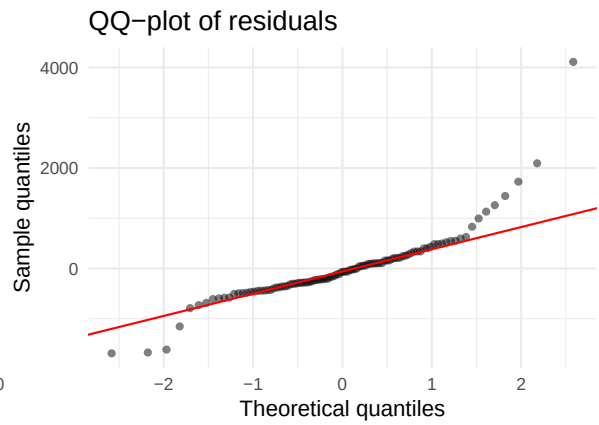
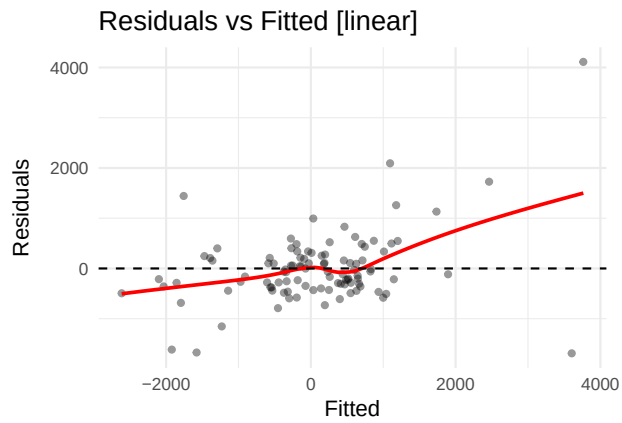


Table 9: Shapiro-Wilk — EGF

Scale	W	p-value	Interpretation
linear	0.9446	3e-04	Strong departure from normality
log-ratio	0.9303	0e+00	Strong departure from normality

5.8 ErbB2 (ERBB2)

AIC linear: 1697 | AIC log-ratio: -13 | Preferred: log



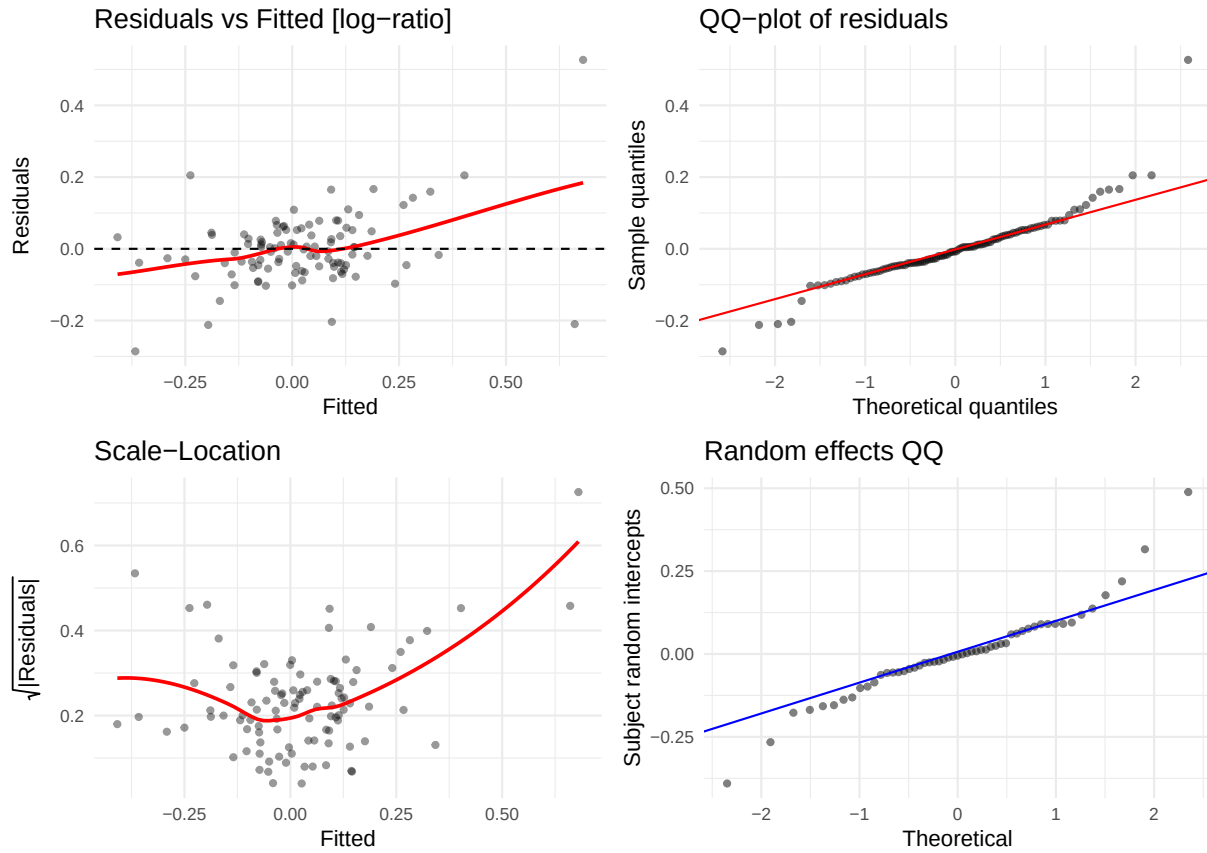
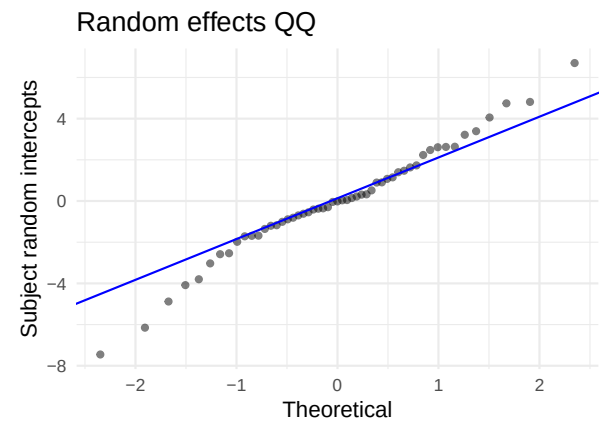
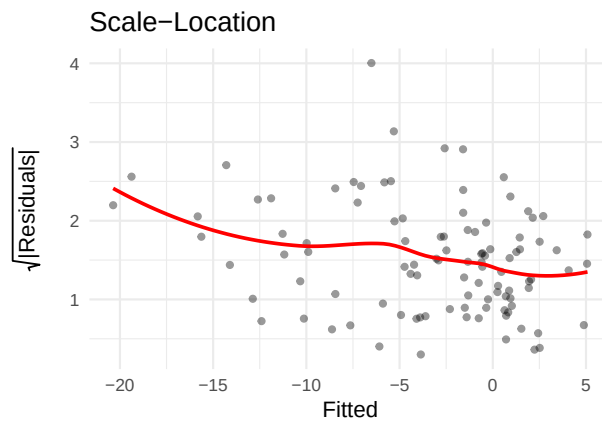
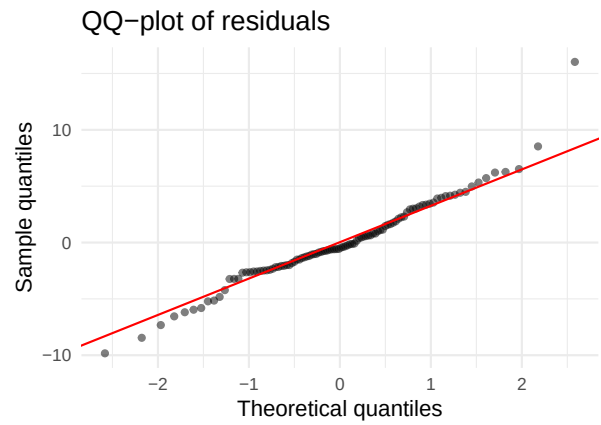
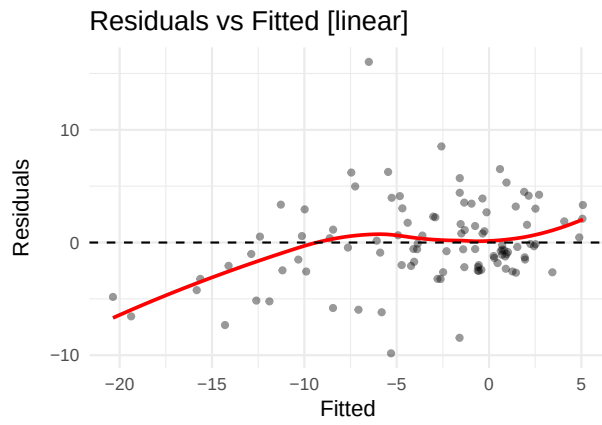


Table 10: Shapiro-Wilk — ErbB2

Scale	W	p-value	Interpretation
linear	0.8283	0	Strong departure from normality
log-ratio	0.8904	0	Strong departure from normality

5.9 FGF Basic (FGFB)

AIC linear: 649.4 | AIC log-ratio: 209.9 | Preferred: log



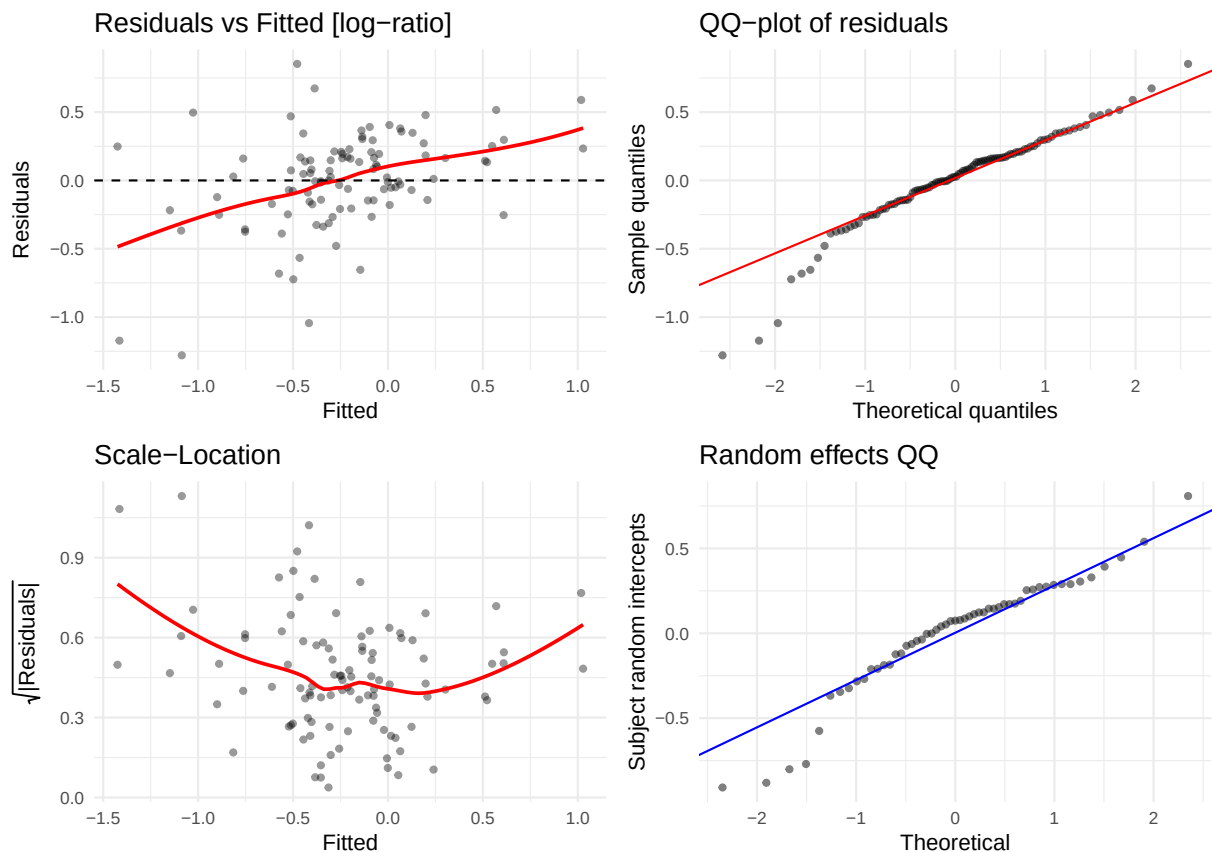
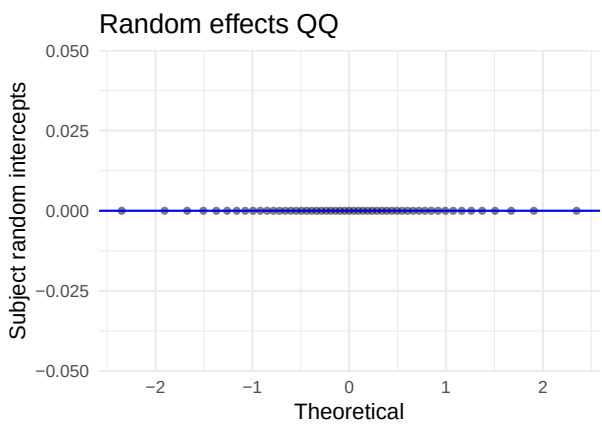
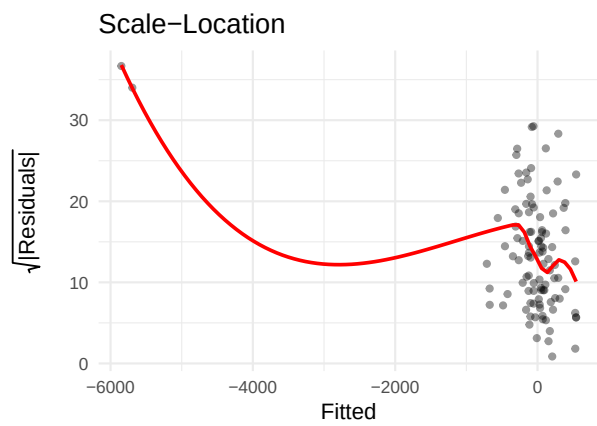
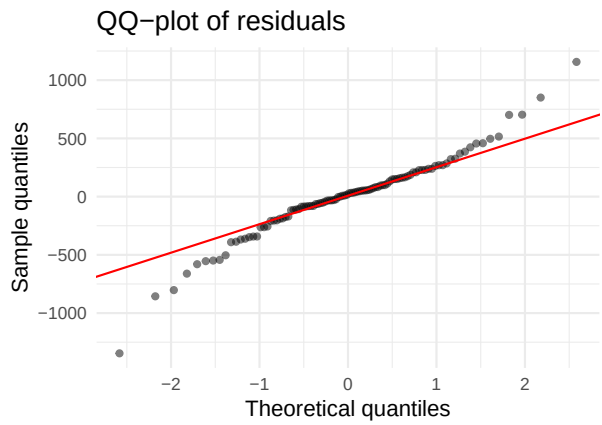
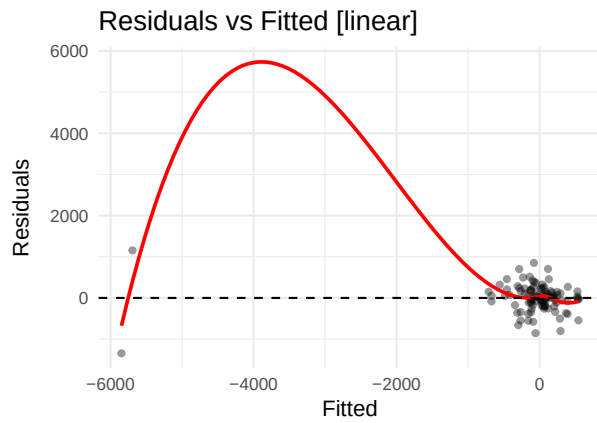


Table 11: Shapiro-Wilk — FGF Basic

Scale	W	p-value	Interpretation
linear	0.9601	0.0036	Strong departure from normality
log-ratio	0.9405	0.0002	Strong departure from normality

5.10 Fractalkine (FRACTKN)

AIC linear: 1458.8 | AIC log-ratio: 219.4 | Preferred: log



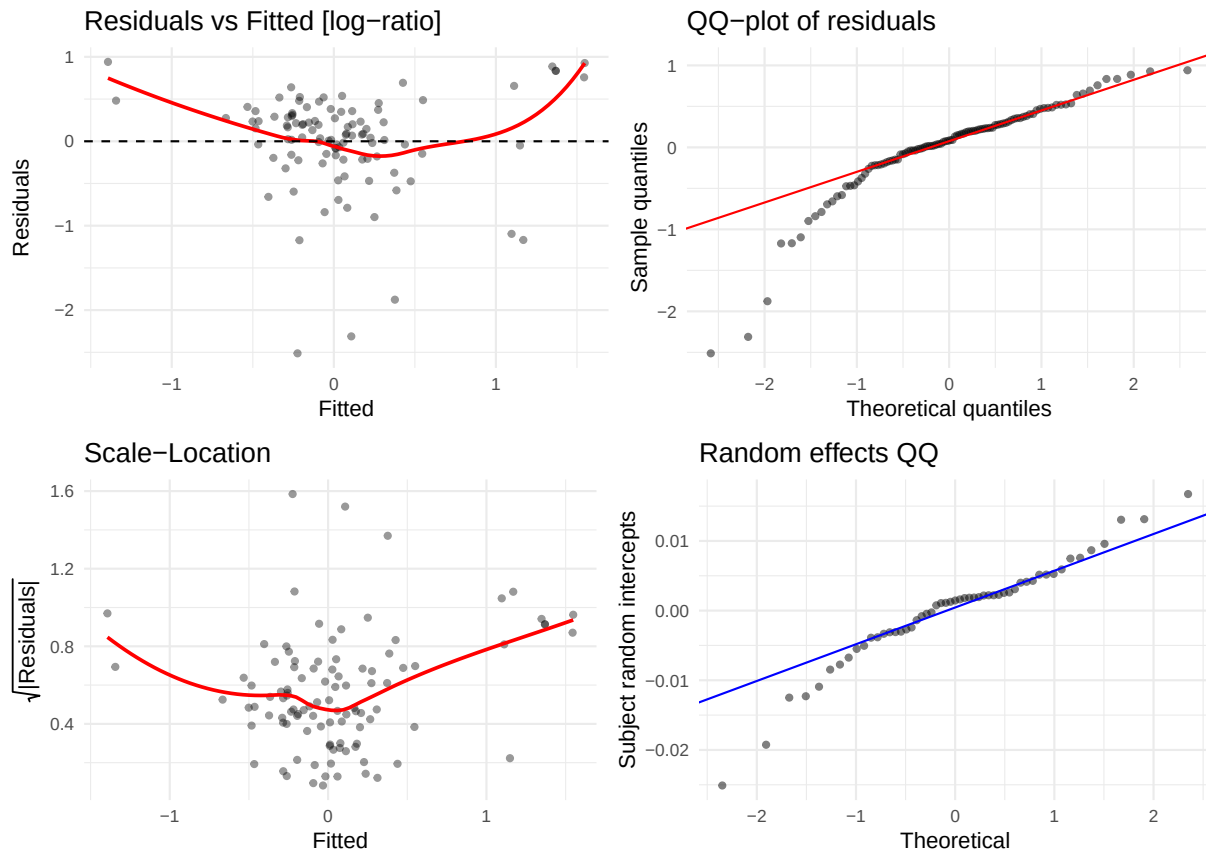
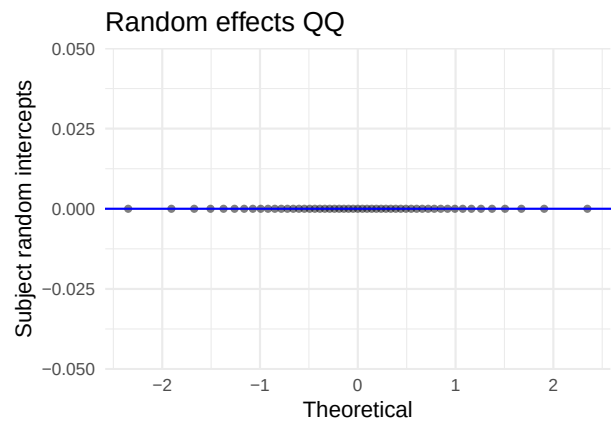
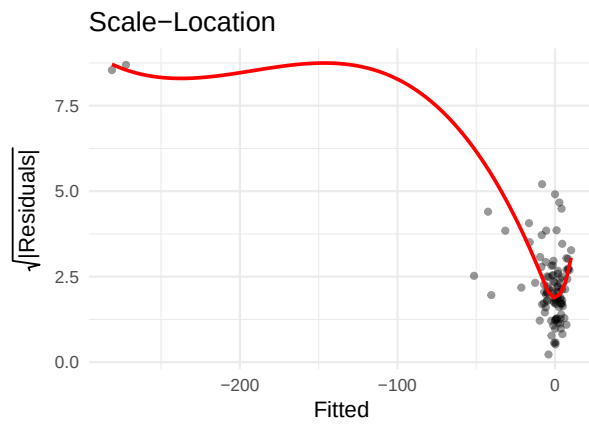
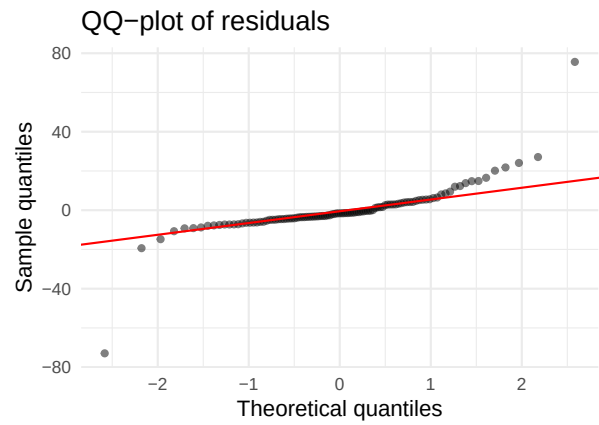
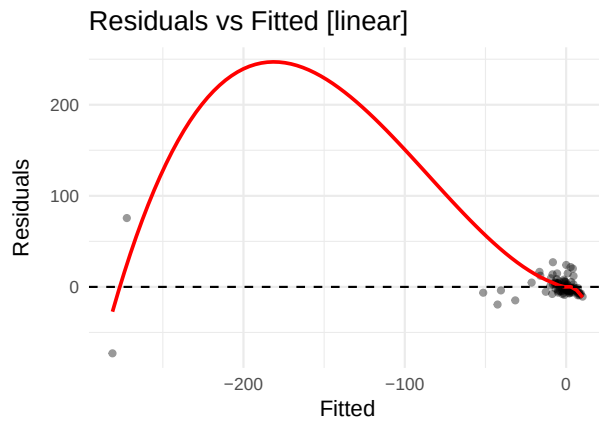


Table 12: Shapiro-Wilk — Fractalkine

Scale	W	p-value	Interpretation
linear	0.9555	0.0017	Strong departure from normality
log-ratio	0.8605	0.0000	Strong departure from normality

5.11 Granzyme B (GZMB)

AIC linear: 820.6 | AIC log-ratio: 150.7 | Preferred: log



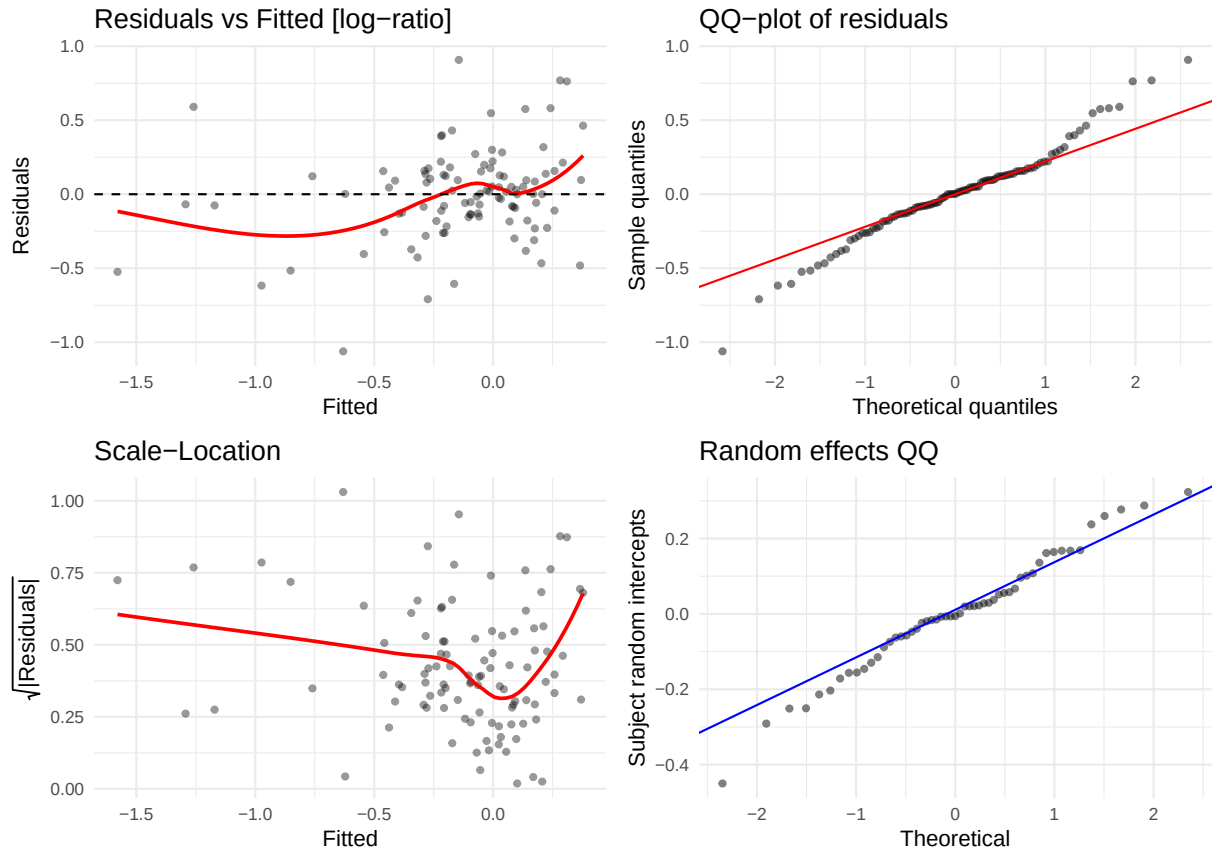
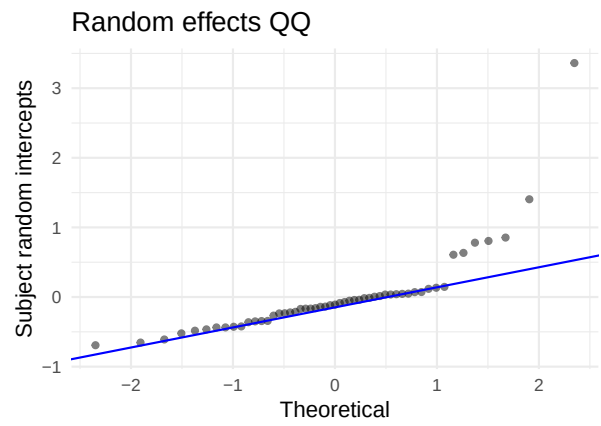
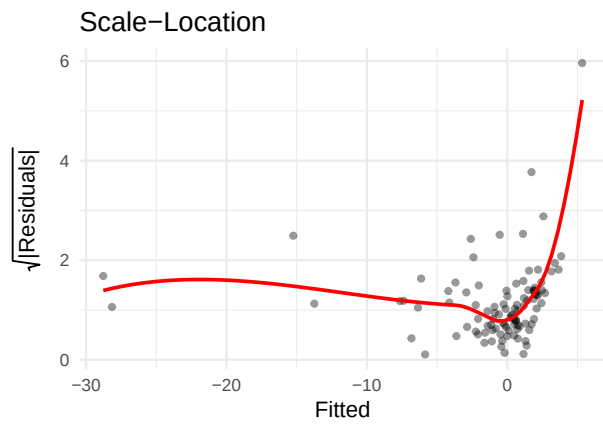
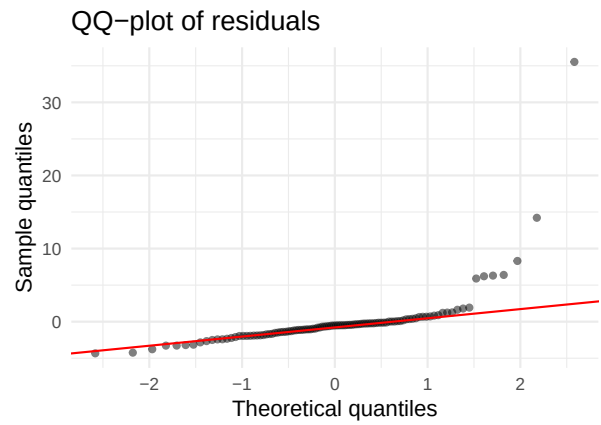
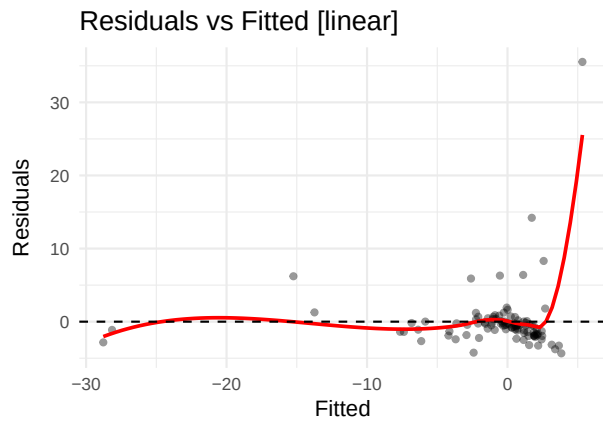


Table 13: Shapiro-Wilk — Granzyme B

Scale	W	p-value	Interpretation
linear	0.6839	0.0000	Strong departure from normality
log-ratio	0.9727	0.0327	Some departure from normality

5.12 IFN-g (IFNG)

AIC linear: 624.1 | AIC log-ratio: 271.7 | Preferred: log



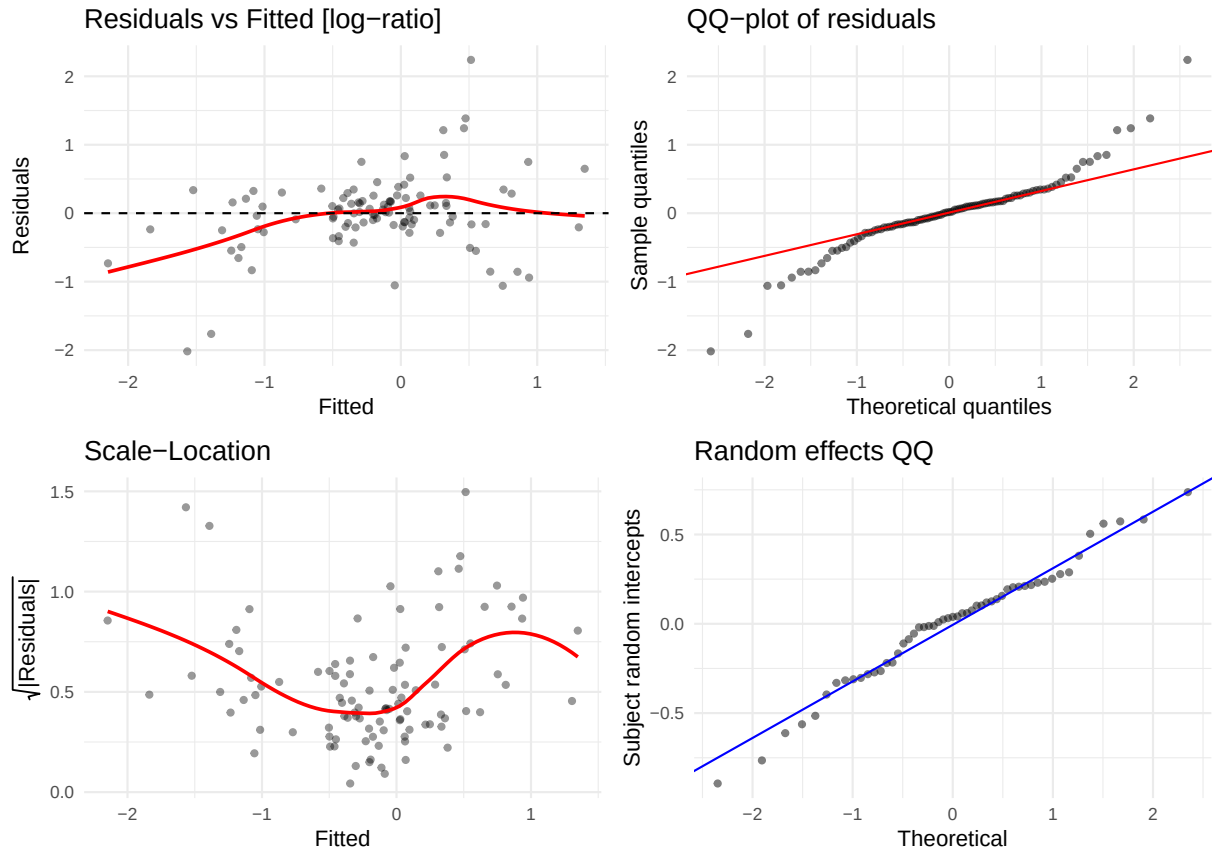
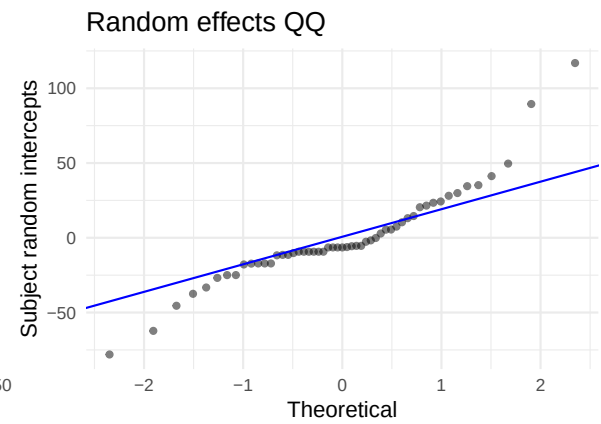
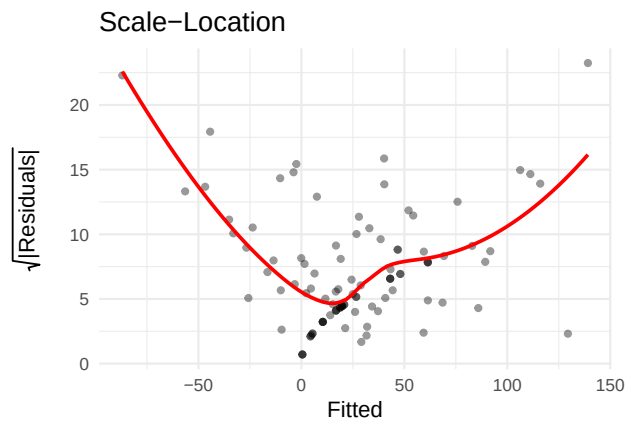
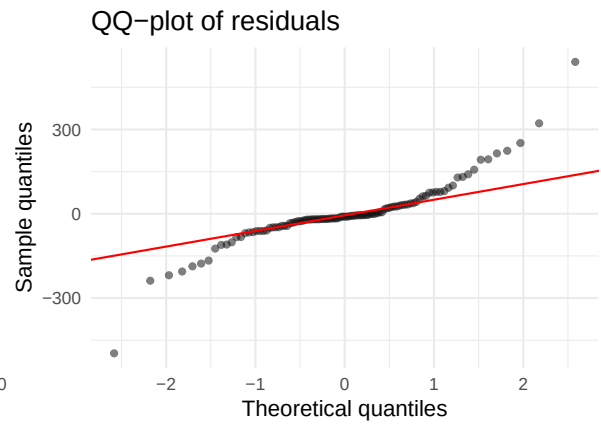
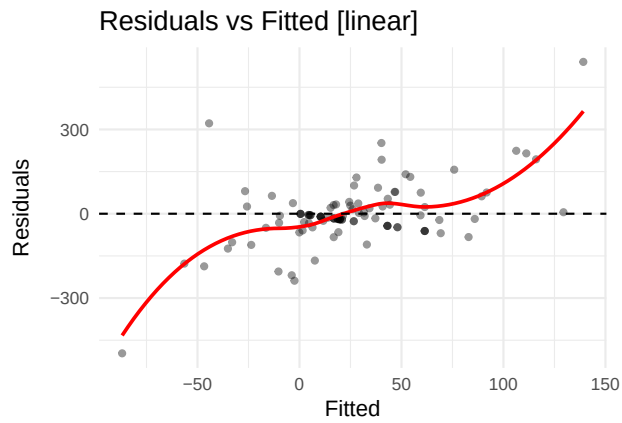


Table 14: Shapiro-Wilk — IFN-g

Scale	W	p-value	Interpretation
linear	0.4711	0	Strong departure from normality
log-ratio	0.9163	0	Strong departure from normality

5.13 IL-12 p70 (IL12P70)

AIC linear: 1279.3 | AIC log-ratio: 248.8 | Preferred: log



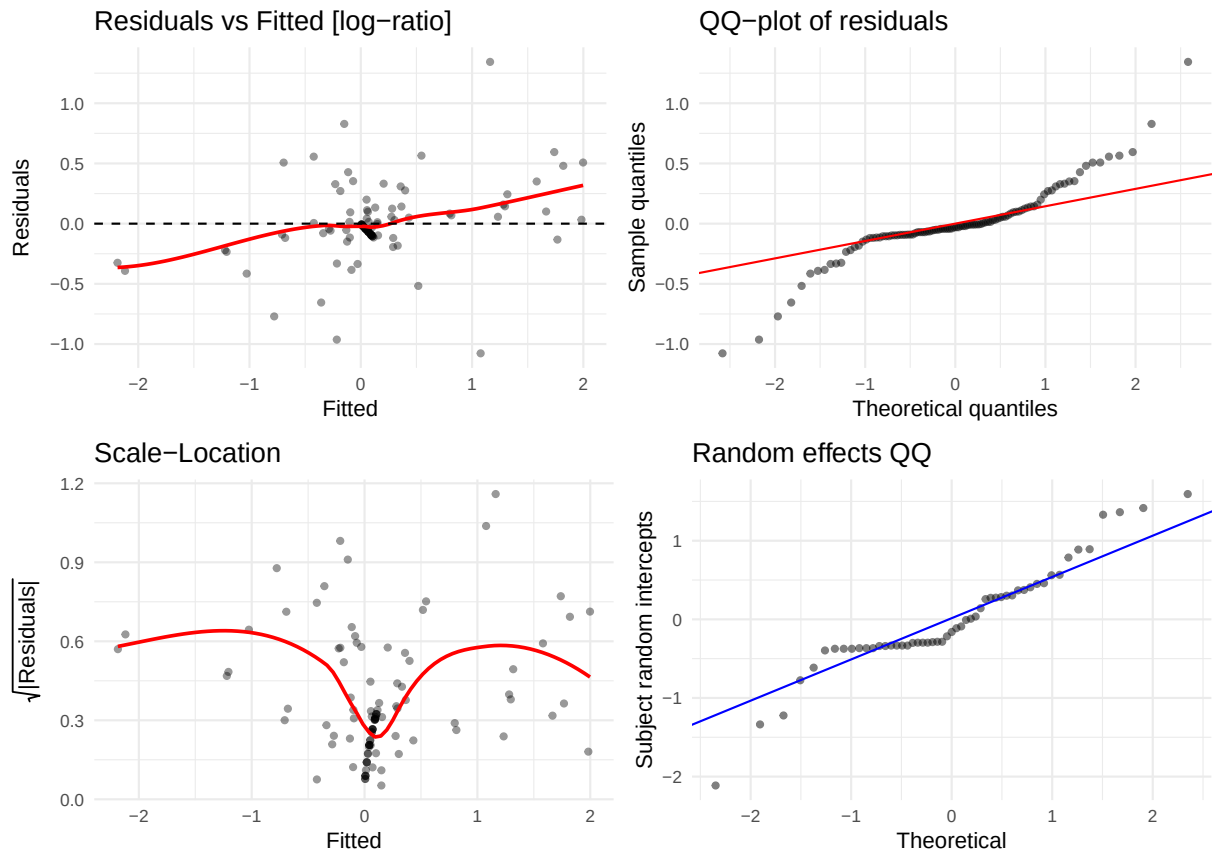


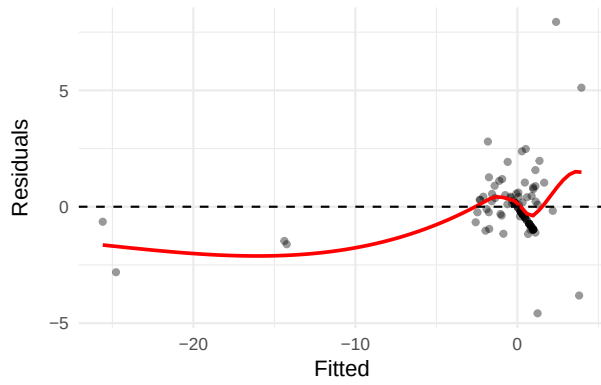
Table 15: Shapiro-Wilk — IL-12 p70

Scale	W	p-value	Interpretation
linear	0.8589	0	Strong departure from normality
log-ratio	0.8829	0	Strong departure from normality

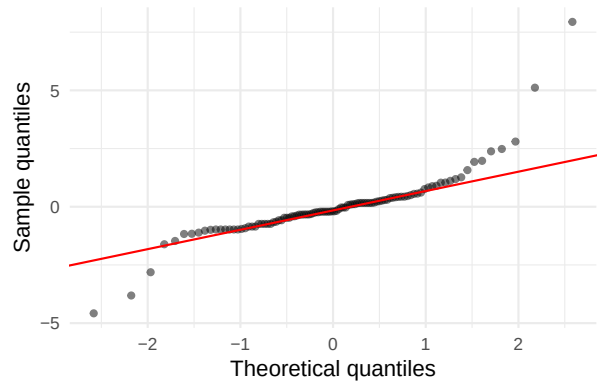
5.14 IL-1b (IL1B)

AIC linear: 512.8 | AIC log-ratio: 288.5 | Preferred: log

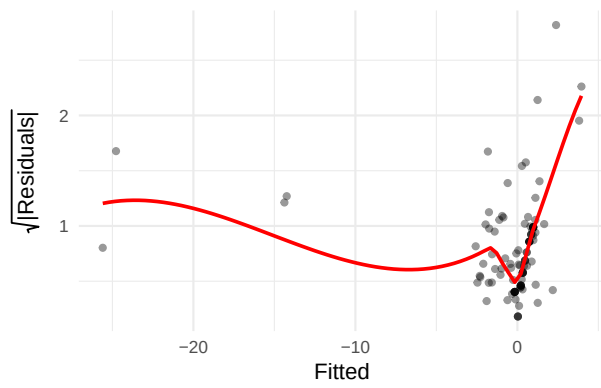
Residuals vs Fitted [linear]



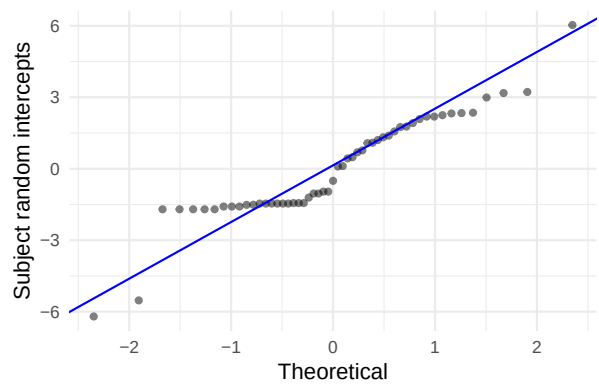
QQ-plot of residuals



Scale-Location



Random effects QQ



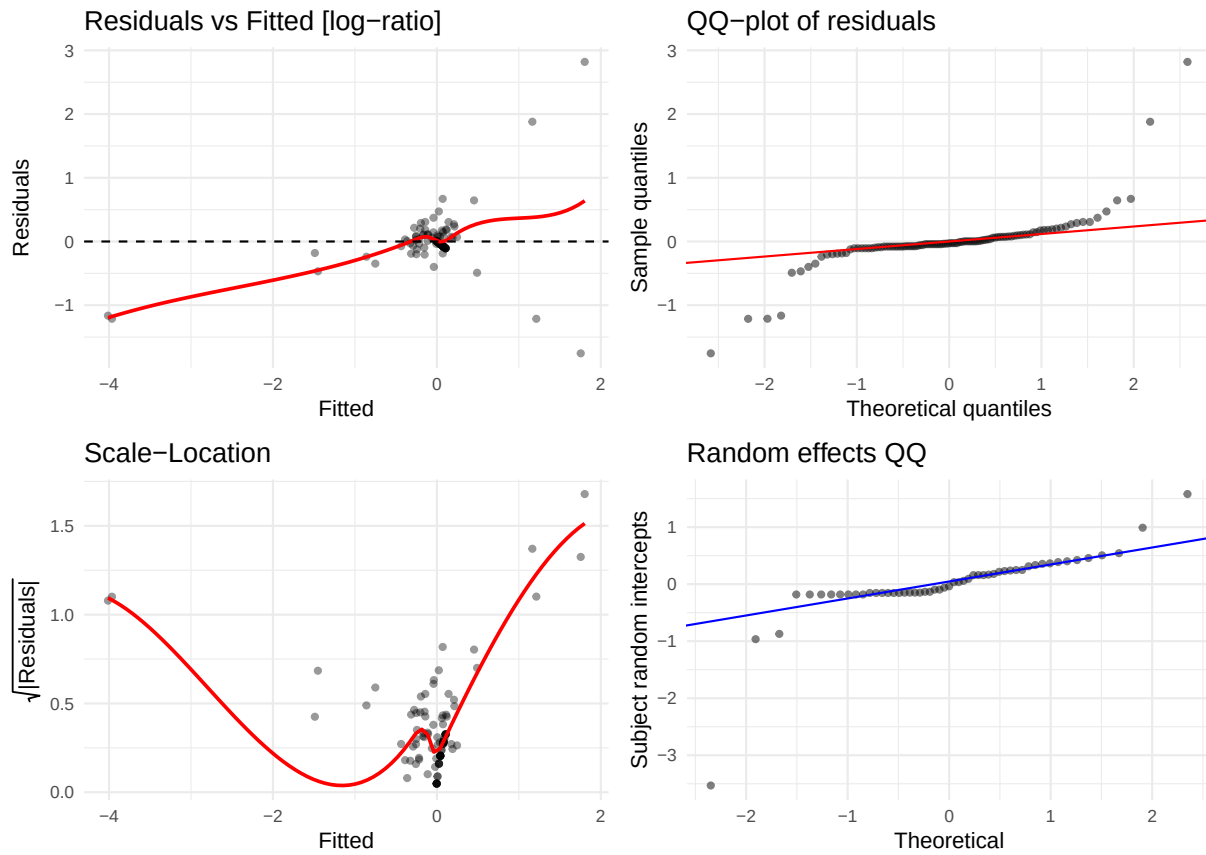
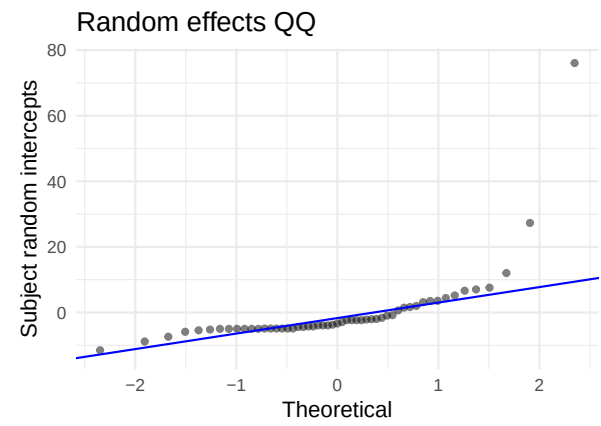
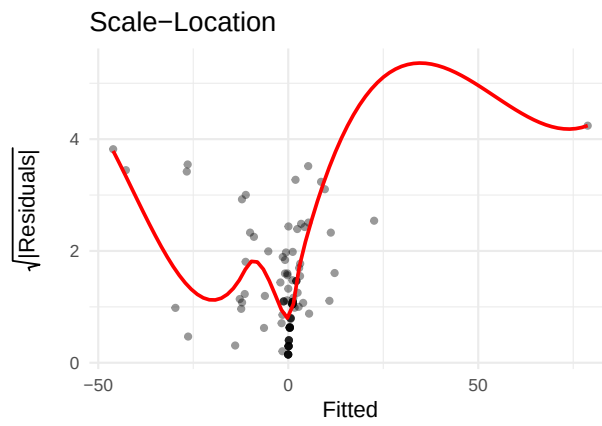
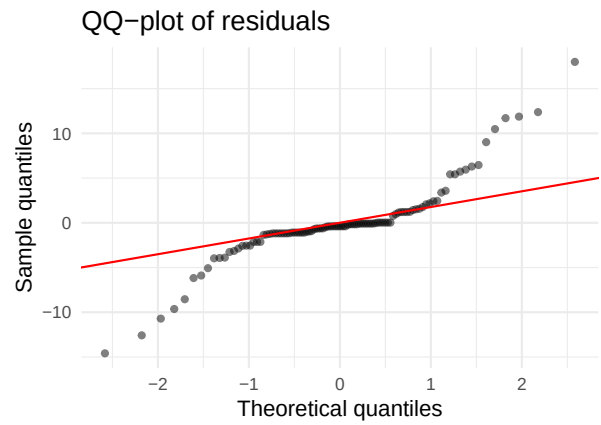
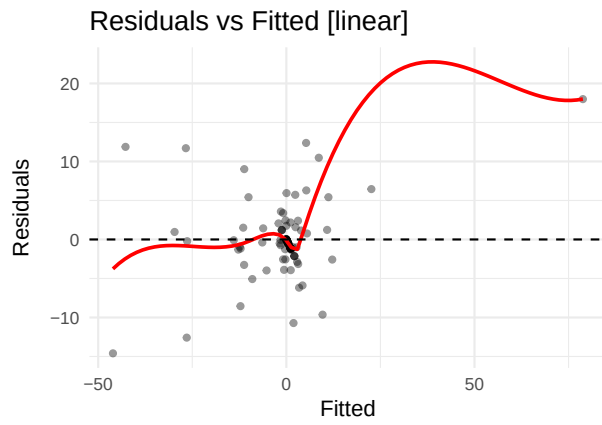


Table 16: Shapiro-Wilk — IL-1b

Scale	W	p-value	Interpretation
linear	0.7959	0	Strong departure from normality
log-ratio	0.6338	0	Strong departure from normality

5.15 IL-2 (IL2)

AIC linear: 786.7 | AIC log-ratio: 360.3 | Preferred: log



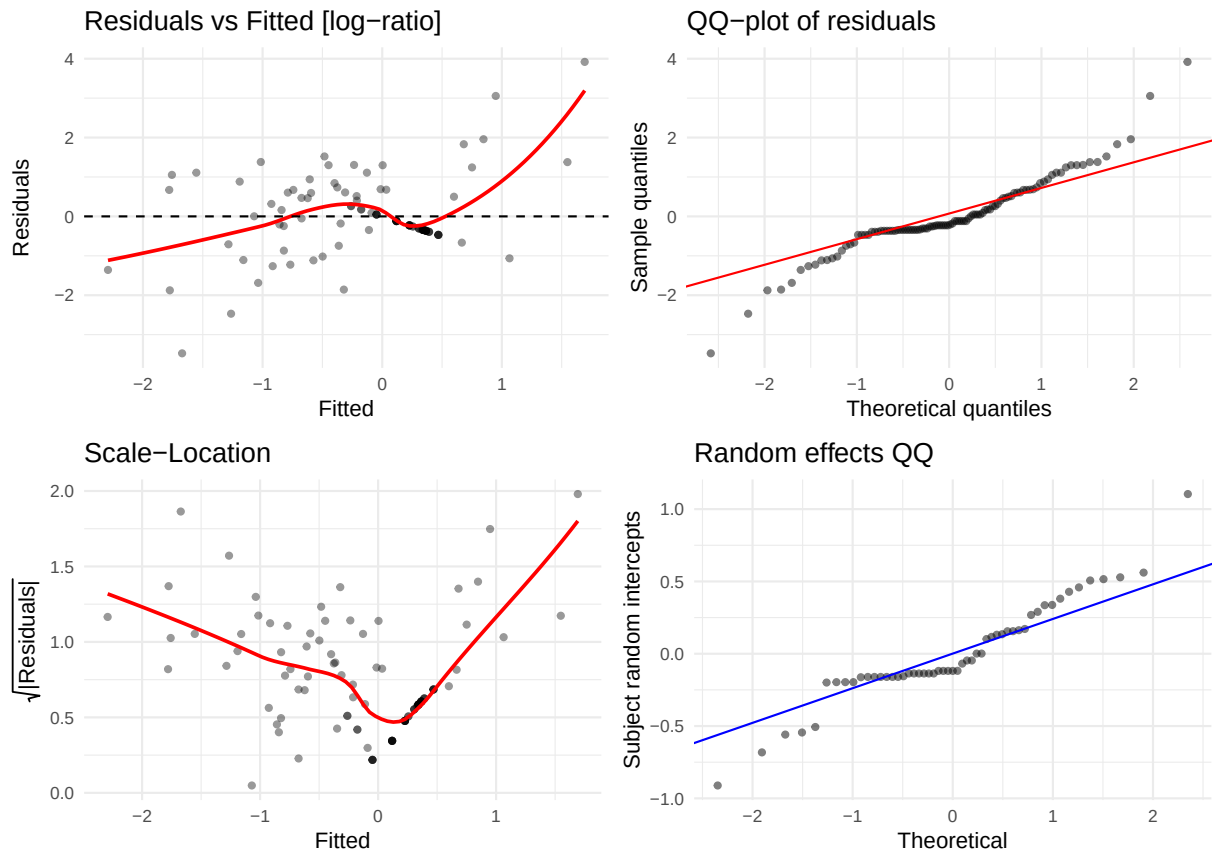
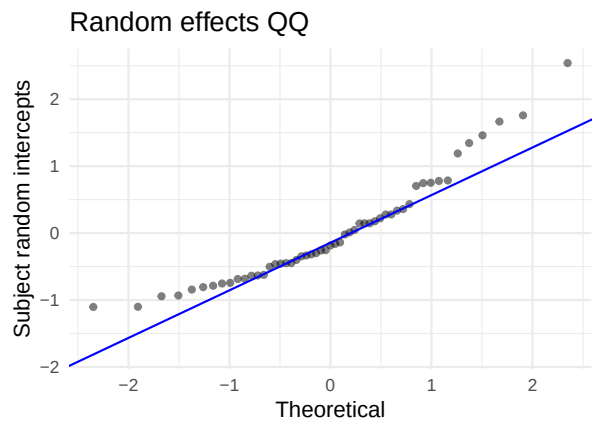
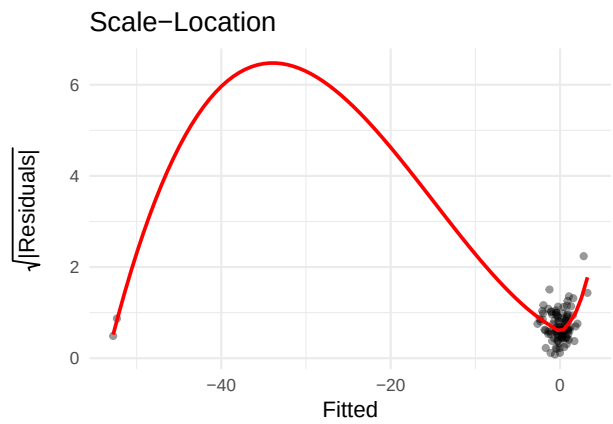
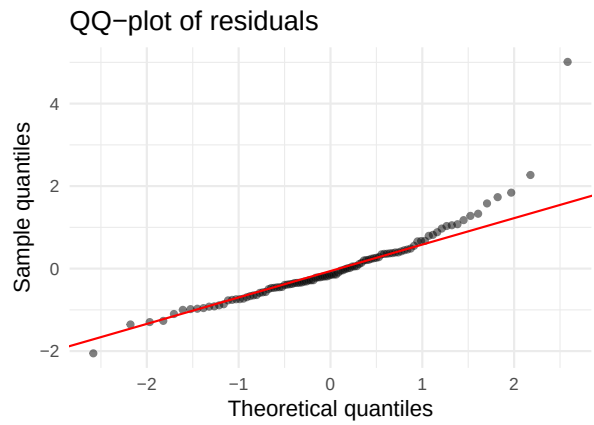
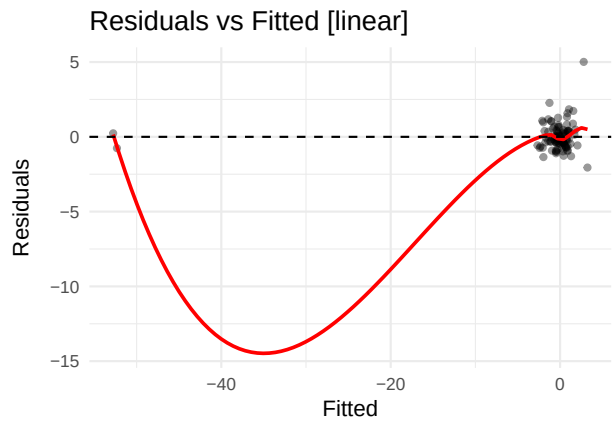


Table 17: Shapiro-Wilk — IL-2

Scale	W	p-value	Interpretation
linear	0.8549	0	Strong departure from normality
log-ratio	0.9257	0	Strong departure from normality

5.16 IL-6 (IL6)

AIC linear: 387.3 | AIC log-ratio: 355 | Preferred: log



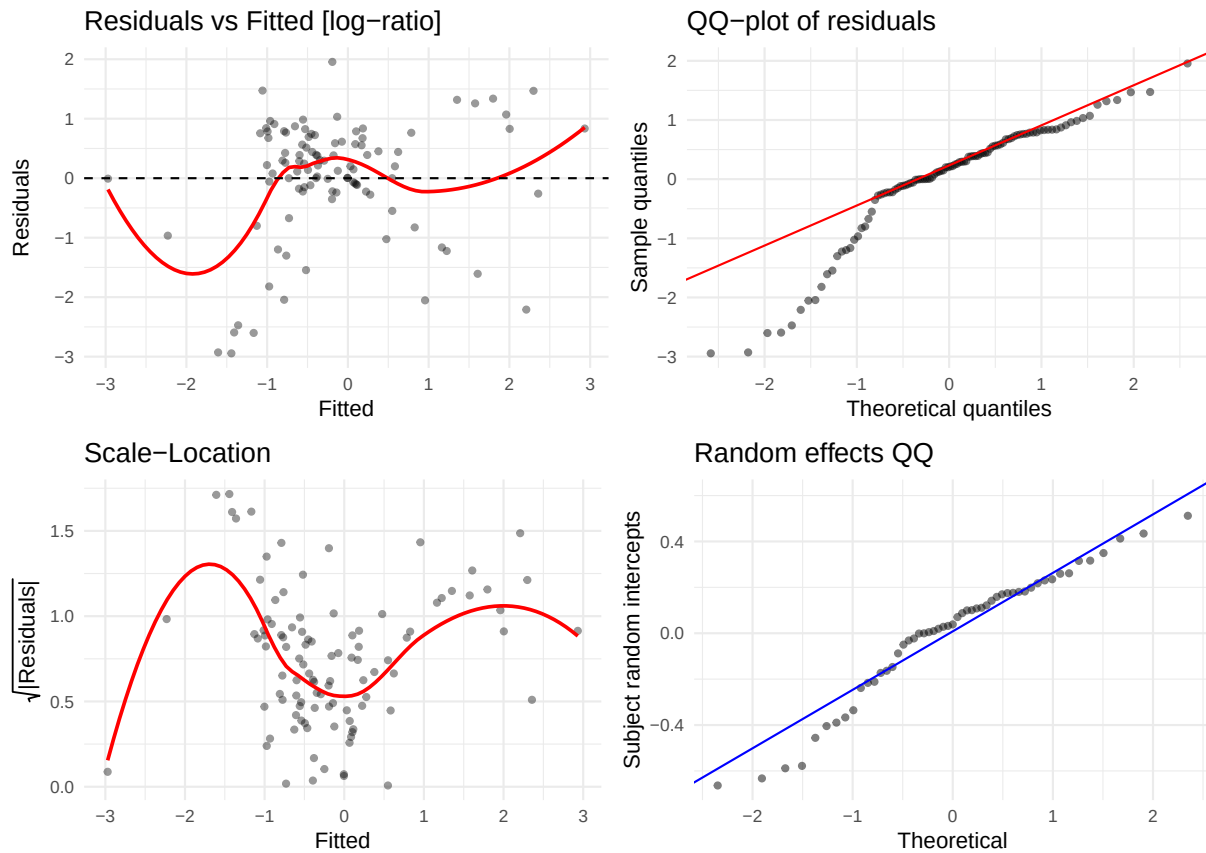
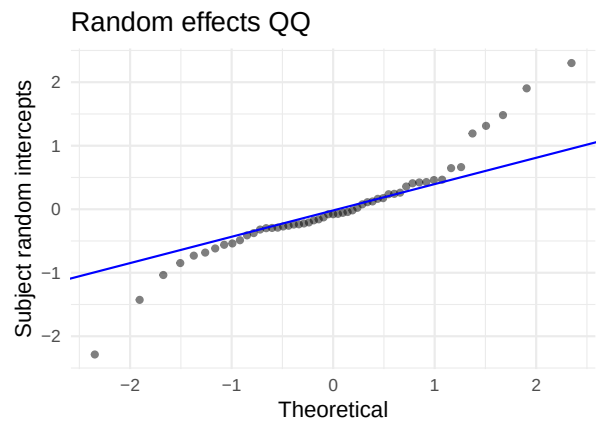
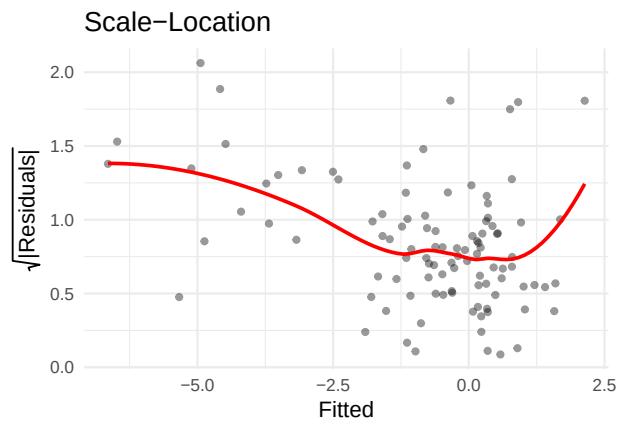
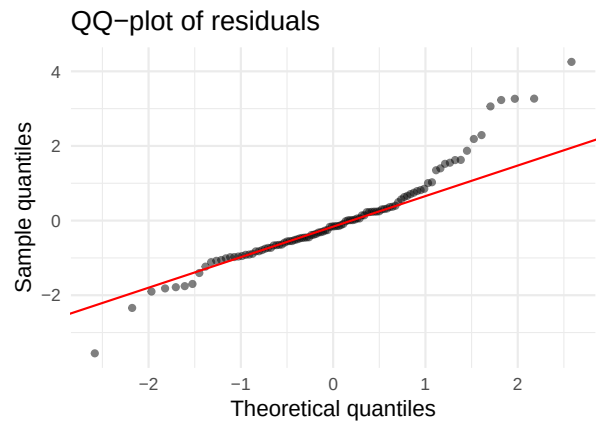
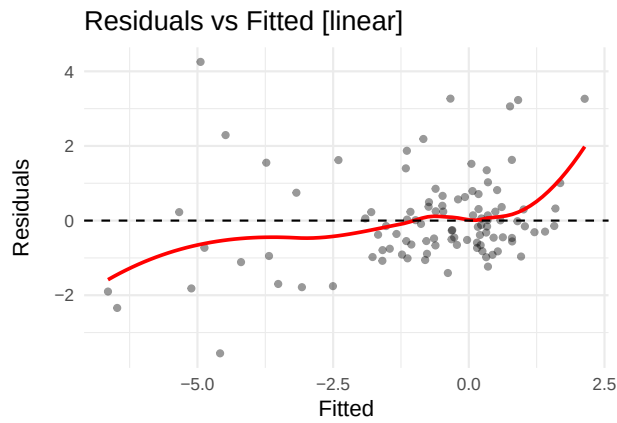


Table 18: Shapiro-Wilk — IL-6

Scale	W	p-value	Interpretation
linear	0.8755	0	Strong departure from normality
log-ratio	0.8969	0	Strong departure from normality

5.17 IL-8 (IL8)

AIC linear: 425.6 | AIC log-ratio: 64.9 | Preferred: log



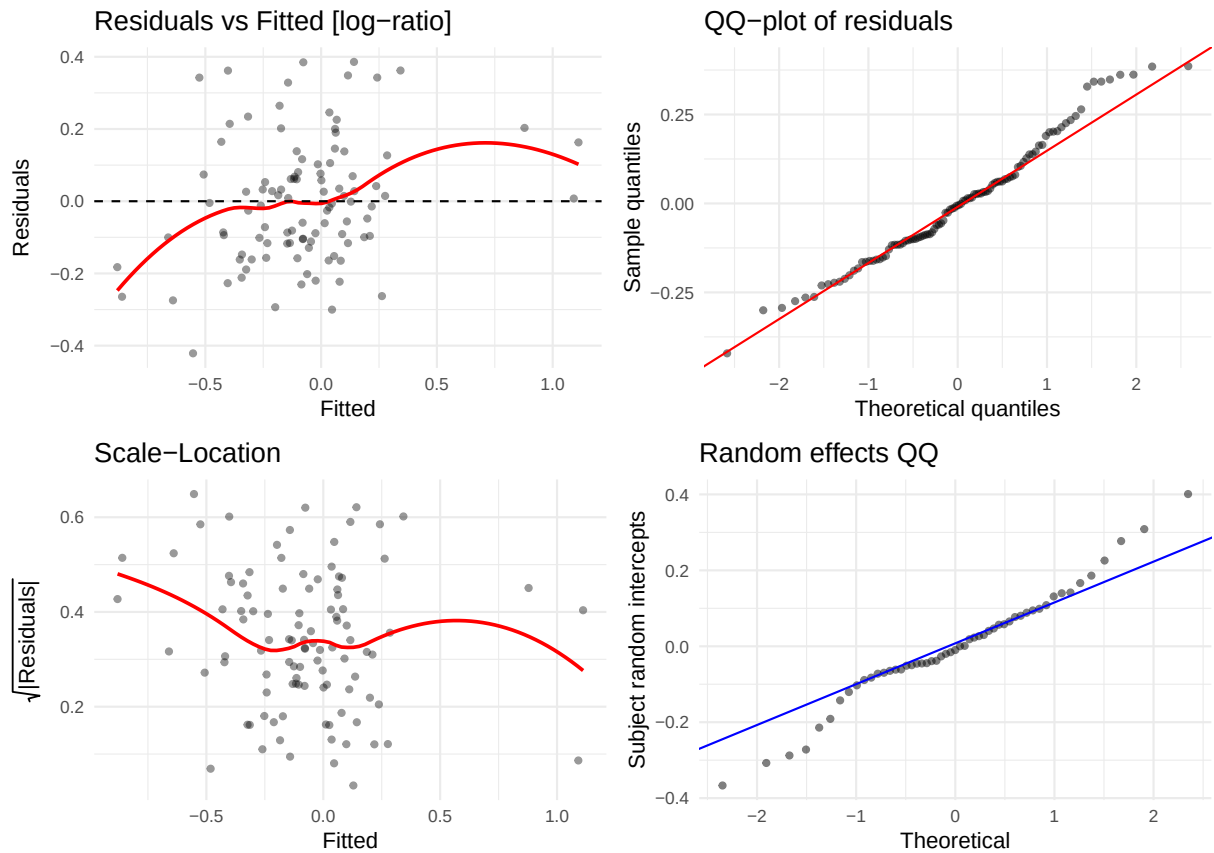
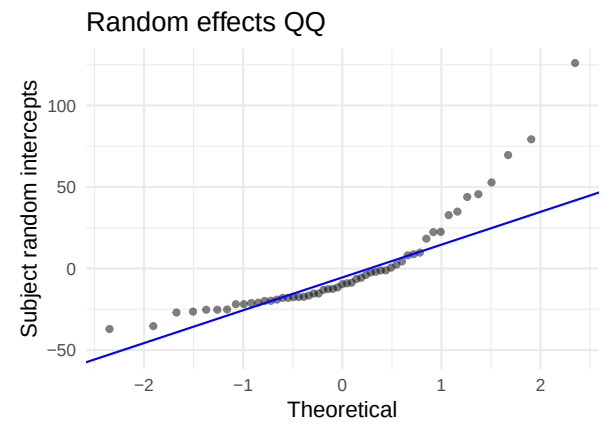
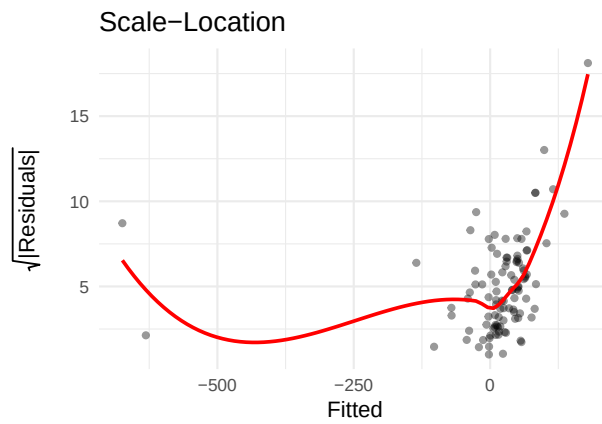
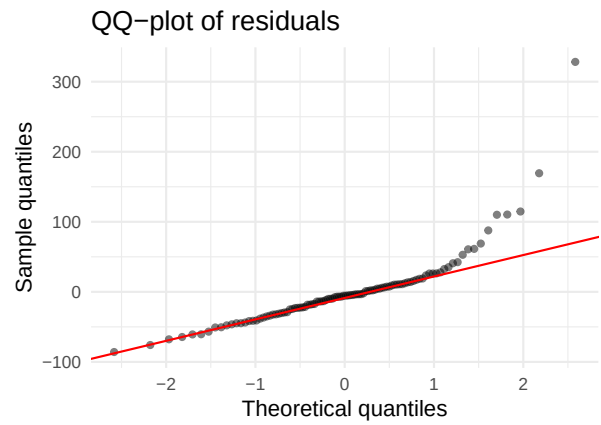
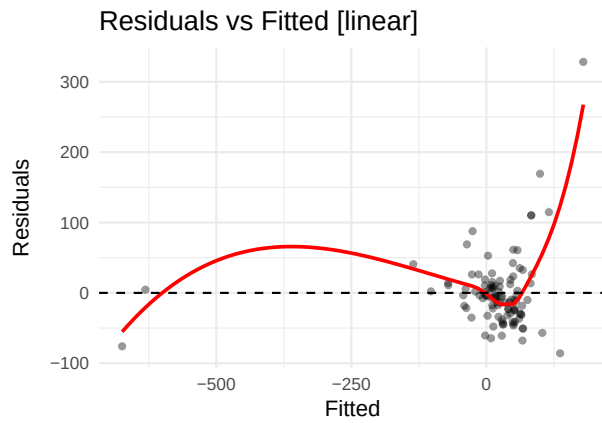


Table 19: Shapiro-Wilk — IL-8

Scale	W	p-value	Interpretation
linear	0.9335	0.0001	Strong departure from normality
log-ratio	0.9781	0.0876	Borderline

5.18 IP-10 (CXCL10) (IP10)

AIC linear: 1152.5 | AIC log-ratio: 174.9 | Preferred: log



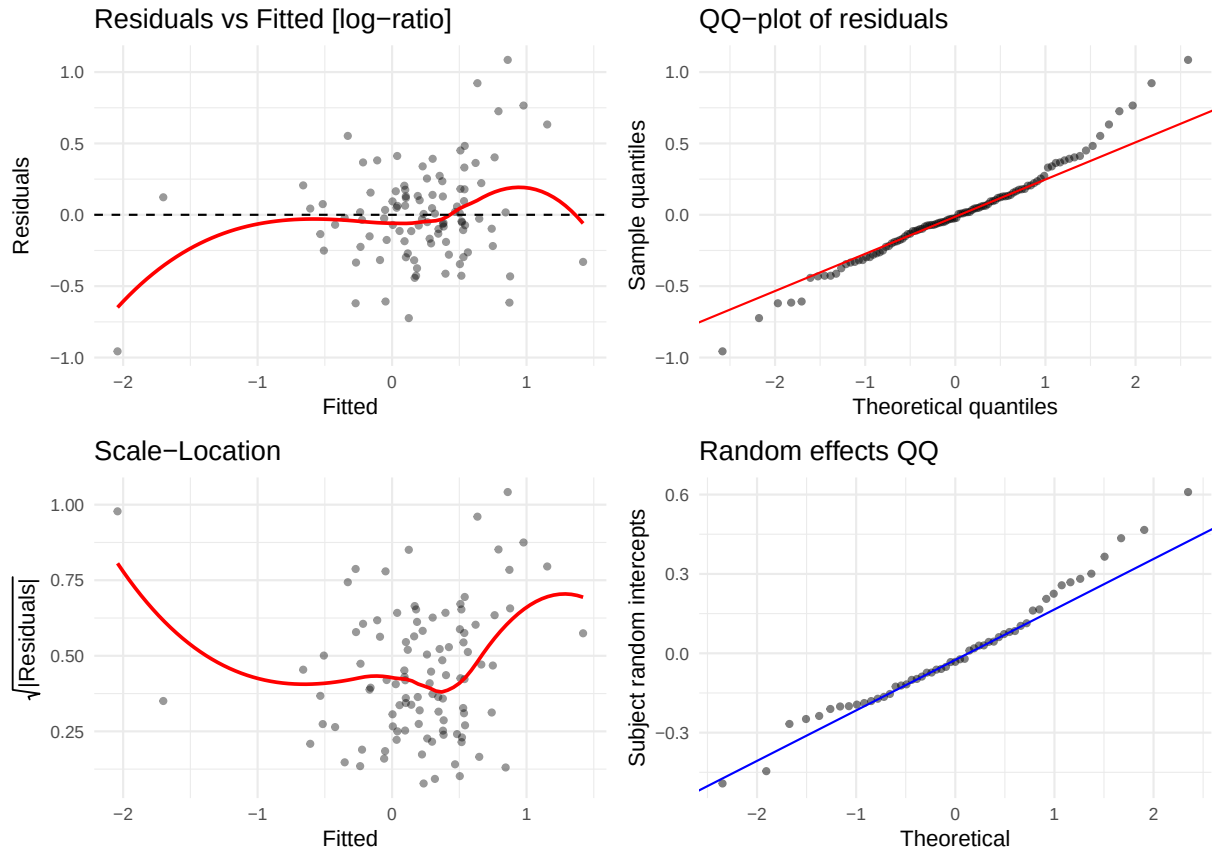
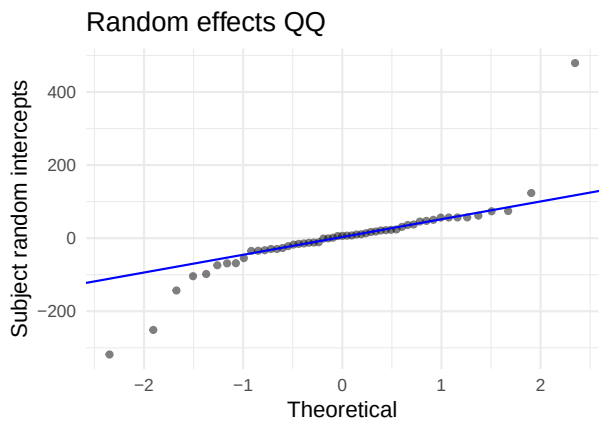
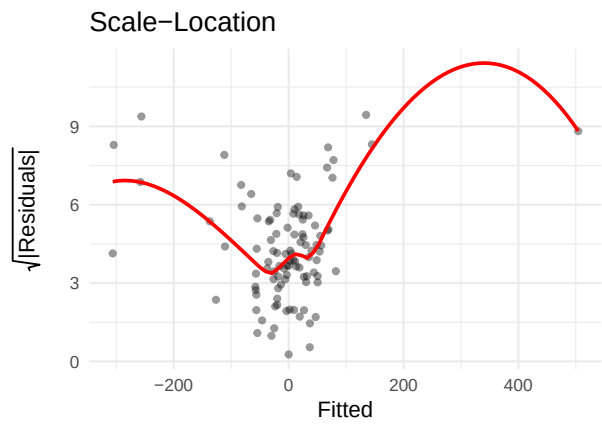
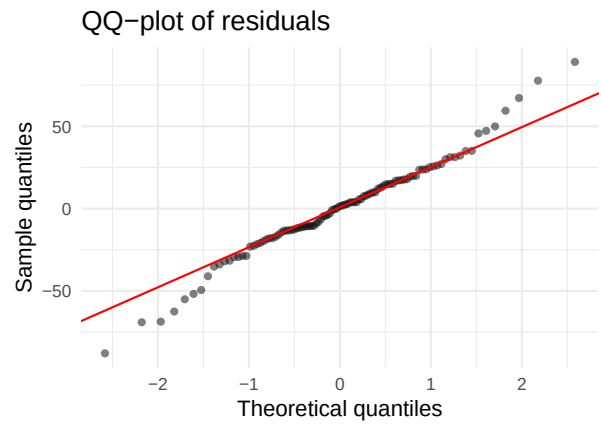
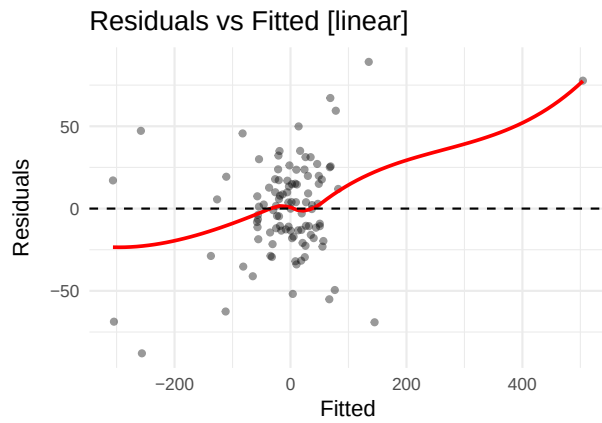


Table 20: Shapiro-Wilk — IP-10 (CXCL10)

Scale	W	p-value	Interpretation
linear	0.7734	0.0000	Strong departure from normality
log-ratio	0.9779	0.0844	Borderline

5.19 I-TAC (CXCL11) (ITAC)

AIC linear: 1171.6 | AIC log-ratio: 33 | Preferred: log



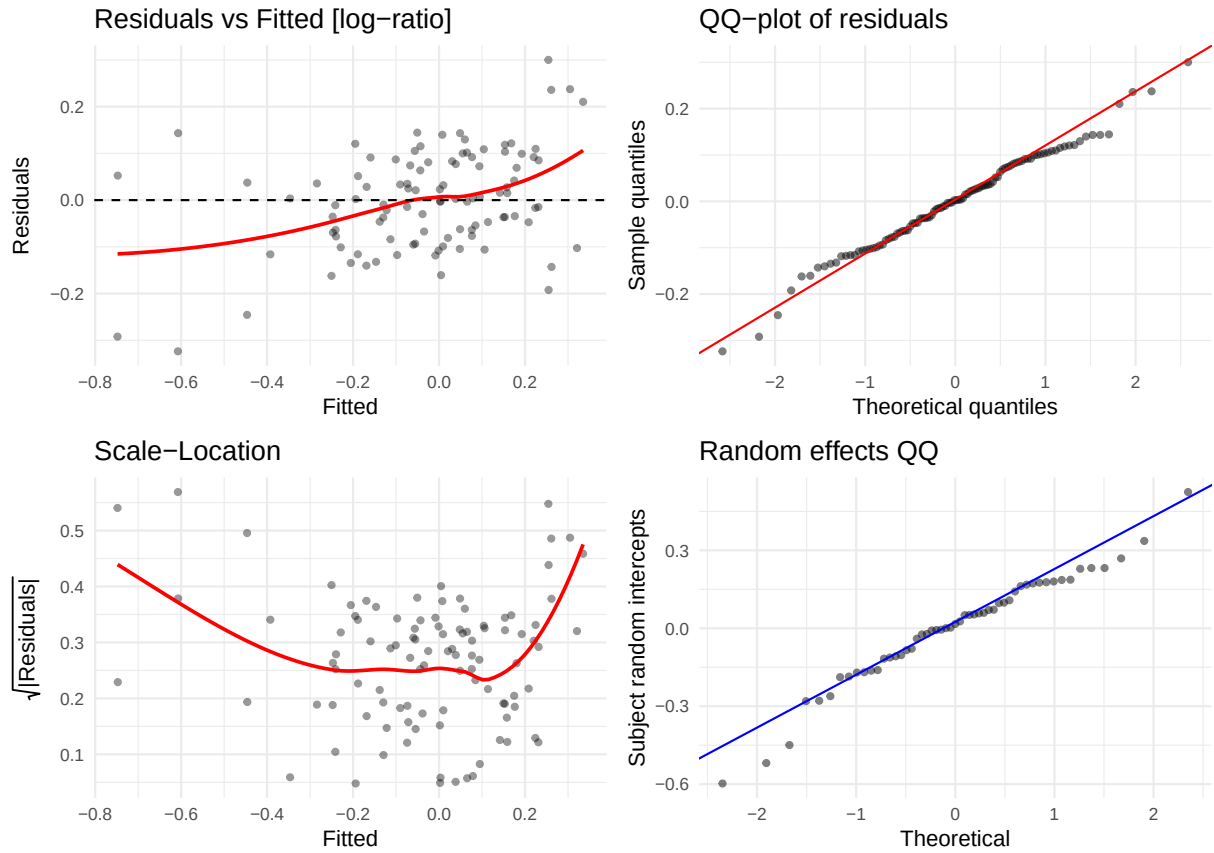
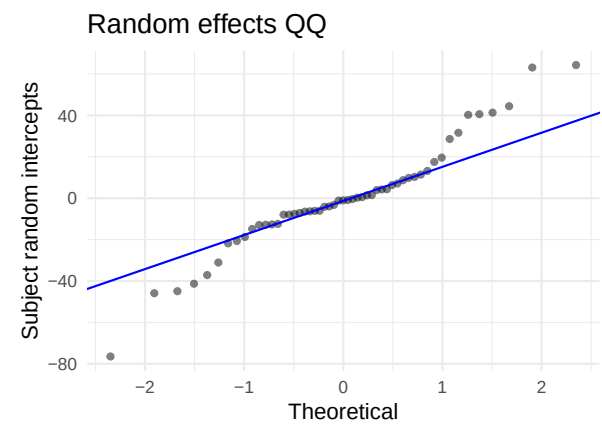
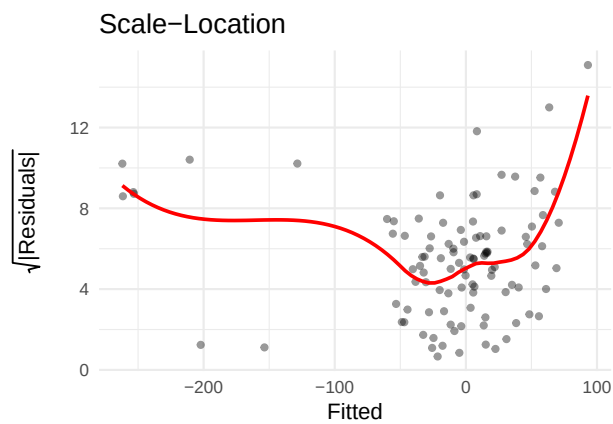
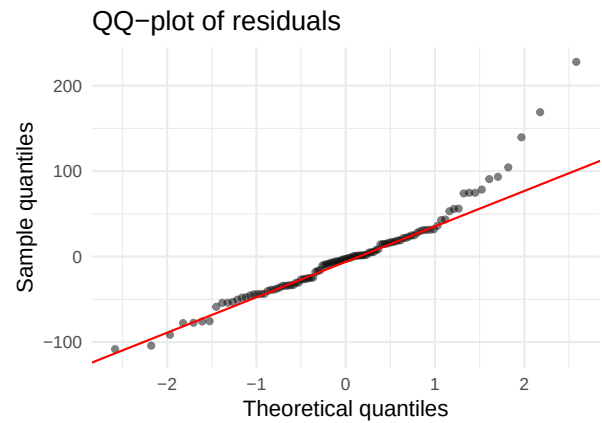
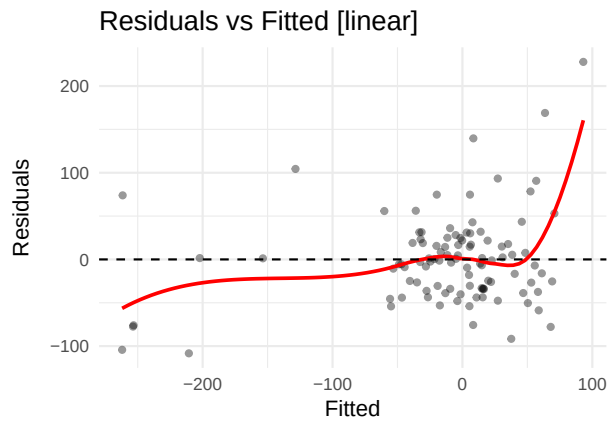


Table 21: Shapiro-Wilk — I-TAC (CXCL11)

Scale	W	p-value	Interpretation
linear	0.9815	0.1639	No evidence of non-normality
log-ratio	0.9878	0.4760	No evidence of non-normality

5.20 MCP-1 (MCP1)

AIC linear: 1144.2 | AIC log-ratio: 192.4 | Preferred: log



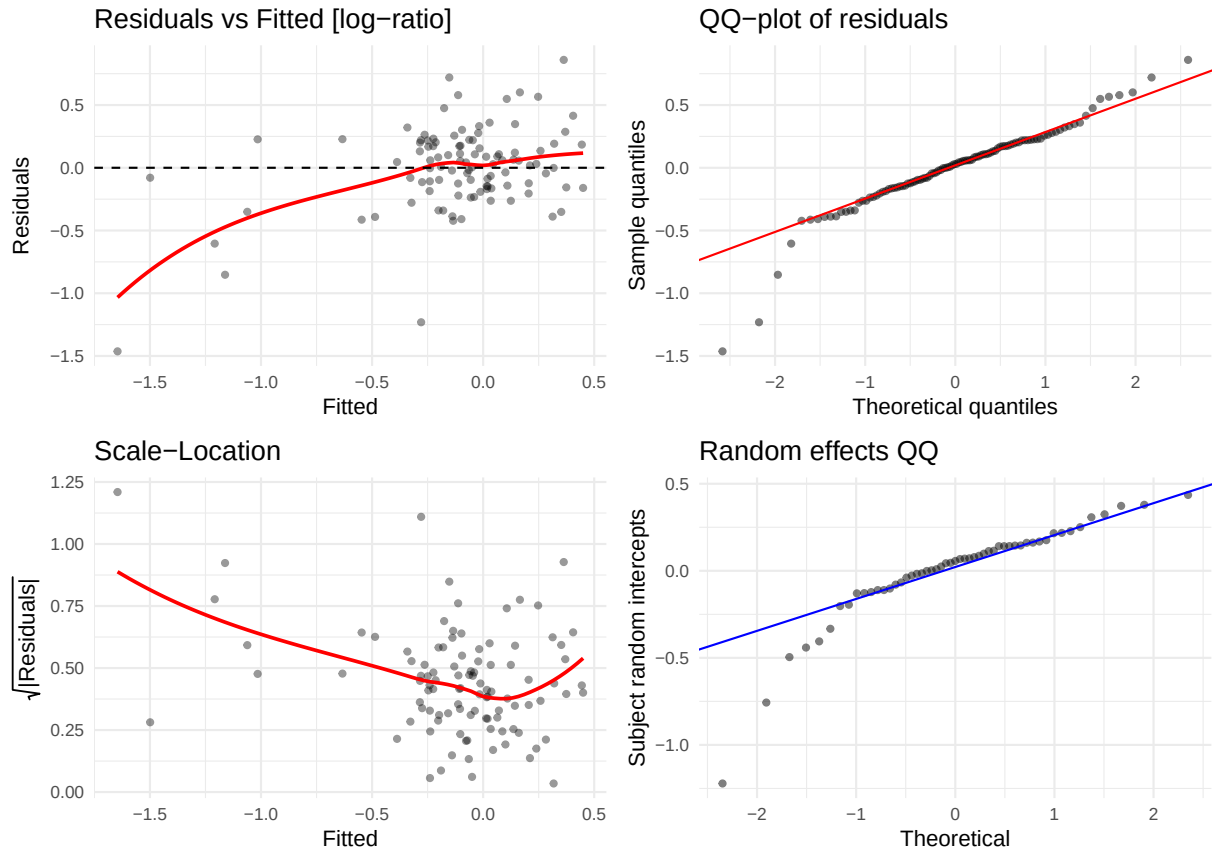
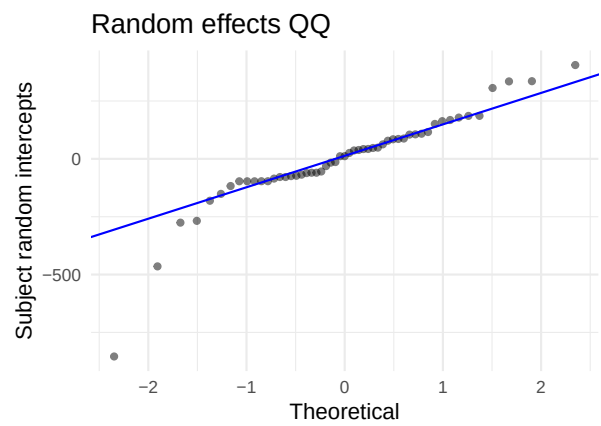
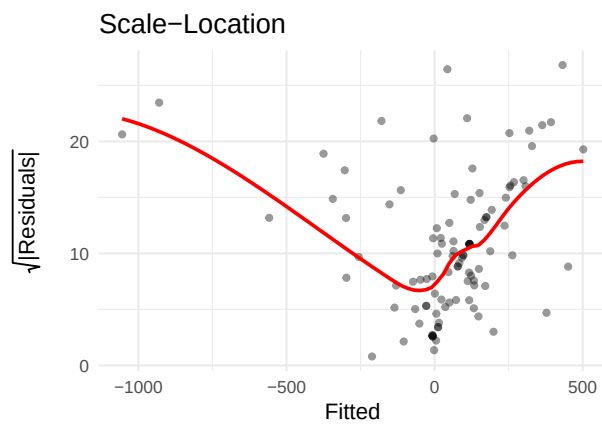
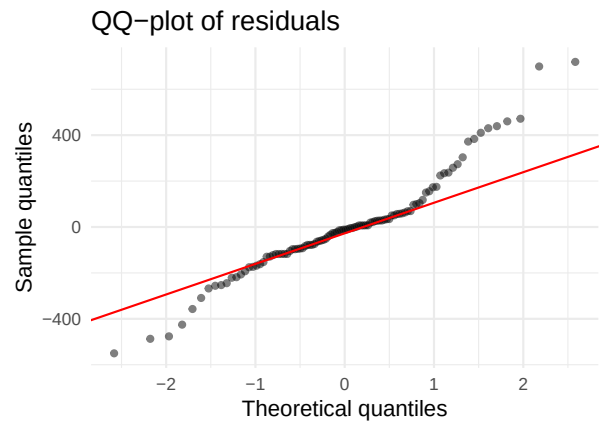
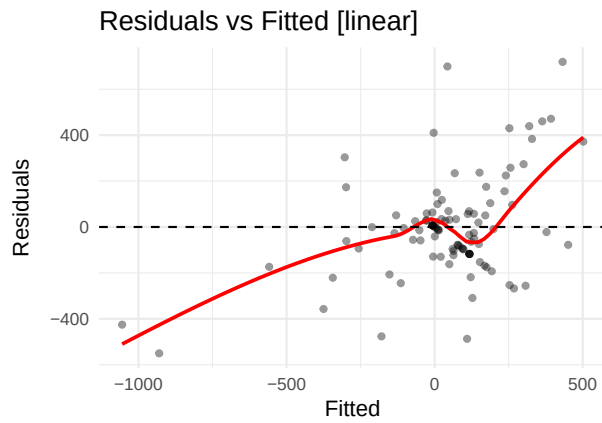


Table 22: Shapiro-Wilk — MCP-1

Scale	W	p-value	Interpretation
linear	0.9265	0	Strong departure from normality
log-ratio	0.9246	0	Strong departure from normality

5.21 MIG (CXCL9) (MIG)

AIC linear: 1453.2 | AIC log-ratio: 338.2 | Preferred: log



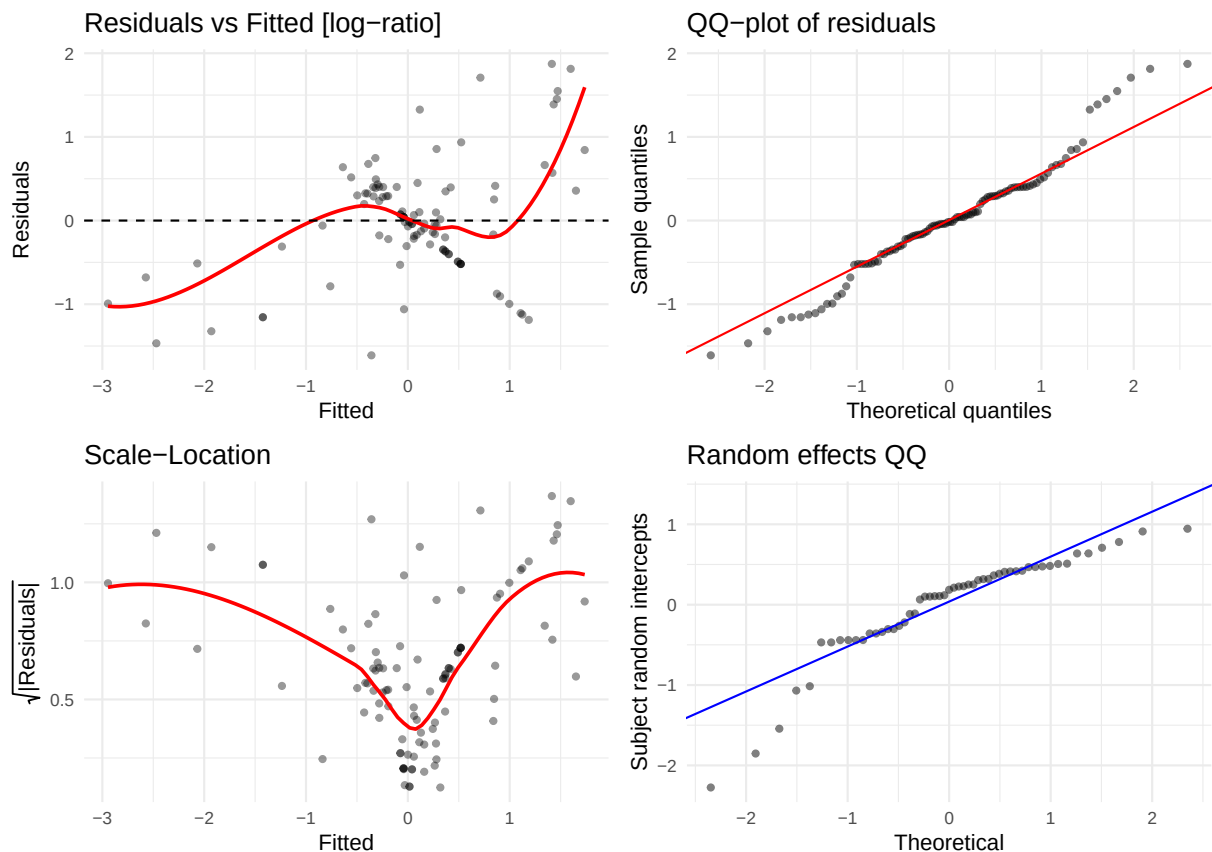
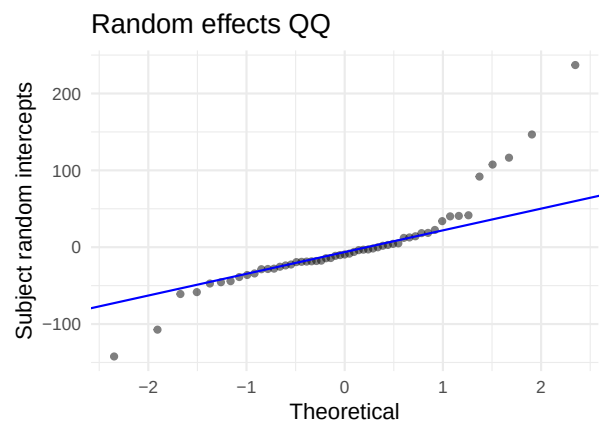
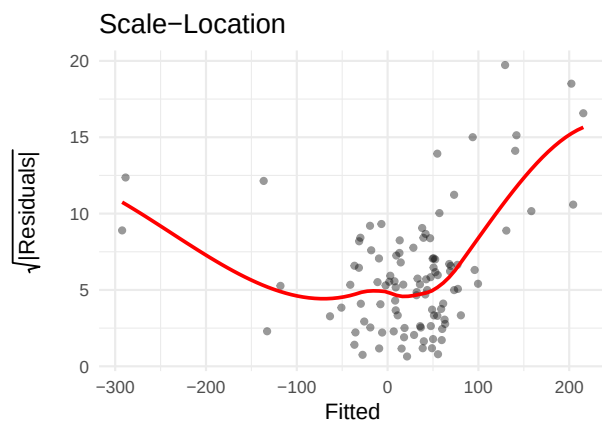
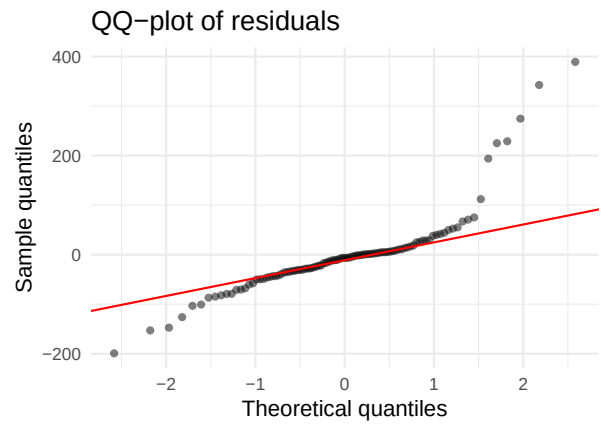
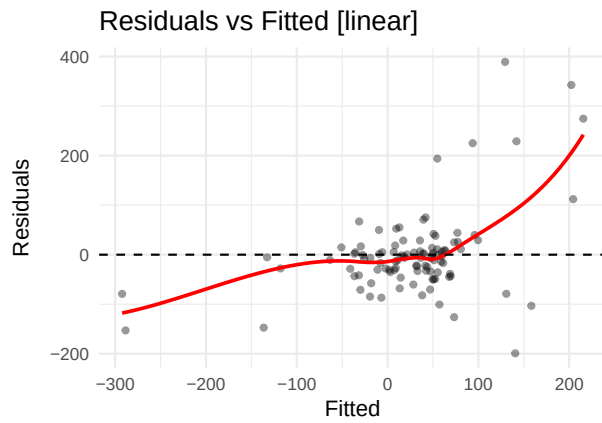


Table 23: Shapiro-Wilk — MIG (CXCL9)

Scale	W	p-value	Interpretation
linear	0.9425	0.0002	Strong departure from normality
log-ratio	0.9709	0.0236	Some departure from normality

5.22 MMP-10 (MMP10)

AIC linear: 1254.5 | AIC log-ratio: 25.5 | Preferred: log



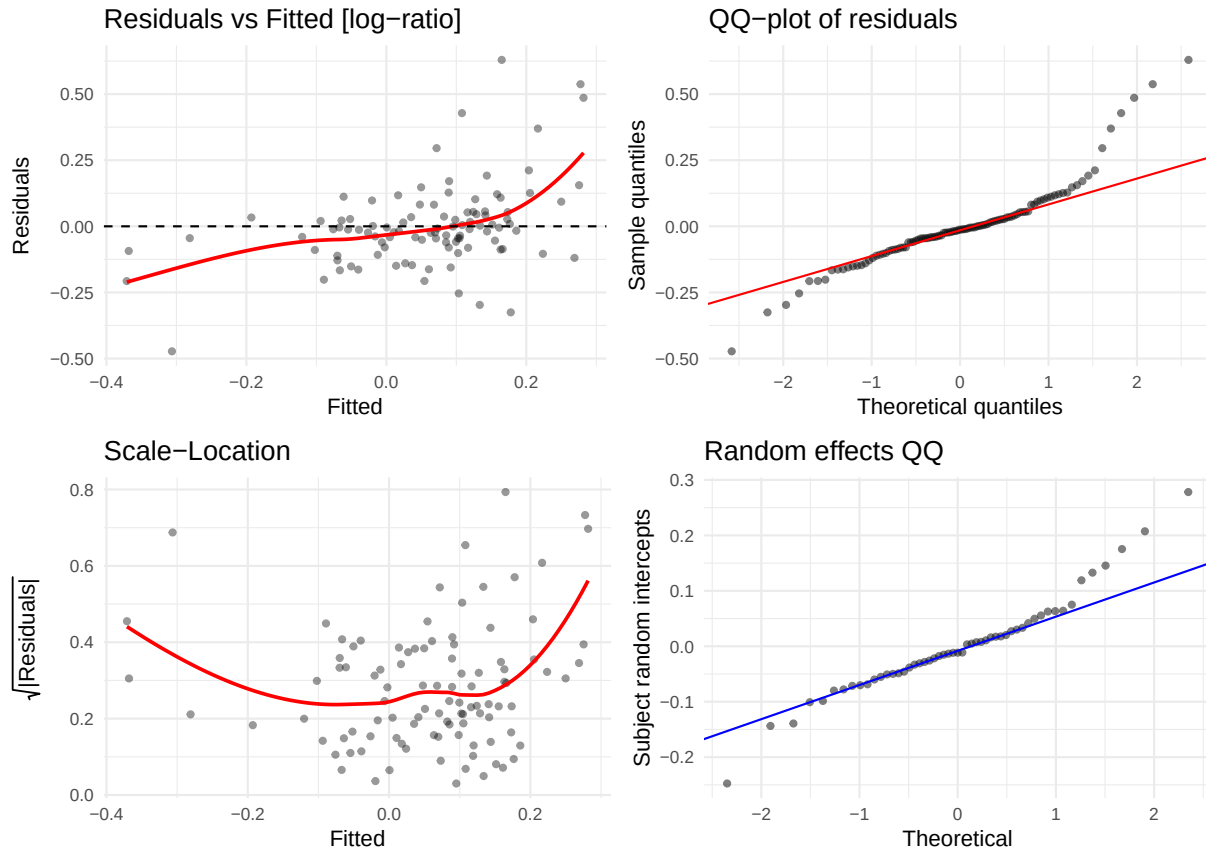
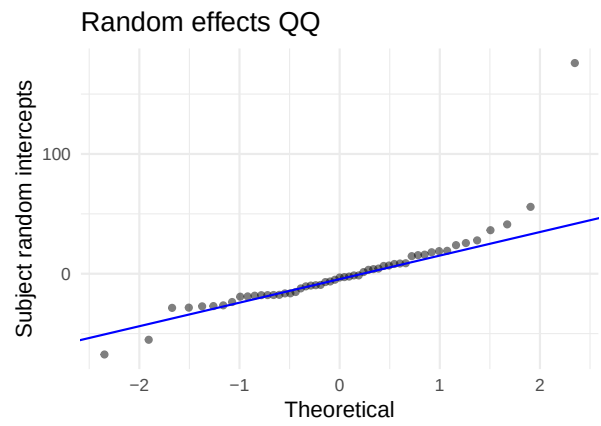
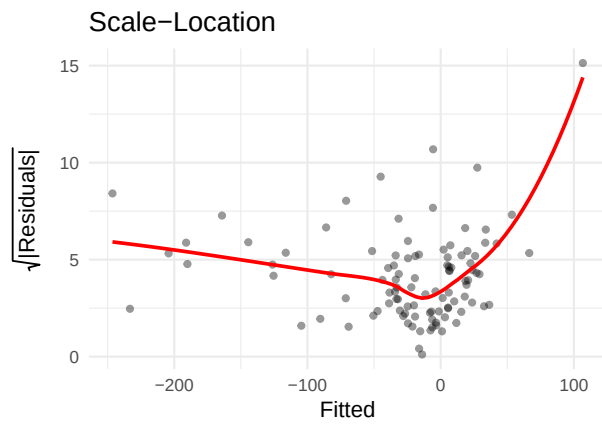
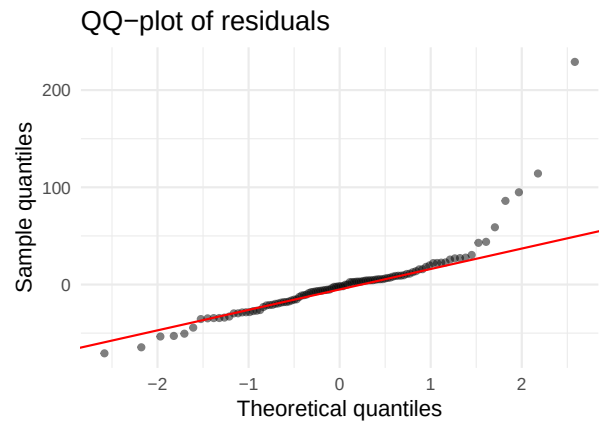
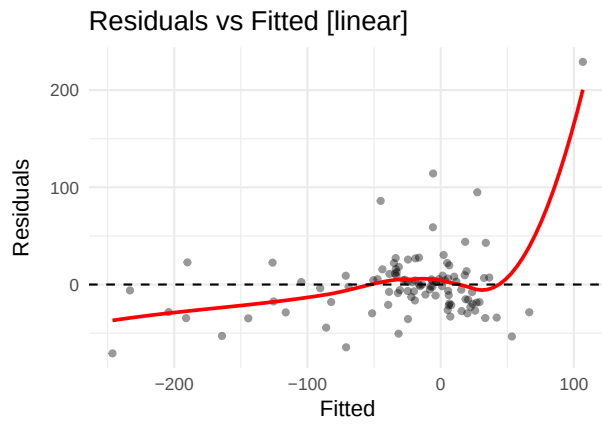


Table 24: Shapiro-Wilk — MMP-10

Scale	W	p-value	Interpretation
linear	0.7911	0	Strong departure from normality
log-ratio	0.8957	0	Strong departure from normality

5.23 PDGF-DD (PDGFDD)

AIC linear: 1106.7 | AIC log-ratio: 195.4 | Preferred: log



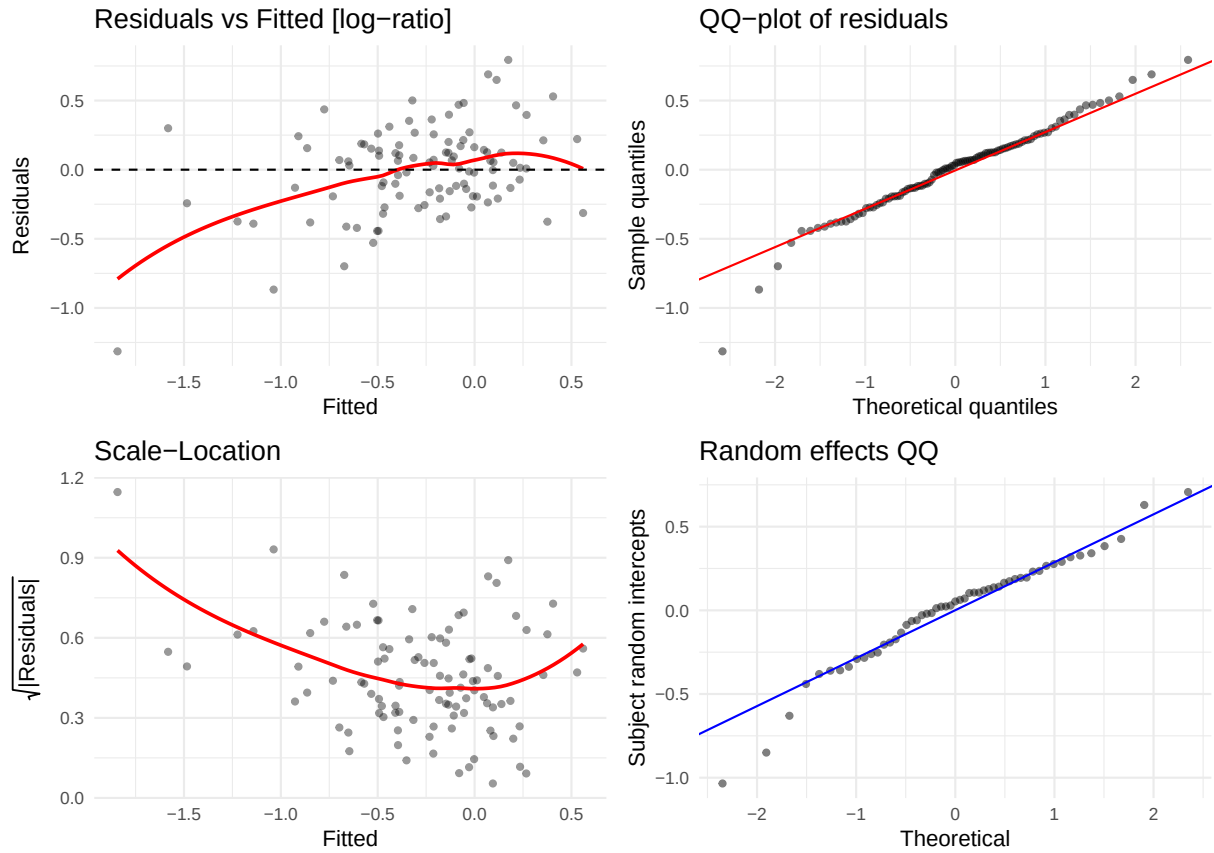
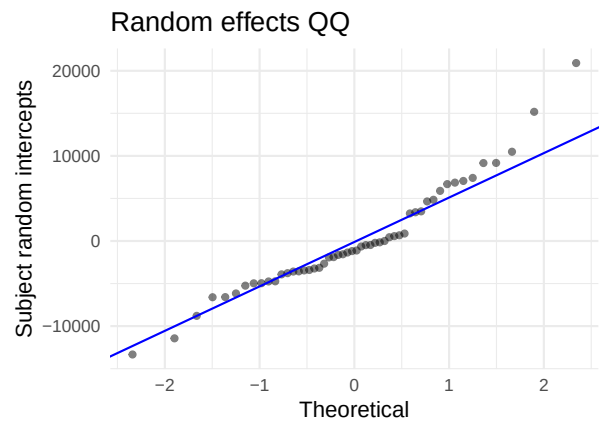
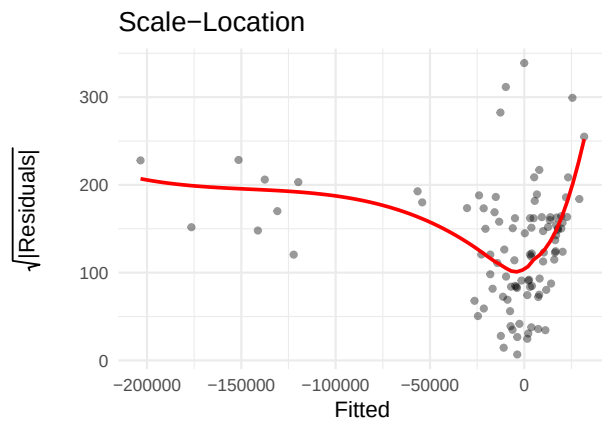
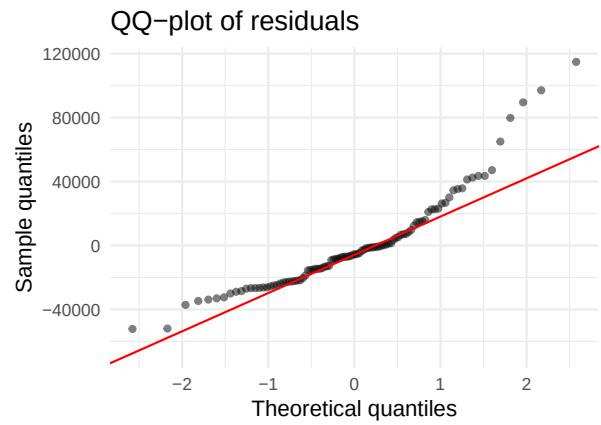
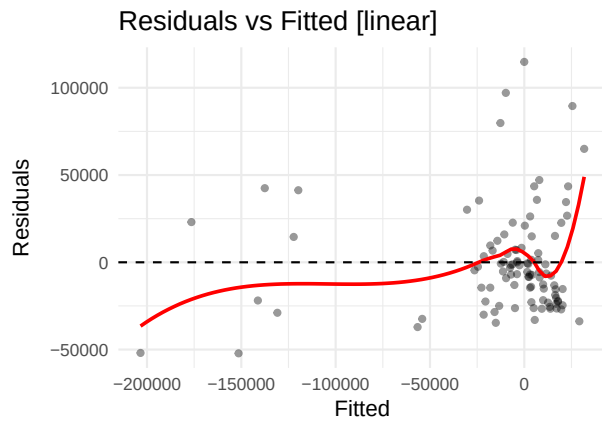


Table 25: Shapiro-Wilk — PDGF-DD

Scale	W	p-value	Interpretation
linear	0.7810	0.0000	Strong departure from normality
log-ratio	0.9684	0.0151	Some departure from normality

5.24 PECAM-1 (PECAM1)

AIC linear: 2298.7 | AIC log-ratio: 248.3 | Preferred: log



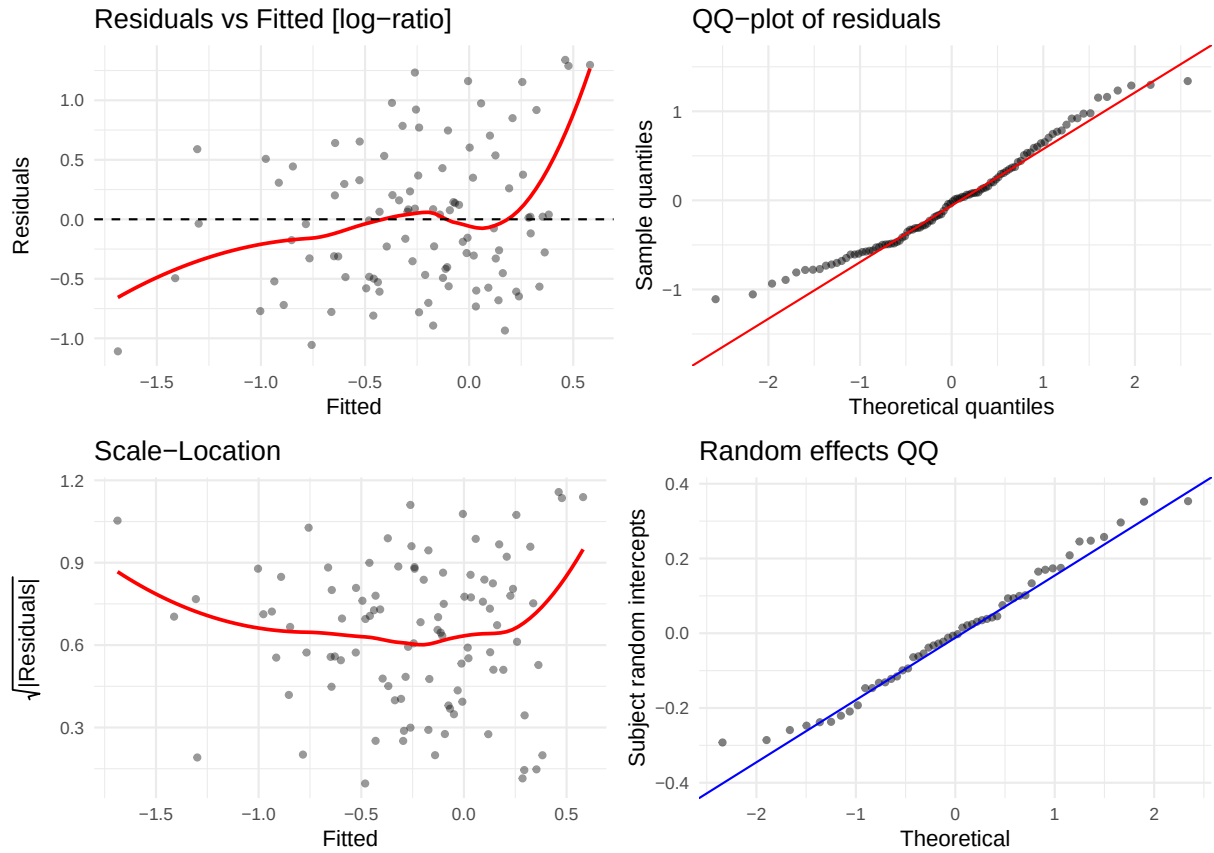
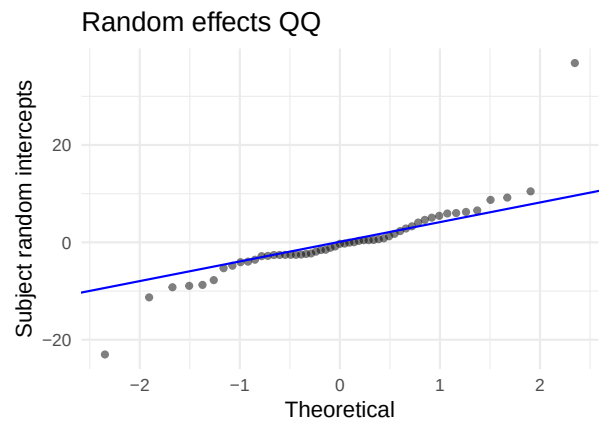
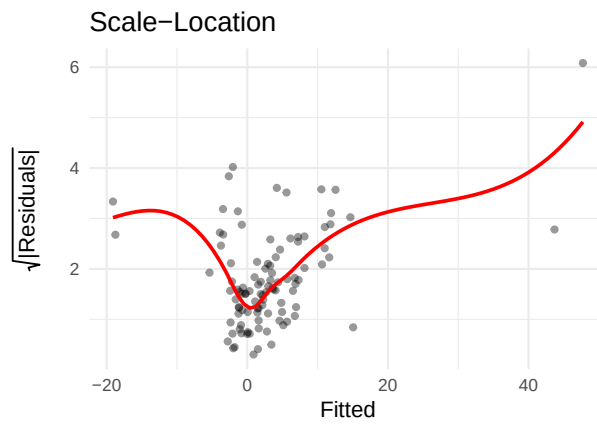
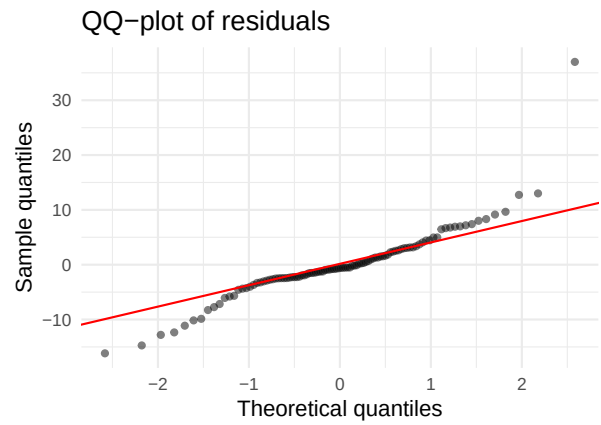
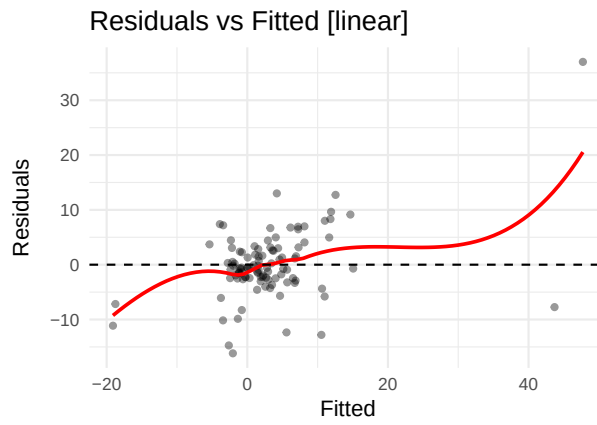


Table 26: Shapiro-Wilk — PECAM-1

Scale	W	p-value	Interpretation
linear	0.8908	0.0000	Strong departure from normality
log-ratio	0.9698	0.0212	Some departure from normality

5.25 PIGF (PIGF)

AIC linear: 789.3 | AIC log-ratio: -25.9 | Preferred: log



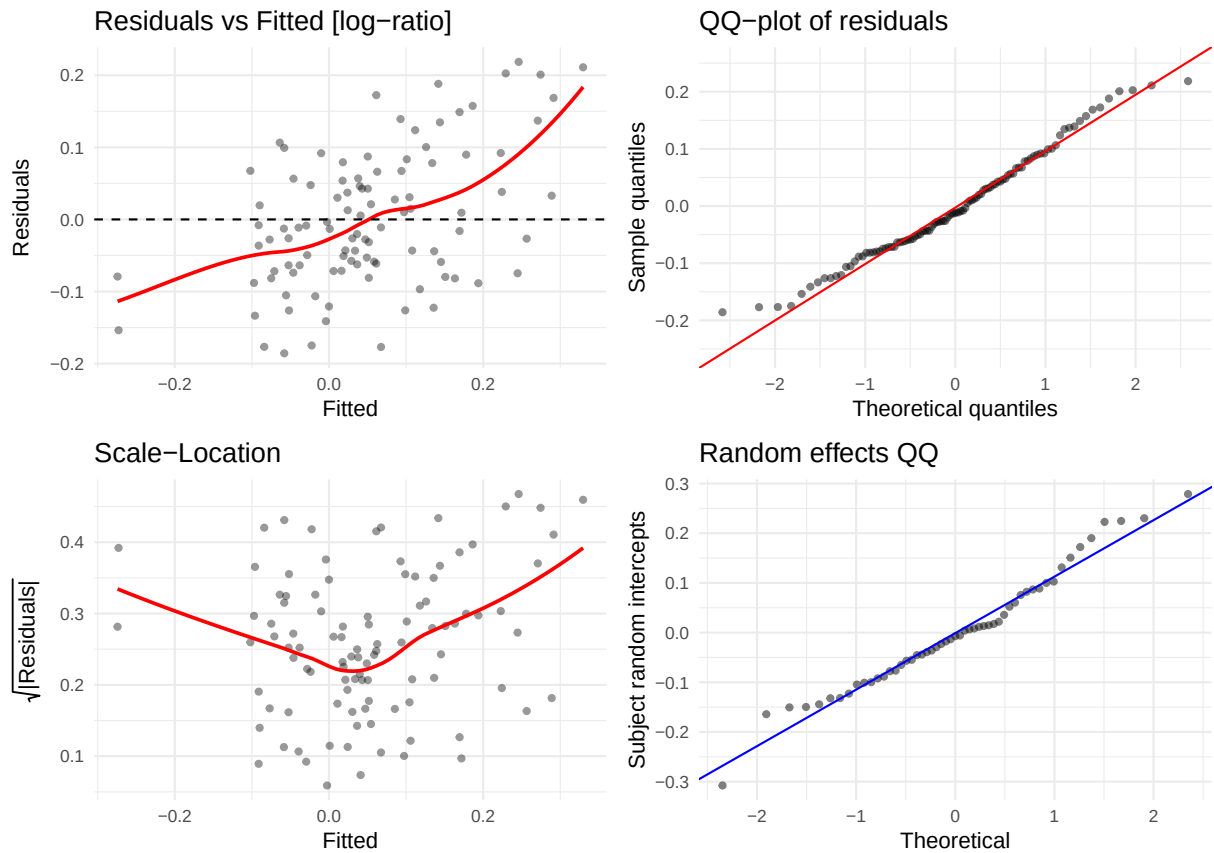
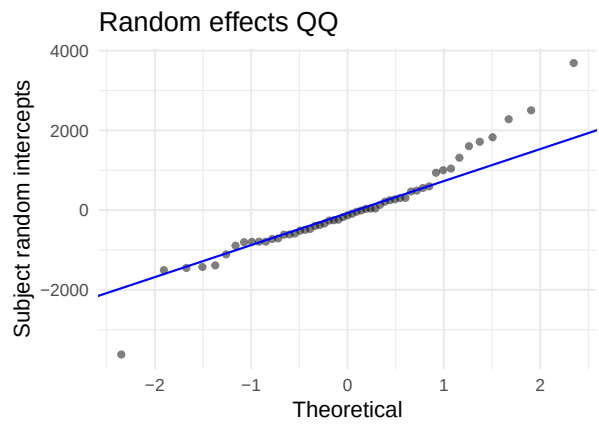
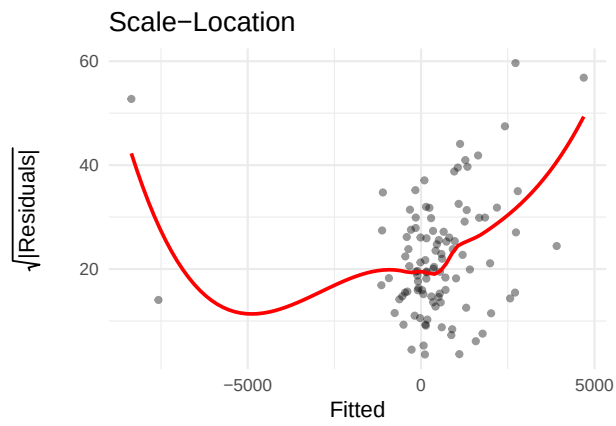
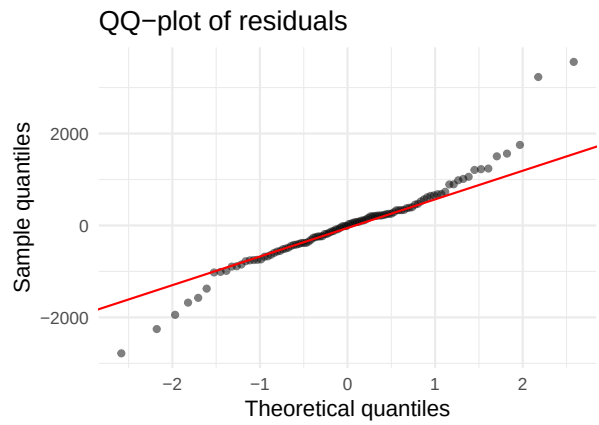
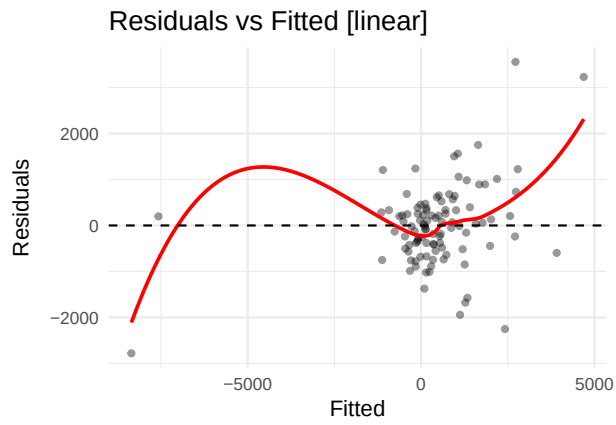


Table 27: Shapiro-Wilk — PIGF

Scale	W	p-value	Interpretation
linear	0.8658	0.0000	Strong departure from normality
log-ratio	0.9798	0.1202	No evidence of non-normality

5.26 Tenascin C (TENASC)

AIC linear: 1752.1 | AIC log-ratio: -115.2 | Preferred: log



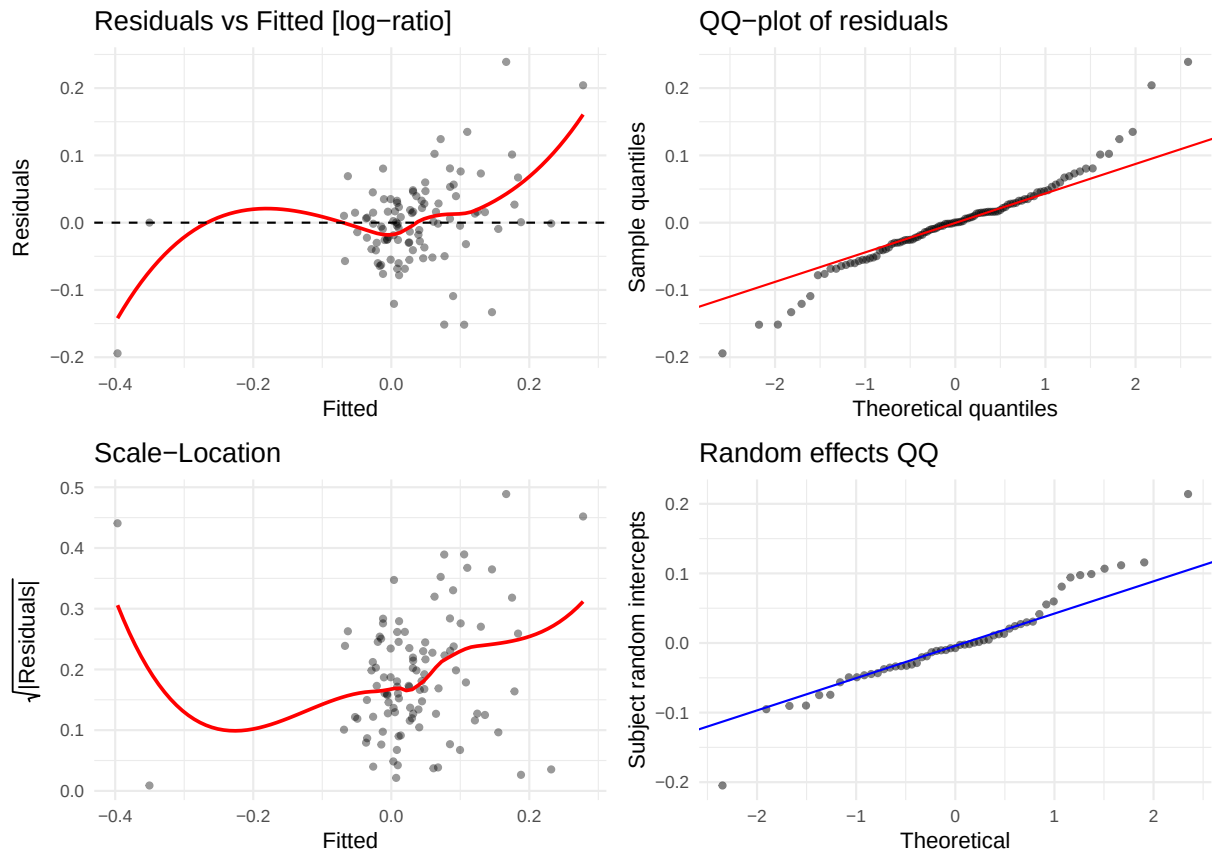
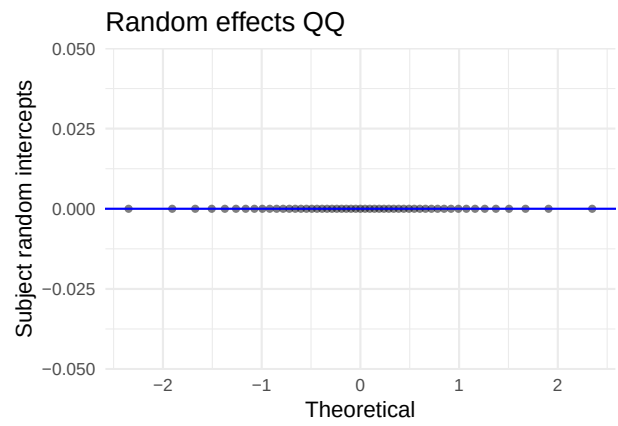
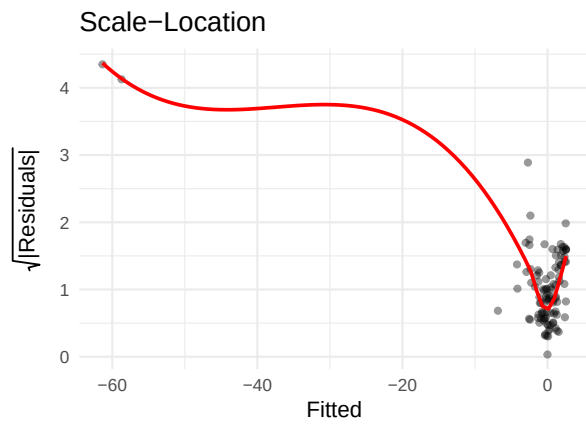
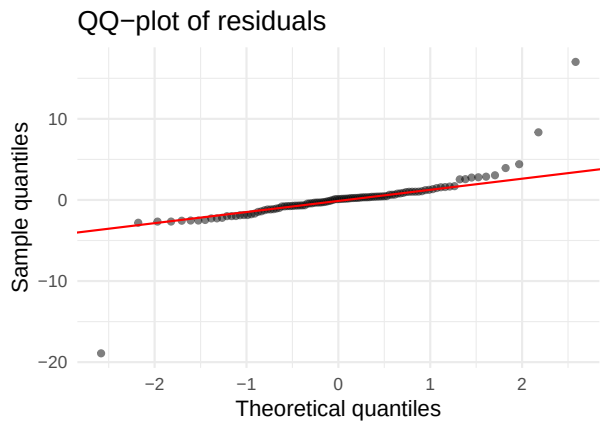
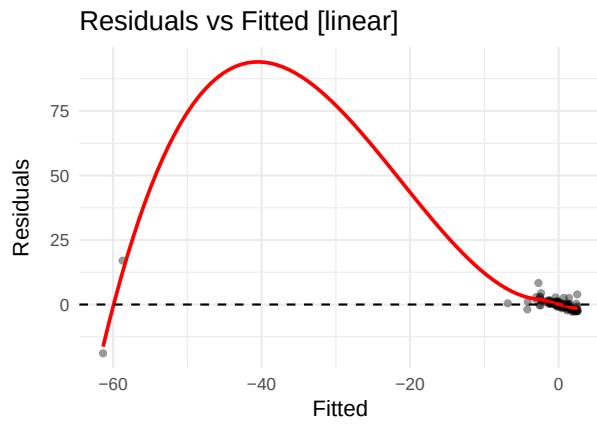


Table 28: Shapiro-Wilk — Tenascin C

Scale	W	p-value	Interpretation
linear	0.9355	1e-04	Strong departure from normality
log-ratio	0.9521	1e-03	Strong departure from normality

5.27 TGF-alpha (TGFA)

AIC linear: 539.2 | AIC log-ratio: 261.7 | Preferred: log



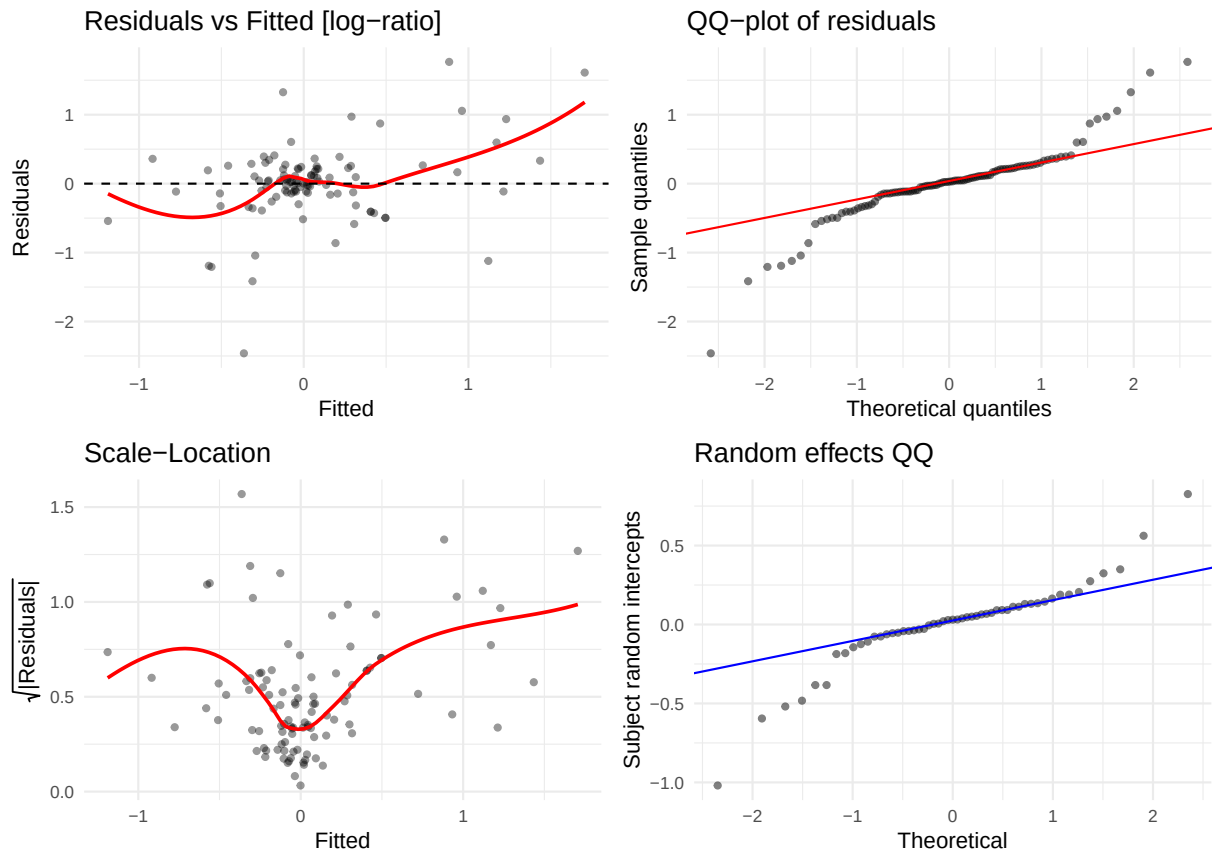
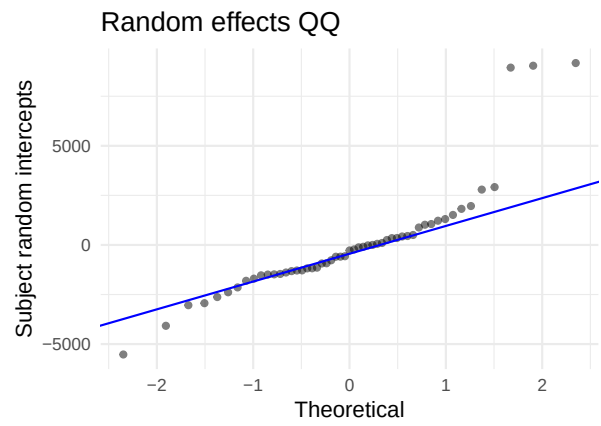
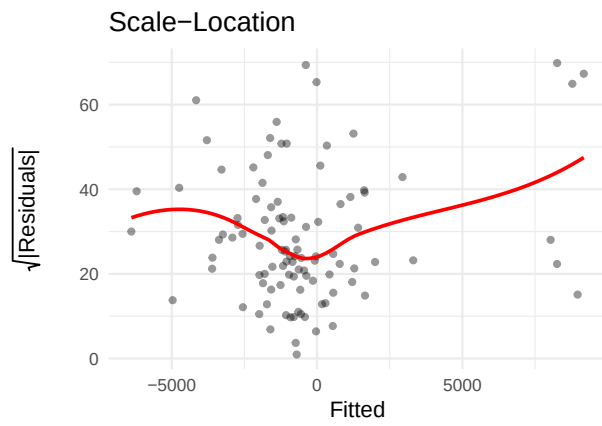
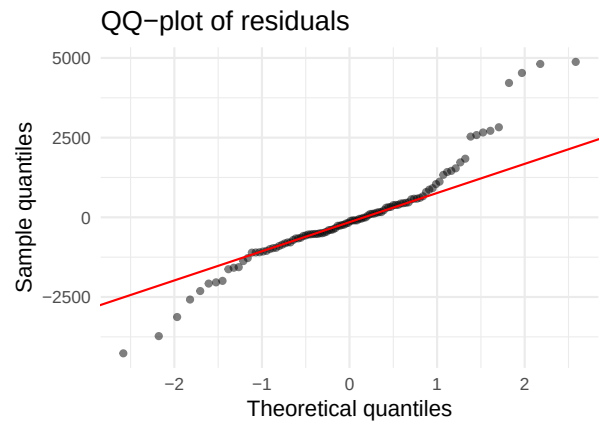
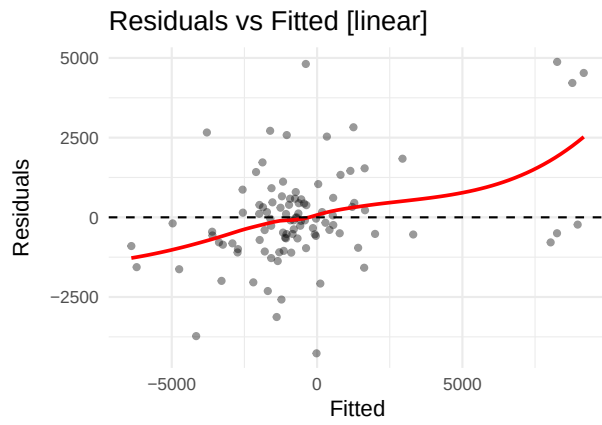


Table 29: Shapiro-Wilk — TGF-alpha

Scale	W	p-value	Interpretation
linear	0.6470	0	Strong departure from normality
log-ratio	0.8851	0	Strong departure from normality

5.28 Thrombospondin-2 (THBS2)

AIC linear: 1880.6 | AIC log-ratio: 83.1 | Preferred: log



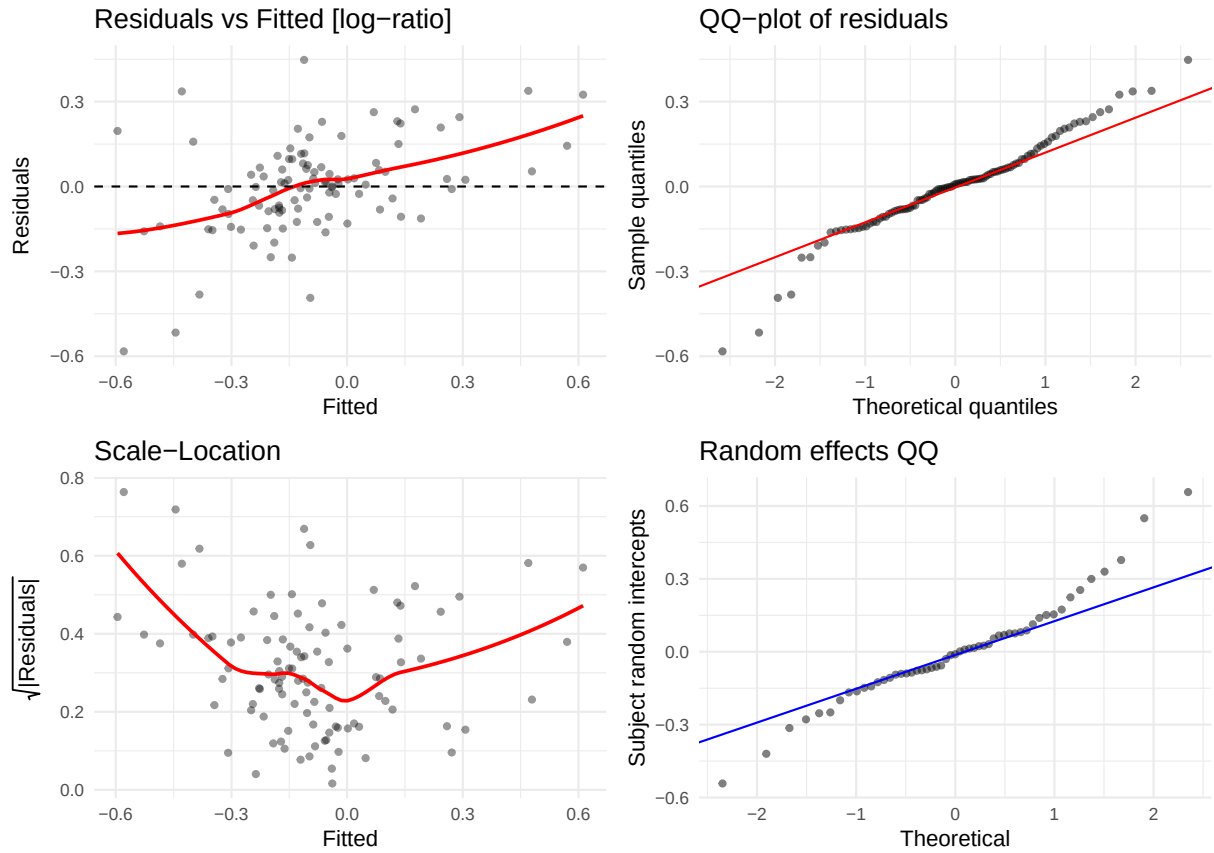
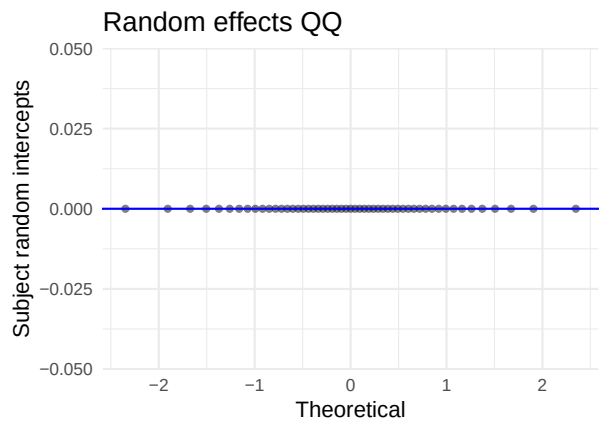
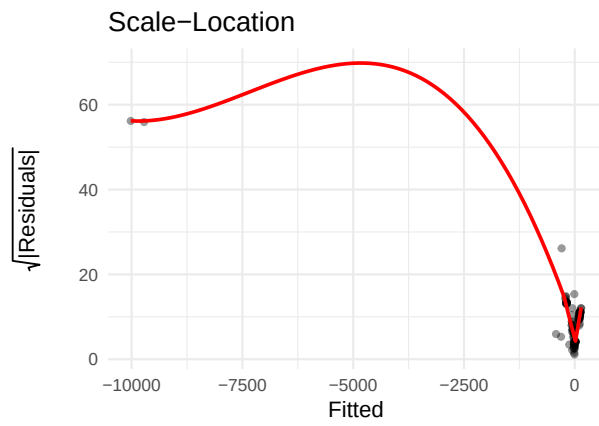
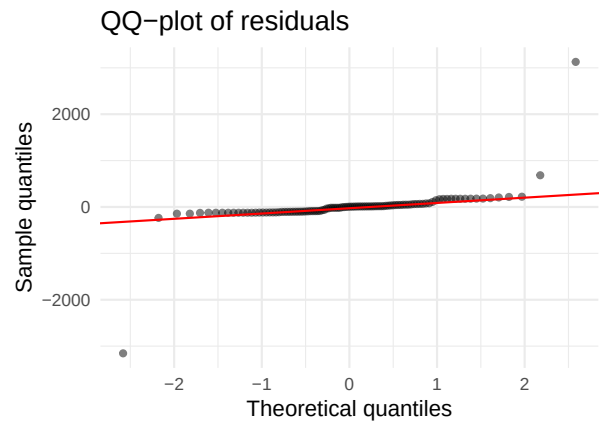
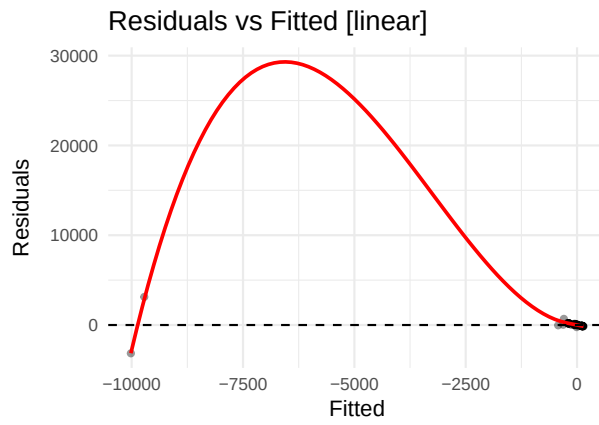


Table 30: Shapiro-Wilk — Thrombospondin-2

Scale	W	p-value	Interpretation
linear	0.9241	0.000	Strong departure from normality
log-ratio	0.9630	0.006	Strong departure from normality

5.29 TNFRSF18 (GITR) (TNFRSF18)

AIC linear: 1513 | AIC log-ratio: 217.5 | Preferred: log



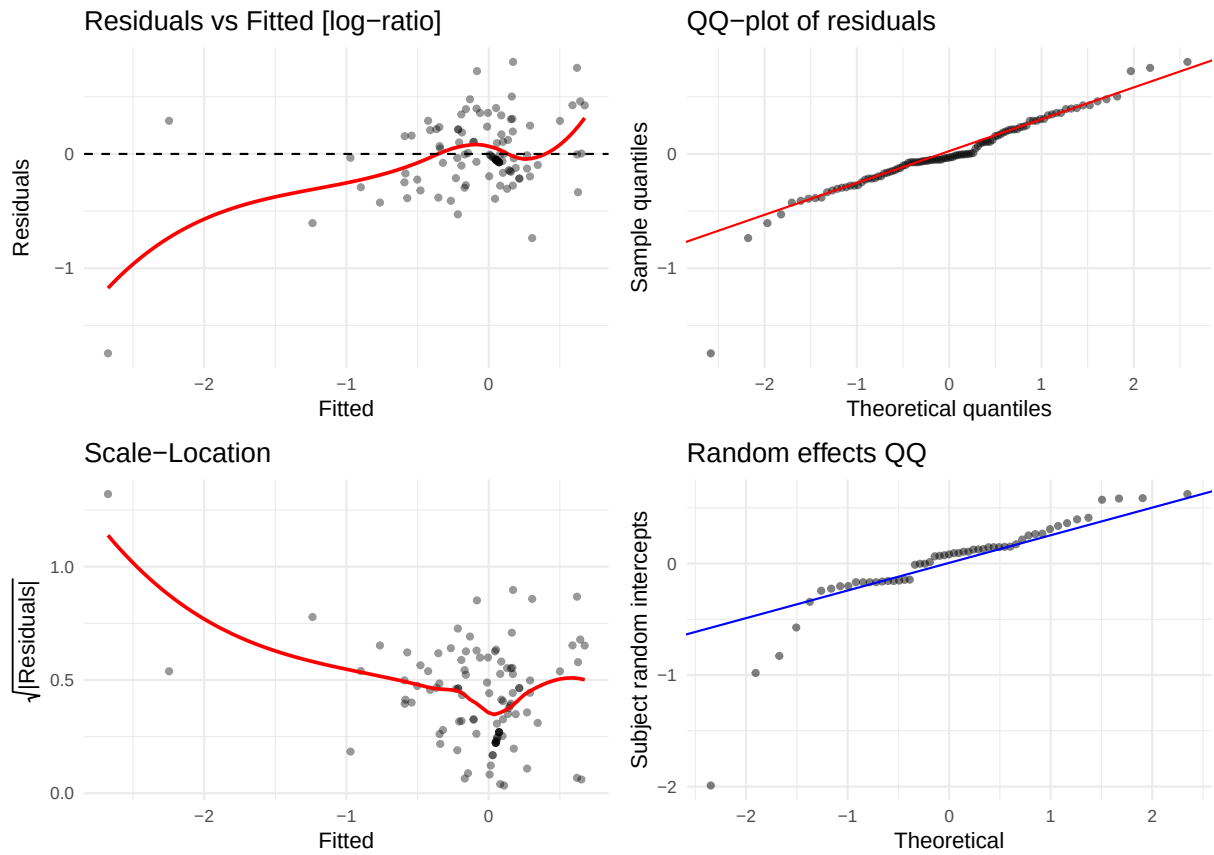
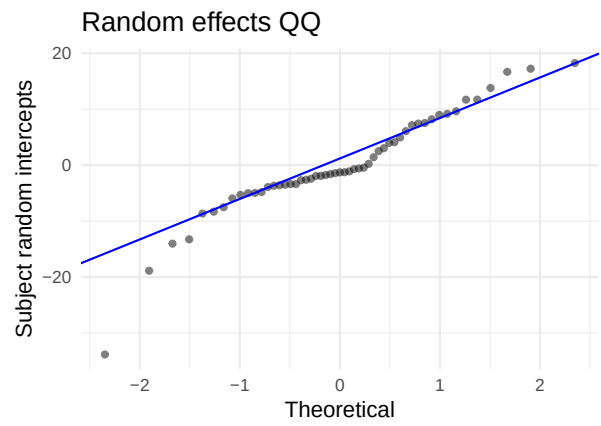
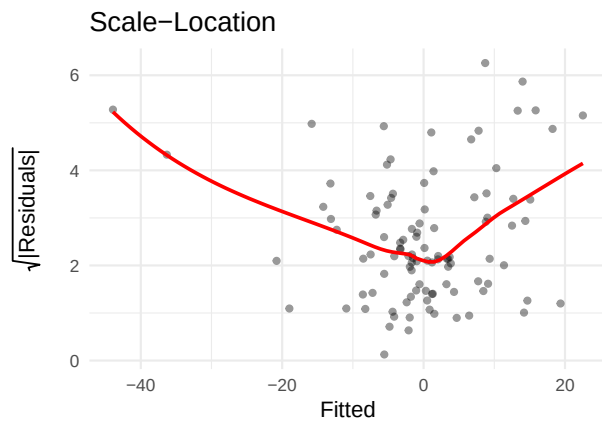
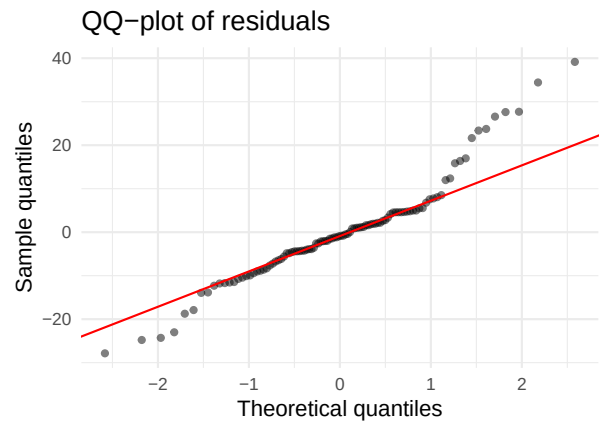
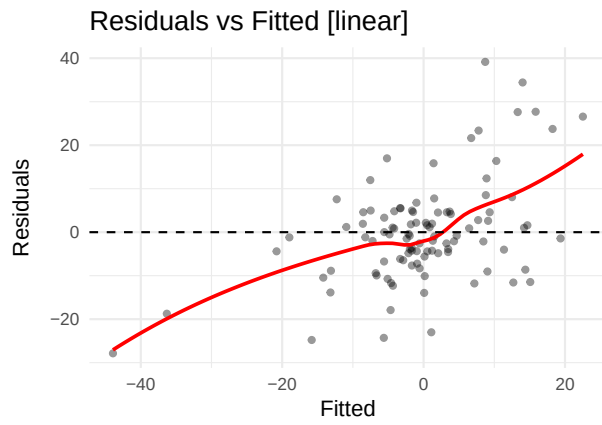


Table 31: Shapiro-Wilk — TNFRSF18 (GITR)

Scale	W	p-value	Interpretation
linear	0.3568	0	Strong departure from normality
log-ratio	0.9114	0	Strong departure from normality

5.30 TNFSF6 (FasL) (TNFSF6)

AIC linear: 876.1 | AIC log-ratio: 62.8 | Preferred: log



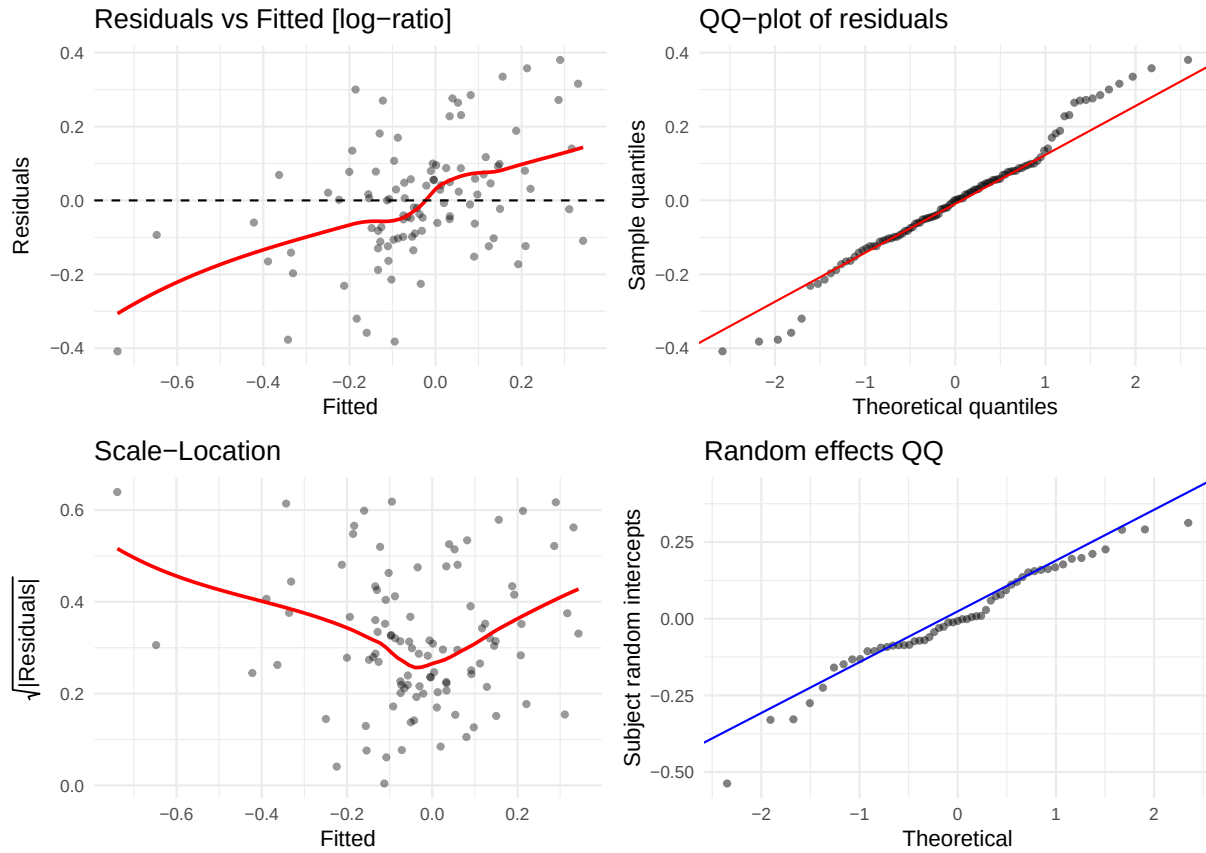
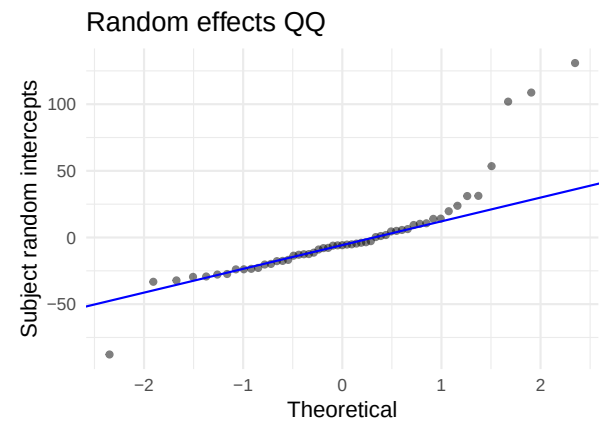
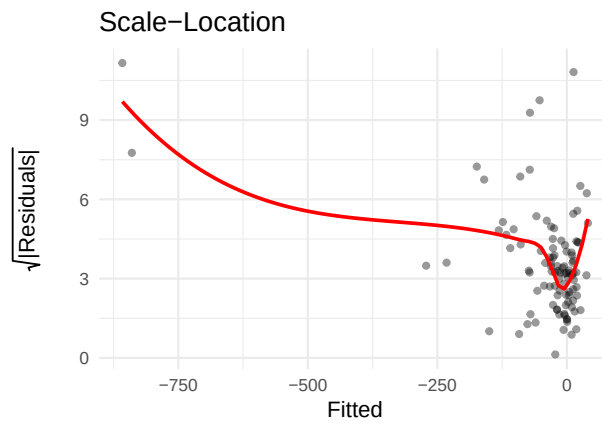
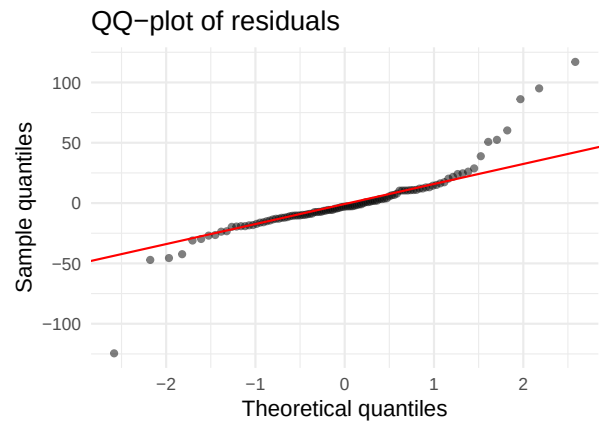
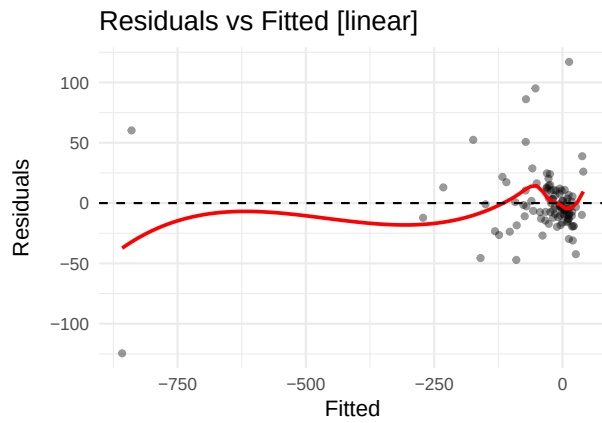


Table 32: Shapiro-Wilk — TNFSF6 (FasL)

Scale	W	p-value	Interpretation
linear	0.9412	0.0002	Strong departure from normality
log-ratio	0.9789	0.1019	No evidence of non-normality

5.31 VEGF-A (VEGFA)

AIC linear: 1077.7 | AIC log-ratio: 192.6 | Preferred: log



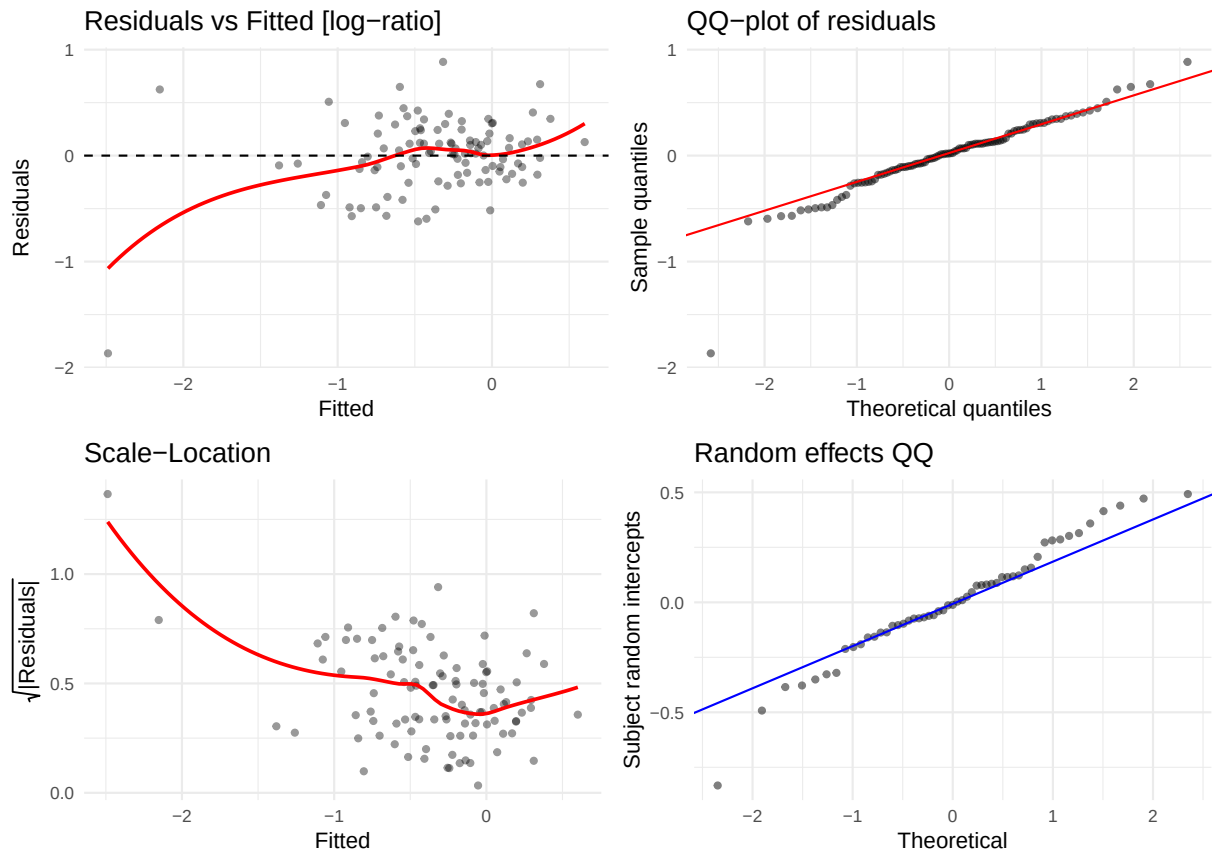
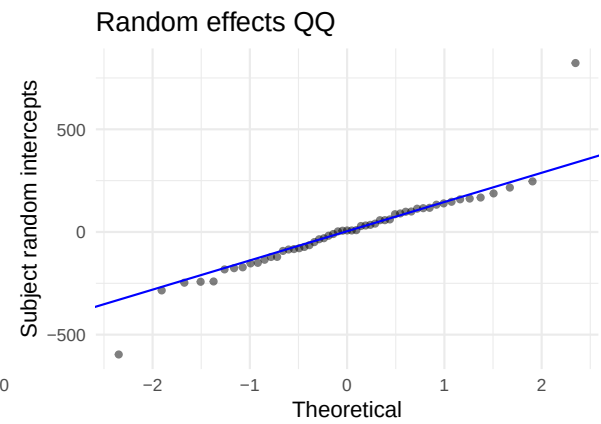
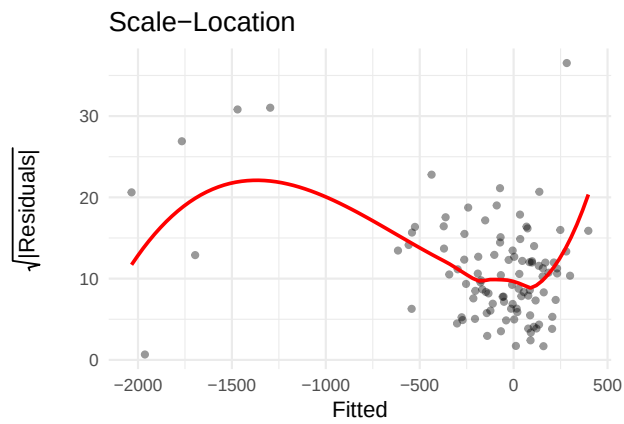
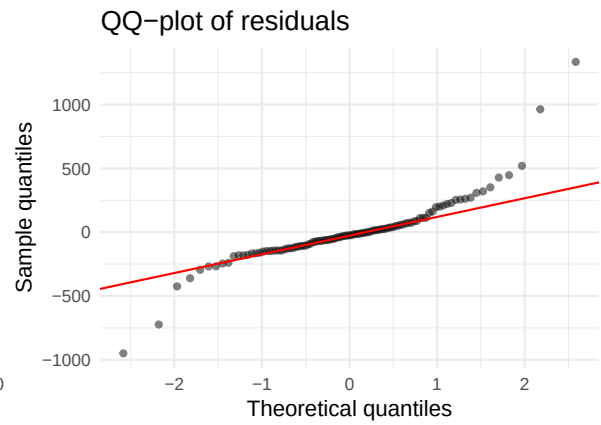
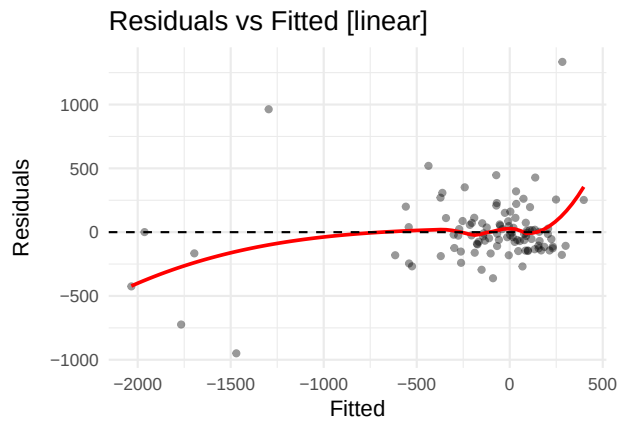


Table 33: Shapiro-Wilk — VEGF-A

Scale	W	p-value	Interpretation
linear	0.8320	0	Strong departure from normality
log-ratio	0.9126	0	Strong departure from normality

5.32 VEGF-C (VEGFC)

AIC linear: 1476.2 | AIC log-ratio: 136.8 | Preferred: log



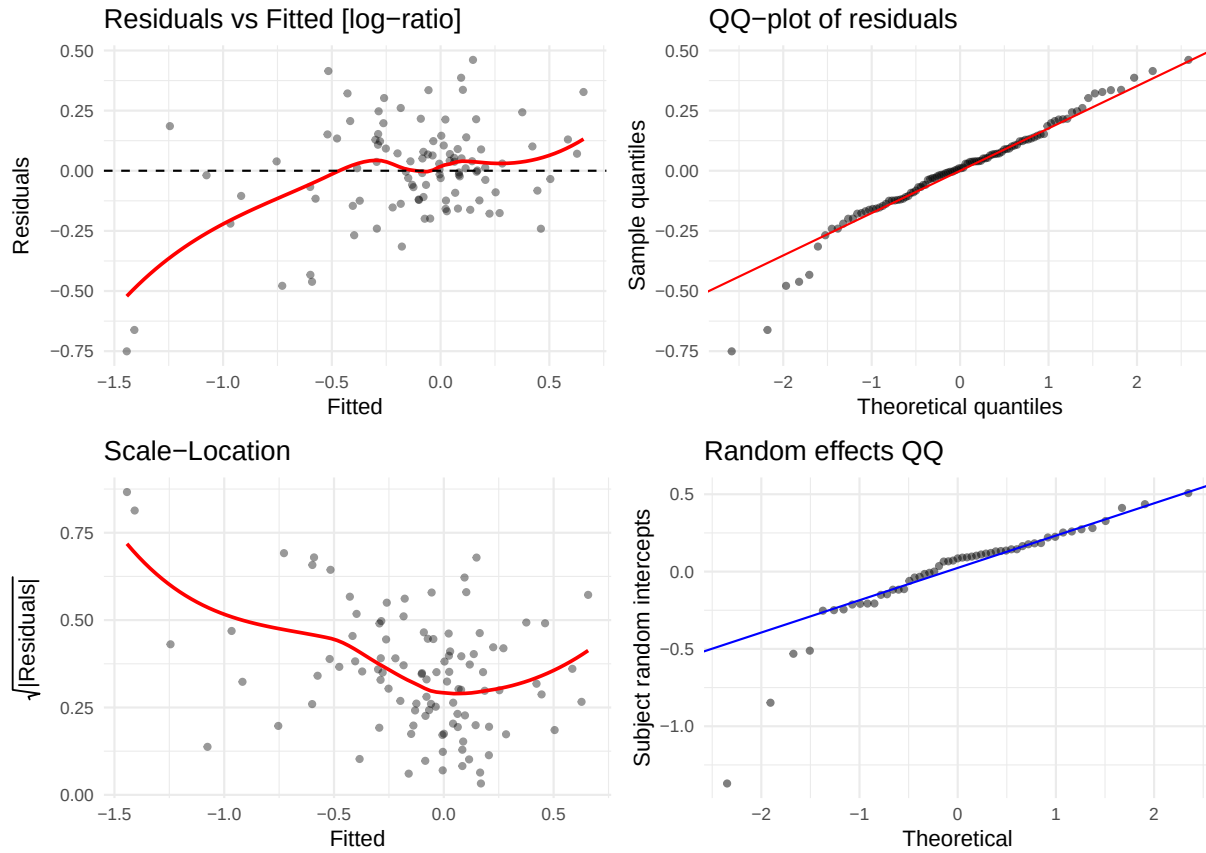
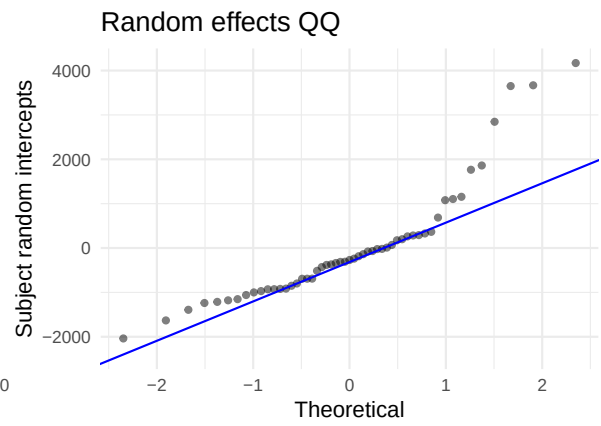
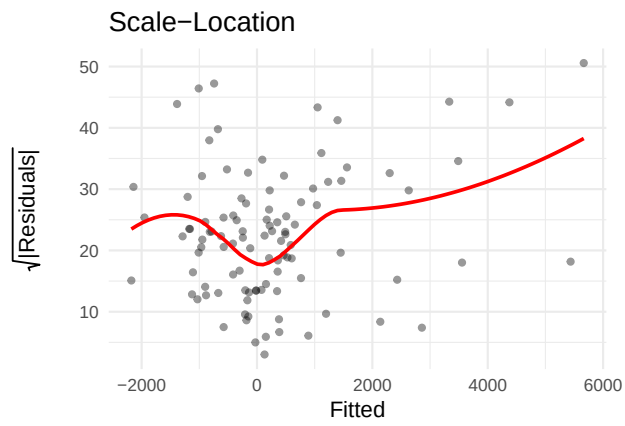
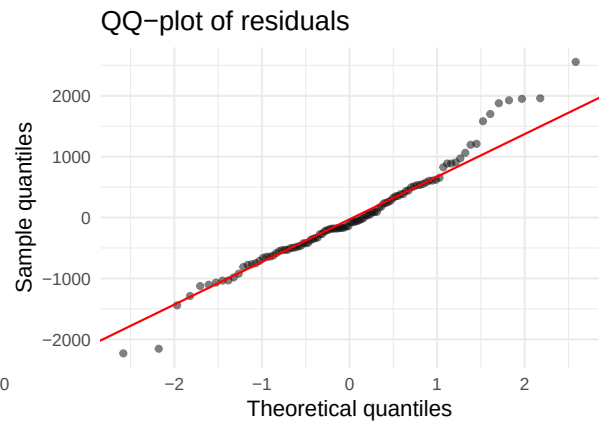
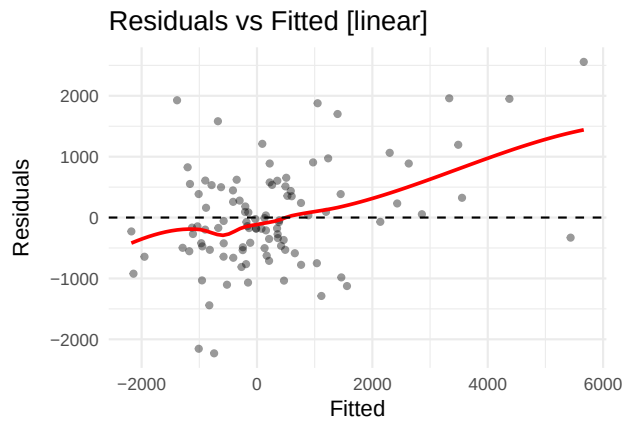


Table 34: Shapiro-Wilk — VEGF-C

Scale	W	p-value	Interpretation
linear	0.8341	0.0000	Strong departure from normality
log-ratio	0.9572	0.0022	Strong departure from normality

5.33 VEGFR2 (VEGFR2)

AIC linear: 1751.4 | AIC log-ratio: -81.7 | Preferred: log



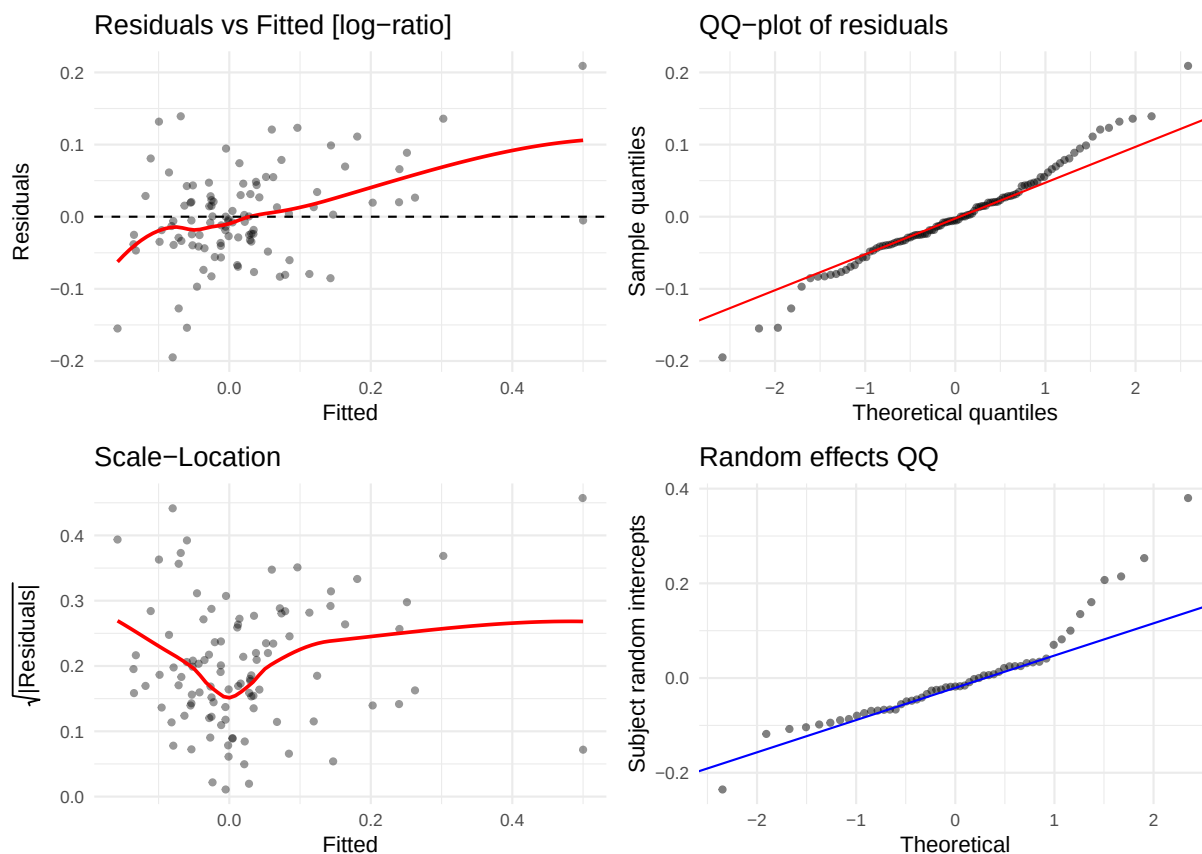


Table 35: Shapiro-Wilk — VEGFR2

Scale	W	p-value	Interpretation
linear	0.9682	0.0147	Some departure from normality
log-ratio	0.9811	0.1526	No evidence of non-normality

6 AIC comparison summary

```
##
##
## **Summary:** 33 analytes favour log-ratio; 0 favour linear; 0 log models failed to converge.
```

7 Recommendations

7.1 Log-ratio model preferred ($\Delta\text{AIC} > 2$)

Back-transform emmeans with `exp()` to obtain **fold-changes** (e.g. `exp(0.30)` 1.35 = 35% increase over baseline). Report geometric means rather than arithmetic means in tables. Document the per-analyte offset used in `lb_log_offset()`.

7.2 Linear model preferred or comparable fit

Retain the linear model. If this is driven by many OOR values collapsing variability on the log scale, flag the analyte and consider a Tobit LMM.

7.3 High >ULOQ proportion (> 20%)

For analytes with >20% of observations censored above ULOQ: the linear and log-ratio models will underestimate concentrations. Consider:

- Substituting ULOQ for >ULOQ values (imputation at upper bound) as a sensitivity analysis.
- A **Tobit mixed model** (`censReg` or `crch` packages) for the primary analysis.

7.4 Non-normality not resolved by log-ratio (Shapiro-Wilk $p < 0.01$)

If both models show strong residual non-normality:

1. Check for outliers: flag observations with $|\text{standardised residual}| > 4$ for data-quality review.
2. Report a sensitivity analysis excluding confirmed outliers.
3. Consider a **robust LMM** (`robustlmm::rlmer`) for the primary analysis.

7.5 Heavy tails (excess kurtosis > 7 in chg)

Common for VEGF-A, IL-6, IL-8 due to extreme responders. These analytes are strong candidates for the log-ratio model and outlier review.

Generated from `reports/model_fit/impress_lb_model_fit.Rmd` — sham-randomised data.